



Physical activity and your mental health

Information about how physical activity can help your mental health, and tips for choosing an activity that works for you, and how to overcome anything that might stop you from becoming more active.

If you require this information in Word document format for compatibility with screen readers, please email: publications@mind.org.uk

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What is physical activity?

Being physically active means sitting down less and moving our bodies more. Many people find that physical activity helps them maintain positive mental health, either on its own, or in combination with other treatments.

This doesn't have to mean running marathons or training every day at the gym. There are lots of different things you can do to be a bit more active. For ideas, see our information on [choosing an activity](#). We also have some tips to help you [get started](#), and information about [how much activity is healthy](#).

However, it can be difficult to be physically active, especially if you are [feeling unwell](#). We have information which you may find helpful if:

- you have particular physical or mental health considerations that you need to think about [before you start getting active](#)
- you feel like, right now, [physical activity isn't working for you](#).

How can physical activity help my mental health?

There are many studies which have shown that doing physical activity can improve mental health. For example, it can help with:

- **better [sleep](#)** – by making you feel more tired at the end of the day
- **happier moods** – physical activity releases feel-good hormones that make you feel better in yourself and give you more energy
- **managing [stress](#), [anxiety](#) or intrusive and racing thoughts** – doing something physical releases cortisol which helps us manage stress. Being physically active also gives your brain something to focus on and can be a positive coping strategy for difficult times.

"I've always found that exercise is the one thing that gets me well again, with depression, it's invaluable. It makes me feel great, just healthy and active. I don't feel as tired or lethargic when I exercise and it makes me happy and content in myself."

- **better [self-esteem](#)** – being more active can make you feel better about yourself as you improve and meet your goals
- **reducing the risk of [depression](#)** – studies have shown that doing regular physical activity can reduce the likelihood of experiencing a period of depression
- **connecting with people** – doing group or team activities can help you meet new and like-minded people, and make new friends.

But physical activity isn't always helpful for everyone's mental health. You may find that it is helpful at some times and not others, or just that [it doesn't work for you](#). For some people, physical activity can start to have a negative impact on their mental health, for example, if you have an eating problem or tend to [overtrain](#).

"Exercise ... can seem like [an] impossible challenge when you're having a tough time."

What if I'm feeling unwell?

When you're feeling unwell, it can be really hard to get started and it can be frustrating when people tell you about the benefits of being more active.

If you're in a really bad place, don't be too hard on yourself if you can't exercise. It can be easy to start feeling guilty or beat yourself up about not exercising, and this can start to contribute to feeling unwell.

You may need to focus on other things for a while, and build some physical activity into your routine once you're feeling a bit better. It's important to find a balance, and figure out what works best for you.

What type of activity might work for me?

Being physically active tends to be easier if you choose an activity that you enjoy, and that fits into your daily life. If you force yourself to do something you don't enjoy, you're much less likely to keep it going and experience benefits to your mental health.

There are lots of different things you can try – not everybody will enjoy or feel comfortable doing all of these activities, so you may need to try a few before you find something you like. You may also find that different things work for you at different times, depending on how you're feeling.

If you think you might find it hard to get going with any of these things, we have [information which may help you get started](#).

Activities at home

- **Try to sit less** – if you spend lots of time sitting down, try to get up and move around a bit every hour. If you're worried you might forget, you could set an alarm to remind yourself.
- **Chair-based exercises** – if you have mobility problems, a physical condition, or find it difficult spending time out of a chair, [the NHS website has activity routines you can try while sitting down](#).
- **Play an active computer game** – there are a few different gaming consoles you could try which involve actively moving your body while playing computer games.
- **Do exercises or stretches at home** – the [NHS website has lots of different routines](#), or you could try an exercise CD or DVD.
- **Do an online activity programme** – there are lots of free, online exercise regimes designed for you to try at home, including everything from chair-based exercises to yoga and cardio workouts.
- **Do active household chores**, like Hoovering, tidying or DIY.

- **Include more activity in your day-to-day routine** – run up the stairs instead of walking, carry your bags of shopping in one at a time or do some gentle stretching while you're watching TV.
- **Dance** – put on some music while you're cooking and dance around your kitchen, or have a mini dance party with your friends or family.

"It calms my mind, it stops me ruminating, it actively lifts my mood and it makes me feel a lot more positive about life."

Activities out and about

- **Walk a bit more** – to work, to the shops, or to the end of the road and back.
- **Play a game in the park** – for example, frisbee, tag or a game of catch.
- **Try a new sport, or join a team, group or exercise class** – the [Be Inspired](#) website has lots of information about what different sports and activities are like, and how to get involved.
- **Volunteer outdoors** – [The Conservation Volunteers](#) and [The Wildlife Trusts](#) run outdoor volunteering projects around the UK.
- **Find your local leisure centre** – leisure centres have a range of sports facilities, such as badminton and squash courts, and run exercise classes and groups, such as Zumba and aerobics. They often feel more inclusive than private gyms, and many have discount schemes and childcare facilities. Check your [local council website](#) to find your nearest centre.
- **Try a dance class** – from Zumba to swing, ballroom or dancercise, the [NHS website has a directory of classes in your local area](#).
- **Walking or running groups** – [Walking for Health](#), [Let's Walk Cymru](#), [Ramblers](#) and [Run Together](#) all organise free, inclusive local groups with trained volunteers.

"I'm not the sporty type, but I love walking. It really lifts my mood."

- **Outdoors gyms** – some local parks have free outdoors gym equipment you can use. You can try your local council website to find the location of any outdoor gyms near you.
- **Cycling** – whether riding to the shops or to work, or going on long bike rides at the weekend, the [Sustrans](#) website has lots of ideas for routes and information about safe cycling to get you started.
- **Adventure gaming apps** – some gaming apps are a great opportunity to explore outside.
- A mindful sport, such as [yoga](#), [pilates](#), [tai chi](#) or [Nordic walking](#) – the NHS website has information about what these involve and how to find classes.
- **Gardening or seated gardening** – the [Carry on Gardening](#) website has information about gardening for emotional wellbeing and with particular disabilities. If you don't

have a garden at home, the [Social Farms & Gardens](#) website has details of community garden and farms around the UK.

- **Be active in nature** – our information on [nature and mental health](#) has lots of ideas for getting active outdoors.
- **Swimming** – [swimming.org](#) has a search tool to find your local pool, information about adult swim classes and water-based sports such as aqua aerobics, aqua Zumba, water polo and synchronised swimming, as well as pool exercises you can do on your own.

“Swimming has helped me. The pool is one of my safe places now and I go twice a week. It means that I’m tired at the end of the day so I can sleep much better, and I feel happier about my body.”

Motivation and extra support

- **Music** – putting music or a podcast on your headphones can help distract, entertain or motivate you while you exercise.
- **Apps and programmes**, such as the NHS's [Strength and Flex](#) and [Couch to 5K](#), give you step-by-step programmes to follow, include information about how to exercise safely and help keep you motivated.
- **Enjoy alone time** – being active alone can provide a good way to reflect on how you're feeling or practise being mindful.
- **Online communities** – you could check in with other people who are also trying to get more active on an online community, such as Mind's community [Elefriends](#). This can help you stay motivated and connect with others in a similar situation.
- **Ask for recommendations** – some activities are more inclusive than others. Try asking your GP, friends or an online community like Elefriends for recommendations and tips.
- **Raise money for charity** – many charities, including [Mind](#), support people who want to do an active challenge, like an organised run or bike ride, and use it as a chance to raise funds and support the charity's work.
- If you **identify as female**, check out the [This Girl Can](#) website for lots of ideas, from trying a new sport to being more active as part of your day-to-day life.
- **Exercise with other people** – many people find that joining a group or getting active with someone they know – like a friend, family member, colleague or support worker – can be motivating and make a new activity more enjoyable.

“The thought of going to a gym on my own terrified me but I started going to various exercise classes with a friend. The difference it made to my mental health was incredible.”

Disability, mental health and inclusive activities

- **Disability sports** – the [NHS](#) and [Disability Sports Wales](#) websites list local organisations that offer disability sports in your area, whatever your disability or level of fitness.
- **Walking sports** – many sports are available in a walking version, such as walking football, walking hockey or walking basketball.
- **Inclusive gyms** – the [Activity Alliance](#) has information about inclusive gyms and leisure centres, which offer welcoming and accessible environments for people with disabilities. This includes a search tool to help you find an accredited inclusive gym in your area.
- **Try an NHS routine** – the [NHS website has tips and routines for people with disabilities](#) who want to get more active, as well as [fitness guides for wheelchair users](#).
- **Specific activities for people with a mental health problem** – some local Minds offer physical activity sessions through Mind's [Get Set to Go programme](#).
- **Ask for a referral to a physical activity scheme** – if you have a mental health problem, your GP may be able to refer you to a physical activity programme.

How can I start getting active?

It can be difficult to start being more active, particularly if you're not feeling well or you feel like there are things getting in the way.

These are some tips to help you get started.

How to get started

- **Start off slowly.** It may take a while to build up your fitness. Doing too much at first will make you feel tired and may put you off.
- **Plan a realistic and achievable routine.** Try to find ways to be active that fit into your day-to-day life around your commitments, or build activity into your daily life. Trying to move a bit more every day can really help.
- **Be kind to yourself.** Sometimes you can't be as active as you would like, and your energy levels will vary on different days. It's fine to slow down or take a break.
- **Try to identify your triggers and work around them.** For example, if you find leaving the house difficult or don't like to exercise in front of other people, you could try doing some exercise at home.
- **Keep trying.** It may take a while to find an activity you like. As well as trying different activities, you may find that you prefer a particular class, instructor or group.

"Although I love many aspects of running, it also brings out my capacity for guilt and self-criticism. There can be quite a lot of "no excuses" and "man up" type messages in the running world and I really struggle to keep these in perspective at times."

- **Work with your highs and lows.** If you take medication that leaves you feeling exhausted in the mornings, let yourself rest and build in some exercise later on. If you find that exercising in the evenings affects your sleep, try doing some activity earlier in the day. You may also have periods of time when you're unable to exercise because of your mental health – that's OK. Let yourself have a break if you need it, and start again once you're feeling better.
- **Have some alternatives.** If you can't be as active as you would like, it's a good idea to have alternative options that will help lift your mood. See our [information on self-care](#) for ideas.
- **Try not to compare yourself to other people.** Set your own goals based on your own abilities and what you would like to achieve. Try to pay attention to how you are feeling and the progress you are making rather than other people.

Get Set To Go

[Mind's Get Set To Go programme](#) has lots of helpful information if you want to get more active but you're finding it difficult. For example, if you:

- feel like you aren't a sporty type
- don't feel very confident doing exercise, or comfortable with your body
- feel low in energy
- don't have much money to spend on exercise.

There are also [personal stories from people who have found ways to be more active and help improve their mental health](#).

"I have done some fantastic activities [with Get Set to Go], had loads of fun and sampled sports that I've never had the chance to try before."

Free and low-cost activities

- **Find activities you can do for free.** The [NHS website has lots of ideas for getting active without spending any money](#).
- **Look for local schemes and discounts.** Some councils offer cheaper leisure centre memberships for people who want to be physically active, especially if you have a health problem or are inactive, so it's worth checking your council's website. Many private gyms also offer free trials or discounts.
- Many **local Minds** also run physical activity sessions at minimal cost – you can [contact your local Mind](#) and ask what they offer.

"Parkrun has given me the confidence to really believe in myself and keep my depression at bay."

If you need a confidence boost

- **Look for groups of like-minded people.** Some leisure centres and sports clubs provide sessions aimed at people with mental health problems. This can be a great way of boosting your confidence levels.
- **Go with someone else.** Some clubs will allow you to attend with someone you know, such as a friend, family member, colleague or support worker, for the first few sessions while you get used to the new surroundings.
- **Consider doing an activity on your own.** There's lots of options if you prefer not to be active with other people. Walking, running or cycling are great physical activity - they can help you clear your head and can be built into your daily routine.

"When I'm feeling low, I crank up the music and hit the streets; it stops me heading to a really dark place."

If you feel conscious about your body

- **Remember that you're not alone.** Most people have worries about their bodies, and other people may well be feeling self-conscious too.
- **Find an inclusive class.** There are lots of classes where you will find people of all shapes and sizes. For example, you may be able to find a friendly Zumba class in your community centre or local walking group.
- **You could look for women- or men-only sessions.** Many leisure centres and swimming pools run women- or men-only sessions, which may provide an environment in which you feel more comfortable being active.

"In my aqua classes there are people of all shapes and sizes – and honestly no one cares."

If you're struggling with your mental health

- Look for **activities specifically aimed at people with mental health problems.** Some sports and leisure providers have sessions for people with mental health problems. Your local Mind may also provides access to different types of physical activity – you can [contact your local Mind](#) and ask what they can offer.

Exercise on prescription

If you have a diagnosis of mild to moderate depression, or if you are physically unwell, your GP may be able to refer you to a physical activity programme to help improve your mental and physical health – this is also called **exercise on prescription**.

In this type of programme, sessions usually last from 45 minutes to 1 hour, and take place at least three times a week over a 10–14 week period. The sessions often take place at a local leisure centre. If you think this might work for you, talk to your GP.

[Read Laura's blog](#) about why, at times, she's found running and training for races difficult mentally.

Want to add your story? [Find out more about blogging for us.](#)

How much activity should I do?

There are lots of benefits to being more active, and any amount of physical activity can help. You don't have to begin a vigorous training plan to start feeling better.

How much activity you decide to do is personal to you. This will depend on your current level of activity and fitness, and what you can fit in with your day-to-day life.

There may also be things that affect how much activity is safe for you to do, such as medication you might be taking, an eating problem, anxiety, OCD, or a physical health condition. See our information on [what to do before you start exercising](#) if you think these things may affect you.

What's important is that you work out what feels realistic for you at the moment. This may change from time to time, depending on how you are feeling, and what you are able to do.

NHS guidelines

The [NHS website has information about how much exercise a person is recommended to do each week](#), and how intense (moderate or vigorous) this activity would ideally be. The NHS's information includes examples of activities which count as moderate or vigorous exercise.

However, it's important to remember that this is just a guide based on the average person, and it's OK if you don't feel like you can achieve this right now. The important thing is to start to try to increase your activity levels, and to find something that works for you.

Overtraining

Doing too much physical activity can be as unhealthy as doing too little. This is called overtraining or overexercise.

Overtraining can have a negative impact on both your mental and physical health. For example, it can:

- cause **injury and long-term physical damage** to tendons, ligaments, bones, cartilage and joints.
- destroy **muscle mass**.
- become **life-threatening** - if your body isn't getting enough nutrition, it is forced to break down muscle for energy.

- have a negative impact on **your relationships, mood and ability to function** in other areas of your life.

How can I spot if I'm overtraining?

It can be really hard to spot that you are starting to overtrain. Some signs that you may be overtraining include:

- **Never having a day off** – including training when injured or unwell.
- Physical activity has started to affect your **relationships, work, hobbies or other responsibilities**.
- You feel **anxious or irritable** if you miss a session.
- You constantly feel you have to **push yourself** to go further, faster or heavier.
- You **no longer enjoy it** but feel it's something you have to do.

How can I stop myself from overtraining?

If you feel like you are overtraining, it can help to:

- **Re-focus** - re-assess why you wanted to be active in the first place and what you enjoy about it.
- **Try a new activity or mix up your routine**.
- **Try a less competitive sport or atmosphere** – like tai chi, yoga or pilates, or a different class or instructor.
- **Limit your exercise time** to a healthy schedule - for example, sessions of 30 minutes, 5 times a week.
- **Listen to your body** - have some time off if you need it, or if you're in pain or unwell.
- **Take a break** - you may need to stop exercising for a while to focus on other things.
- **Seek help** - you may need to get support if you feel like overtraining is starting to have a negative impact on your mental or physical health - for example, by [visiting your GP](#).

What should I consider before I start getting active?

If you have a mental health problem, or if you're physically unwell, there may be certain things that might affect the type and amount of activity you can do. It's important to think about these before you start getting active, to make sure what you're doing is safe.

Medication

Some [medication](#) can cause side effects that affect the type and amount of physical activity that is safe for you to do. Always check with your GP before you start a new routine, or if you change your medication or dose.

For example:

- Some [antidepressants](#) can cause dizziness, high or low blood pressure, or affect your heart rate.
- [Antipsychotics](#) can cause muscle spasms, disturbed heart rhythm and palpitations, drowsiness, blurred vision or dizziness.
- [Beta-blockers](#) slow your heart rate and lower your blood pressure, so your heart will work harder when exercising. You may need to adjust how you exercise to avoid becoming exhausted by this.
- If you take [lithium](#), you should check with a GP before doing any physical activity. This is because losing fluid from your body during exercise (for example, by sweating) can sometimes increase the concentration of lithium in your blood to a harmful level.
- [Tranquillisers](#), for example [benzodiazepines](#) like Diazepam, can slow your reaction times, or cause drowsiness, dizziness or unsteadiness.

Anxiety or panic attacks

If you experience [anxiety or panic attacks](#), you might find that some of the physical sensations you get while exercising, such as raised heart rate, feeling shaky or dizzy, breathlessness or feeling hot, can feel similar to a panic attack. This can then cause you to feel anxious, and may cause a panic attack.

If you experience this:

- **Start off slowly.** This may help you spot the difference between the effects of physical activity and a panic attack.
- **Do a gentler activity.** An activity that focuses on strength and stretching, such as yoga or tai chi, may work better for you than one that requires more intense exercise.
- **Take deep, slow breaths.** This may help stop you hyperventilating. Focus on breathing out.
- **Avoid triggering situations.** For example, if you want to avoid crowds or travelling, you could go jogging or walking in a local park, or try exercising at home.

Eating problems

Many people with [eating problems](#) have a complex relationship with exercise, and [overtraining](#) can become an unhealthy part of your condition. However, physical activity can still be a positive part of your recovery – you may just need to be more careful about the type and amount of activity you do.

If you have, or are recovering from, an eating problem, it is a good idea to talk this through with your GP before you start an activity.

[Read Angela's blog about taking on a difficult trek to Iceland while living with an eating problem, depression and anxiety, and how the challenge helped her mental health.](#)

Compulsive or addictive feelings

Some people experience compulsive or addictive feelings about physical activity (sometimes called an exercise addiction), which can lead to harmful [overtraining](#). These feelings can be a form, or a symptom, of [obsessive-compulsive disorder \(OCD\)](#), or part of an [eating problem](#).

If you tend to experience compulsive or addictive feelings about exercise, or start to experience them once you start doing more physical activity, it is a good idea to talk to your GP about how to manage this or think about [seeking help](#).

Physical health conditions

For many people with a physical health condition, doing an appropriate amount of physical activity can be an important part of managing your condition and avoiding future health problems. However, depending on your condition, you may need to be more careful about the type and amount of activity you do, to make sure what you are doing is safe and won't have a negative impact on your health.

You should be particularly careful if you have:

- high blood pressure
- chest pains
- a heart condition
- diabetes
- are pregnant or have recently given birth
- an injury.

Always check with your GP about what is safe for you before you start any physical activity.

Physical Activity Readiness Questionnaires

Some sports clubs, gyms and leisure centres may ask you to complete a short health questionnaire before letting you use the facilities and equipment (often referred to as a PAR-Q, or physical activity readiness questionnaire). These questionnaires focus mainly on making sure you are physically healthy enough to use the facilities and activities that the club, gym or centre offers, so you shouldn't be asked to disclose any mental health problems as part of the questionnaire.

But if you are concerned that a mental health problem may affect your ability to do any activities, you can still mention this on the questionnaire. You could also ask to discuss this with somebody who works at the club, gym or centre, if you think that will put you at ease about starting an activity.

Depending on your answers, you may be asked to have a check-up with your GP before participating in any activities. This may seem overcautious, but it's really important to make sure if it's safe for you to do more physical activity, and if so, what type and amount of activity is appropriate for you.

What if getting active doesn't work for me?

While many people find physical activity helpful, not everyone does. You may find that there are times when it is helpful, and times when it isn't.

For example:

- **You may not always be able to exercise** – if you are unwell, you may need to focus on looking after your mental health in other ways.
- **It may not help** – there may be days, weeks or months when physical activity doesn't make you feel better, and you may need other types of support.
- **For some people, exercise can make mental health worse** – it can trigger [anxiety](#), be part of a mental health problem like [obsessive-compulsive disorder \(OCD\)](#) or an [eating problem](#), or you may start to [overtrain](#).
- **You may be taking medication or have a physical health condition** that means you can't exercise or need to take particular care when doing any physical activity, either for a while or longer term.

If you find that physical activity isn't working for you right now, there are a few things you can do:

- **Try changing your routine, or doing a different type of activity.** Different things work for different people at different times – [there are lots of activities you can try](#).
- **Do what you can when you can.** It's completely normal to have days when you wake up excited about going for a run, and other days when walking upstairs feels like a challenge. It's OK to adapt your physical activity to how you're feeling.
- **Be gentle with yourself.** If you don't manage to do what you were planning, that's OK. Have a break, and try again when you're feeling better.
- **Try out some [other ways of caring for yourself](#),** like relaxation, mindfulness and getting into nature.

- If you're struggling to manage your mental health on your own, [seek help](#). You might want to talk to your GP about possible treatments, such as [medication](#) or a [talking therapy](#).
- If you're finding that exercise is having a negative impact on your mental health, **you may need to take a longer-term break** until you're feeling better.

If you've tried being physically active and it hasn't helped, **it's important not to blame yourself**. Looking after your mental health can be really difficult, especially when you're not feeling well. It can take time, but many people find that when they have the right combination of treatments, self-care and support, it is possible to feel better.

Useful contacts

Mind's services

- **Helplines** – all our helplines provide information and support by phone and email. Our Blue Light Infoline is just for emergency service staff, volunteers and their families.
 - Mind's Infoline – 0300 123 3393, info@mind
 - Mind's Legal Line – 0300 466 6463, legal@mind
 - Blue Light Infoline – 0300 303 5999, bluelightinfo@mind
- **Local Minds** – there are over 140 local Minds across England and Wales which provide services such as [talking treatments](#), [peer support](#), and [advocacy](#). [Find your local Mind here](#), and contact them directly to see how they can help.
- **Elefriends** is a supportive online community for anyone experiencing a mental health problem. See our [Elefriends page](#) for details.

Who else could help?

Active Partnerships

activepartnerships.org

A network of 43 local organisations across England that support sport and physical activity opportunities for participants, coaches and volunteers. Has a search tool to find your nearest partnership, with opportunities to be active in your local area.

Activity Alliance

activityalliance.org.uk

Supports disabled people to become more active. Provides information about how to start, where to find inclusive programmes and gyms, and personal stories about getting more active with a disability.

Be Inspired

beinspireduk.org/play

Guides on how to get started with a range of different sports, including how to find a local organisation and clubs, as well as ideas for exercising on a budget, and with a disability.

Carry on Gardening

carryongardening.org.uk

Practical tips for gardening for emotional wellbeing, especially for anybody with a disability.

Disability Sports Wales

disabilitysportwales.com

Provides a search function to help you find organisations and classes in Wales offering inclusive activities for people with disabilities, including behavioural and learning difficulties.

Get Set to Go

getsetto.go.mind.org.uk

Mind offers physical activity programmes for people with mental health problems through local Minds and online.

Let's Walk Cymru

letswalkcymru.org.uk

Organises groups walks and provides walking information for Wales.

NHS Live Well

nhs.uk/live-well

Offers resources on physical activity, including tips on being more active, exercise programmes and the Couch to 5K app, and activity ideas if you have a physical disability or if you're on a budget. Includes information on finding classes for walking, cycling, dance, yoga, pilates and Tai Chi.

parkrun

parkrun.org.uk

Organises free, weekly, 5km timed runs across the UK, which are open to everyone.

Ramblers

ramblers.org.uk

Organises group walks in cities and the countryside, and has information about walks and getting started.

Run Together

runtogether.co.uk/groups

A directory of running groups across the UK. Supports mental health via a network of Mental Health Champions and #runandtalk events.

Social Farms & Gardens

farmgarden.org.uk

Has a directory of community farms and gardens around the UK.

Sustrans

sustrans.org.uk

Provides information about taking up cycling, including information about cycle safety, and ideas and maps for local rides.

Swimming.org

swimming.org

Has a search tool to find your local swimming pool, information about adult swim classes and water-based sports, as well as pool exercises you can do on your own.

The Conservation Volunteers (TCV)

tcv.org.uk/greengum

Runs free outdoor sessions working on conservation projects around the UK.

The National Trust

nationaltrust.org.uk

Outdoor spaces around the UK.

The Wildlife Trusts

wildlifetrusts.org/closer-to-nature/volunteer

Runs outdoors volunteering opportunities around the UK.

The Woodland Trust

woodlandtrust.org.uk/visiting-woods

Has information about woodland walks and activities around the UK.

This Girl Can

thisgirlcan.co.uk/activities

Ideas to help anyone who identifies as female become more active, providing information about a wide range of sports and other activities, including ways to add more activity in your day-to-day life.

Walking for Health

walkingforhealth.org.uk

Supports a network of walking groups across England.

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References are available on request.

Chapter 10

Social Support, Relationships, and Physical Activity

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Chapter Overview

We live in a social world where our interactions with others influence our physical activity behavior. A parent may take their child to the park to play. A spouse may encourage their partner in reaching their goal of running a half marathon. A person may keep going to an exercise class because they enjoy the people in that class. An older adult may be more likely to go for a walk if they have someone to walk with them. In this chapter, we will explore the different ways that other people influence our physical activity. First, we will explore social support, which is probably the area that has received the greatest attention in the research. We will also look at other types of social influences including social norms and social control. We will also explore the role of groups in our physical activity. We will examine how these social influences relate to physical activity and ways they have been used to promote physical activity. At the end of this chapter, we will learn about some of the many ways we measure these social influences. While we are focusing on physical activity, these social relationships also influence our health and well-being. For more discussion on social support in the context of rehabilitation from sport injury, see Chapter 31 (Griffin et al., 2021).

Social Relationships and Health and Well-being

Social Networks and Health

The recognition of how our social relationships influence our health and well-being has been around a long time. Durkheim in 1897 published a book called *Suicide* that described how a lack of social integration or a lack of connection with others was associated with increased rates of suicide (Berkham et al., 2000). When looking at risk of dying (mortality), there is a decreased risk of dying when someone has a strong social network compared to a poor social network (Holt-Lunstad et al., 2010). The reduction in mortality risk associated with high levels of social support versus low level of support is similar to that reduction of risk by quitting smoking, and larger than risks associated with other conditions, such as obesity (BMI; Holt-Lunstad et al., 2010). There are many pathways in which social networks or social contacts can influence one's health, including influencing one's behaviors (e.g., diet, exercise, smoking), how one feels (e.g., decreasing depression, improving self-esteem, helping us cope) and even improving one's physiological functioning (e.g., improving immune function; Berkham et al., 2000).

The effects of social networks on health are often separated into (a) the *structural aspects* (i.e., how many people in your social network and how often you interact with them) and (b) the *functional aspects* (i.e., the content of those interactions in terms of social support provided; Umberson et al., 2010). Examples of this functional support include providing social support, creating social norms or expectations for how one is to act, and helping to regulate behavior (e.g., social control). These interactions with others can influence both mental health as well as health behaviors (Umberson et al., 2010). Most of the social influences covered in this chapter will look at the functional aspect of social support.

What is Social Support?

Social support is described as the resources that one has available or perceives available to them from their social relationships (Gottlieb & Bergen, 2010). These resources are the ways that other people can help a person. One example of social support would be realizing there is a friend that someone could invite to be physically active with, while another person may rely on a spouse to watch their children, so they can be active themselves. Another person might value having a person motivate them so they can reach their goals. Different people may be able to provide different types of support for an individual to help them be physically active.

There are two general ways to view social support: (a) *perceived support* and (b) *received support* (Gottlieb & Bergen, 2010). Perceived social support is the support one perceives available from others regardless of whether they use it or not. Someone who knows that they could call a friend would perceive that support is available to them if they were to need it. Received social support is the support that others provide to a person. An example of this would be when a teen reports that their parent encouraged them to keep going on a run. This is the support that has been provided often in response to stressful situations. While both types of social support have been related to health behaviors, perceived support may help individuals develop positive coping skills and subsequently decrease the development of chronic disease (Uchino, 2009). Knowing that support is available if needed, regardless of whether one uses it or not, is important for engaging in health behaviors such as physical activity.

Models of Social Support

When looking at how social support may influence health behaviors and health, several different models have been suggested. These models include a main-effect model, stress-buffering model and more recently the thriving through relationships model. Each model specifies a different path through which physical activity may be influenced by our social relationships.

Main-Effect Model

The view that social support has a direct effect on well-being is called the main-effect model (Cohen & Wills, 1985). This model is closely tied to the idea that large social networks are associated with positive outcomes such as well-being and health. According to this model, the social support received has a protective effect on an individual as a whole. In this model, social support directly influences physical activity regardless of whether a person is experiencing stressful events. For example, someone driving a friend to the gym would be having a direct effect on their physical activity. In this example, the social support directly helps someone engage in physical activity. The direct effect on behavior and well-being occurs directly as a function of the social support provided.

Stress-Buffering Hypothesis

The stress-buffering hypothesis focuses on the role of social support in helping individuals cope and manage stress (Cohen & Wills, 1985). Stress is experienced when a person does not think they will be able to meet the demands of the situation (Lazarus et al., 1984). This perception of a threat leads to the experience of stress including feelings of worry and helplessness. It also may lead to negative changes in behaviors including less exercise, alcohol abuse, and poor diet (Cohen & Wills, 1985). Social support may help a person in these potentially stressful situations in two ways. First, with support, they may not perceive that situation as stressful. Someone who knows that they have a friend that they can call on for help may not find a situation as stressful. When in the potentially stressful situation of juggling the demands of being physically active and their job, a friend providing encouragement may help someone realize they can handle the situation. The second way that social support could help in these potentially stressful situations is that support may provide added resources that will help them cope with the situation. For example, someone balancing being active and other demands finds this situation stressful. By having a co-worker help them balance the job demands, they may now have the resources to cope with the situation and be able to fit exercise into their schedule. In this instance, social support helps a person cope with the situation by providing resources. In this model, social support plays a role in protecting someone from the negative effects of stress.



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Thriving Through Relationships

A newer model of social support put forward by Feeney and Collins (2015) focuses on how to help people thrive. This model looks at how social support plays a role in someone's overall well-being. This model focuses on two pathways in which social support influences one's well-being: (a) in the face of adversity and (b) seeking opportunities.

In the first pathway, the focus is on helping an individual deal with life adversity. This pathway is similar to the stress-buffering pathway. It looks at how social support can help someone cope with challenges they are facing. In this case, the support person serves as a *source of strength* and provides support in a way that may help someone deal with challenges. Think about someone who is trying to return to being active following an injury. In this case, they might need the encouragement and support from someone as they are just starting out. The support may take the form of emotional comfort such as listening to someone's fears about returning to being active. The support also could take the form of helping the individual build up their abilities by assisting them with activity modifications that they could do. Another way that support could occur during this return following injury would be to help motivate the person as they are returning physical activity. In these instances, the support provided is helping someone cope with the stress of returning from injury and the support person serves as a source of strength to the recipient. Also see Chapter 31 (Griffin et al., 2021).

The second pathway involves support that encourages people to seek out opportunities. In this manner, social support from others would encourage someone to set goals and push someone to grow, develop and thrive. In this pathway, the person is called a *relational catalyst* as they are helping someone seek out new opportunities. Many of the examples in the physical activity context may relate to this pathway of support. The support provided could take the form of encouraging someone to reach goals they might not usually set for themselves, such as running a half marathon, or helping a person to push themselves outside of their comfort zone and try something new. Other examples of this support include helping the individual reach their goals by helping them plan (e.g., creating an exercise program with them), and providing assistance (e.g., driving someone to a fitness facility or watching their children while they exercise). Within this pathway, the support involves helping someone to develop and grow as they strive to achieve their physical activity goals.

Learning Exercise One

What are the main differences between a person serving as a "source of strength" and a "relational catalyst" when providing social support?

Social Support and Physical Activity

Social support has frequently been examined as a correlate of physical activity. Several reviews have suggested that there is a positive relationship between social support and physical activity (Scarapicchia et al., 2017; Smith et al., 2017). In a review of 20 prospective studies that examined the role of social support for future physical activity participation, there was a small positive effect for social support, meaning that those who reported more social support at one time tended to be more active at a later point in time (Scarapicchia et al., 2017). Two of the most common sources for social support examined were from family, friends, and significant others. However, in this review, there was a lot of variation in results. For instance, support from family appeared to be related to physical activity, but the studies looking at friend social support were inconsistent. In a review of 27 studies examining social support in older adults, family-provided social support showed a positive association with physical

activity participation (Smith et al., 2017). In a review of 75 articles examining social support in adolescents, an overall positive association with physical activity was reported for social support provided by parents, family, and friends (Mendonca et al., 2014). These reviews support the positive effects of the provision and receipt of social support on physical activity across the lifespan (e.g., Mendonca, et al., 2014; Scarapicchia et al., 2017, Smith et al., 2017).

Several studies have started to look at social support that is provided on a day-to-day basis. Using the thriving model of social support, Berli and colleagues (2018) examined how daily support from a spouse influenced one's physical activity level that day. Couples who were previously inactive participated in the study with the goal of becoming more physically active. In this study, people completed an average of 25 more minutes of physical activity on days when their spouse provided support towards their goals than on days when they did not provide support. They also reported more physical activity on days when both partners engaged in physical activity together. This joint activity may reflect a type of companionship support where the partners have someone to be active with. In another study, daily support by spouses was related to increased perceptions of confidence and physical activity in individuals with type 2 diabetes (Khan et al., 2013).

One population in which social support for physical activity is particularly influential is cancer survivors (McDonough et al., 2020). Physical activity settings provide the opportunity for social support both for being active as well as coping with cancer. This is an example of how—in the case of adversity, surviving cancer—the social support is vital for coping with stress and overall well-being (McDonough et al., 2020). In a review of 39 qualitative studies, cancer survivors described many ways their social relationships supported their physical activity, such as through encouragement, accountability, and companionship (McDonough et al., 2020). Support for coping with cancer was also commonly described across several studies; this included strategies such as understanding the cancer experience, being able to talk about cancer, normalizing cancer, and modeling living after cancer.

Social support, especially if it is from family, appears to have a positive influence on physical activity behavior (Scarapicchia et al., 2017; Smith et al., 2017). Most of the research has looked at social support as a whole and has not differentiated the types of social support (e.g., Scarapicchia et al., 2017; Smith et al., 2017). However, as people support each other in many different ways, researchers look at the role of specific types of social support, such as instrumental support (e.g., tangible aid), emotional support (e.g., encouragement and caring), and companionship (Golaszewski & Bartholomew, 2019).

Types of Social Support

Given that social support has big potential to influence physical activity behavior, what does it involve? There are several different ways of classifying social support. These different types of social support pertain to what is the actual content of the social support. Is someone providing information or encouragement, or are they someone to be active with? While there are different ways of conceptualizing the types of social support, (Gottlieb & Bergen, 2010; Golaszewski & Bartholomew, 2019), some common types are described below in Table 10.1.

Research in the physical activity area often combines social support into a summary score and does not differentiate the types of support (Beets et al., 2010; Scarapicchia et al., 2017). Among studies examining social support and youth physical activity, social support is sometimes separated into tangible support (e.g., instrumental and companionship) and intangible support (e.g., emotional and informational; Beets et al., 2010). Both types of support have been related to youth's physical activity. Using a newly developed measurement tool, Golaszewski and Bartholomew (2019) reported that five different types of support (emotional, companionship, instrumental, informational, and validation support) were positively related to self-reported physical activity behavior, with companionship and informational support showing the strongest relationships. While social support can influence physical activity in many ways, social support is not the only social influence on physical activity.

Table 10.1

Examples of Different Types of Social Support (Gottlieb & Bergen, 2010; Golaszewski & Bartholomew, 2019)

Type of support	Definition	Examples
Emotional	Provision of caring and encouragement	Helping someone keep going when they are struggling to reach a goal Cheering a person on or listening to challenges that someone is facing
Instrumental	Tangible aid such as the provision of resources or services	Someone helping a friend by driving them to the gym. A parent purchasing sport equipment for their child Spotting another person as they lift weights
Informational	Provision of advice, feedback, or help problem-solving	Advice from a friend on how to overcome a plateau in their fitness goals Enlisting the help of a partner to problem-solve how to deal with a particular barrier for physical activity
Companionship	Having someone to be active with and spend time with	Having someone to go for a walk with or lift weights with
Esteem	When others make someone feel like they matter, are valued, and accepted	Someone commenting on how much they value being active with a person Helping another feel like they are needed
Validation	Provides social comparison in terms of behaviors and norms	An individual who feels sore after just starting to work out, may seek out validation support from others who also were working out to see if this “soreness” is normal

Learning Exercise Two

Think about your own physical activity and provide an example of each type of social support that you have experienced. Which type do you think made the biggest difference in your own physical activity?

Other Types of Social Influences

Social support is just one of the ways that social relationships may influence physical activity. Several other common social influences include *social norms* and *social control*. These are described below.

Social Norms

Social norms reflect perceptions surrounding what is appropriate behavior (i.e., either typical or desired behavior; Cialdini et al., 1991). There are two primary types of social norms that have been investigated relative to physical activity behavior. Descriptive norms are based on what other people do (Cialdini et al., 1991). For example, if a person's friends are all going on a hike, they likely would feel pressure to go with them. Injunctive norms are norms that are based on what is viewed as desirable behavior by others (i.e., what behavior is acceptable; Cialdini et al., 1991). An example of this would be engaging in physical activity because someone thinks it is what their family thinks they should do. Previously an overlooked type of social influence, social norms seem to influence physical activity behaviors more than previously thought (Priebe & Spink, 2011). When asked reasons for being active, people typically report health- or appearance-related reasons and are less likely to report being active because others are too (Priebe & Spink, 2011). However, perceiving others to be active was related to physical activity participation; this leads the suggestion that descriptive norms are important for influencing physical activity behaviors (Priebe & Spink, 2011).

The evidence for the effects of norms on our physical activity behavior has continued to grow. In a study looking at social norms for using the stairs among university students, both descriptive and injunctive norms were related to stair use (Crozier, 2019). Using data from an online exercise-focused database, people were more active when they had more friends in the database and their friends reported more physical activity (Carpenter & Amaravadi, 2019). In efforts to predict physical activity and healthy eating in a sample of African American church members, researchers found that a descriptive norm for physical activity was positively associated with physical activity engagement, even after controlling for individual variables (e.g., sex, employment status, risk perceptions) and neighborhood variables (e.g., access to parks, presence of sidewalks; Heredia et al., 2020). These three studies illustrate the potential influence of descriptive norms on physical activity behavior.

In addition to being related to physical activity behavior, social norms have been related to other important outcomes. Researchers have reported that performance on a muscular endurance task (i.e., a maximal plank hold task) was greater when people were provided information that others held the second plank for longer (Priebe & Spink, 2014). With this descriptive normative message, people also felt more confident that they could hold their second plank longer (Priebe & Spink, 2014). Another study showed that participants receiving normative information (e.g., "97% of those tested improved their balance") had better standing balance than those who received no information (Spink et al., 2019). Studies such as these have used experimental designs and manipulated the normative messaging, leading to changes in perceptions and behavior. With the potential to manipulate norms, others have examined the effects of using norms to promote physical activity on a larger scale.

Using norms to promote physical activity is receiving increasing attention. For example, messages that focused on norms have been used to prompt stair usage (e.g. "We're doing it! Are you? Join staff and students in taking the stairs today" and "Most staff and students think we should increase stair use"; Crozier, 2019, p. 103). During the period when the norm messages were displayed, stair use increased by 3.1%. To examine the effects of changing physical activity norms in a care center for older adults (age range: 80–95 years), researchers used a news report that highlighted a descriptive norm: "Increasing numbers of exercising older adults in the local community" versus a control story that just referred to "older adults in the local community" (Koeneman et al., 2017). Compared to the control group, older adults exposed to the normative message reported greater intention to take part in the physical activities offered at the care center and reported participating in more physical activity three months later (Koeneman et al., 2017). In another study, pedometers were used to provide descriptive norms by giving information about the group average step count for participants, along with indicating if the person was above or below the average (Wally & Cameron, 2017). Compared to the no-message control group, the group that received this messaging reported more steps (Wally & Cameron, 2017).

The potential impact of norms such as what others are doing has been shown to be related to physical activity across a wide variety of populations (e.g., University students, general population, African American church members, and older adults) and a wide variety of behaviors (e.g., stair use, step counts, muscular endurance and balance tasks). While people may not recognize the role of norms in their own behavior, the research suggests that perceptions of norms (i.e., number of people active) and normative messaging are related to physical activity behaviors.

Learning Exercise Three

Describe the descriptive norms and injunctive norms that you see from your family and friends as they relate to physical activity behaviors, such as exercise or active choices (e.g., taking the stairs).

Social Control

Another way that social relationships influence physical activity is through the provision of social control, which reflects the efforts by one person to monitor, prompt, or persuade another to engage in a desired behavior (Craddock et al., 2015). Social control is viewed as a more active influence where the person providing the influence is attempting to change or control the behavior of the recipient. Much of the research on social control in the physical activity domain has been with older adult populations (e.g., Newsom et al., 2018) or adults at risk or with chronic conditions such as obesity (Scholz et al., 2021) and diabetes (Kahn et al., 2013).

While social control has not received as much attention as social support with respect to physical activity behaviors, it does appear to have the potential to exert both a positive and negative impact depending on the type and other contextual factors surrounding the influence (Craddock et al., 2015; Scholz et al., 2021). Factors such as social support (Kahn et al., 2013) or experiences of emotions (Newsom et al., 2018) may lead to different responses to social control. Positive types of social control (e.g., more encouraging forms that prompt the behavior) appear to be related to positive outcomes (Craddock et al., 2015; Scholz et al., 2021). In contrast, the more pressuring form of social control, often called negative social control, may have a negative or no effect on the behavior (Craddock et al., 2015; Newsom et al., 2018). In a sample of couples who were overweight and intending to become more active, positive social control was associated with more physical activity and feeling more positively while negative social control was related to feeling more negatively and reporting doing the opposite behavior or hiding that they were not active (Scholz et al., 2021). *Behavioral reactance* is a common negative response in which an individual performs the opposite behavior or hides the unwanted behavior in response to the pressuring type of social control (Scholz et al., 2021).

Another relationship where social control has been examined is the parent–child relationship (e.g., Wilson & Spink, 2011). Parents have been identified as key influences for children’s physical activity (Gustafson & Rhodes, 2006). As parents guide their child’s physical activity behavior, they may exert social control. In a qualitative study, parents and adolescents described parents using social control to regulate their child’s physical activity level during a physical activity lapse (Wilson et al., 2014). When parents viewed their child spending too much time in front of a screen or when parents viewed a lack of motivation for physical activity, they described acting to prompt their child to engage in physical activity through the use of social control strategies. This indicates that parents may use social control as a type of regulation to prompt children to be active when they are perceived as not meeting the desired amount of physical activity (Wilson et al., 2014). In another study, adolescents (age 12–18 years old) reported an increase in their physical activity after an activity lapse when they perceived their family exerting more collaborative social control (e.g., offering to be active with them; Wilson & Spink, 2011).

Parental Influences

Parents have been identified as playing a vital role in their children's physical activity. Previously, we explored the role of parental social support (Beets et al., 2010) and social control (Wilson & Spink, 2011; Wilson et al., 2014). A parent may provide opportunities for their child to participate in physical activity by, for example, providing a bicycle or ball for them to play with. A parent may encourage and support their child as they are participating in physical activity. A parent may also serve as a model for the child's behavior by being active themselves. Several reviews have highlighted the positive relationship between parental encouragement and physical activity in children (Beets et al., 2010; Gustafson & Rhodes, 2006). Across these different reviews, the potential of parents to influence their child's physical activity has been consistent. Given the diversity of ways that parents may influence their children's physical activity, there has been an effort to conceptualize the influences. Mâsse and colleagues (2017; 2020) grouped parental influences into three broad types: (a) autonomy promotion, (b) structure, and (c) neglect and control.

Autonomy promotion reflects efforts by parents to support their child's independence as well as promote physical activity through encouragement (Mâsse et al., 2017; Mâsse et al., 2020). This classification also has been called autonomy support (Mâsse et al., 2017). Many aspects of these parental influences focus on guiding the child to choose to engage in physical activity. Encouragement has been reported as a strategy that has consistently been related physical activity behavior (Gustafson & Rhodes, 2006; Beets et al., 2010). Often this encouragement from parents has a stronger effect among younger children than adolescents as it is thought adolescents may be more influenced by peers than parents (Gustafson & Rhodes, 2006). The other aspect of autonomy promotion appears to be praise and rewards provided by parents (Mâsse et al., 2020). This type of parental influence often has been combined with parental support measures and not examined on its own (Beets et al., 2010). Across several studies, the encouragement and reinforcement provided by parents appears to be a key type of parental influence for encouraging children to engage in physical activity (Beets et al., 2010).

Structure refers to parents creating an environment that promotes physical activity (e.g., co-participation, facilitating physical activity, modeling; Mâsse et al., 2017). When parents engage in physical activity, their children are also more likely to participate in physical activity (Beets et al., 2010). In school-aged children, both mothers' and fathers' physical activity was related to their child's physical activity on both the weekend and weekdays (Fuemmeler et al., 2011). In their review of parental influences, Gustafson and Rhodes (2006) identified modeling as a potential type of parental influence; however, this form of influence was not always related to children's behavior. In preschool-aged children, parents' activity was related to their child's activity when they reported being together as opposed to times when they were apart (Keyes & Wilson, 2020). This suggests that parents being active on their own (e.g., going to the gym without their child) may not have the same influence as parents being active with their child (e.g., going for a walk together). Another example of this type of parental influence may come from parental provision of collaborative social control that involves co-participation and helping children learn the skills to be active (Wilson & Spink, 2011). This type of social control is the type that appears to be most promising in prompting children to participate in physical activity (Wilson & Spink, 2011). The beneficial effects of parents and children co-participating in physical activity has been seen across several studies (Beets et al., 2010). Children were more likely to meet the recommended physical activity standards when their parents were directly involved in the physical activity, co-participated, or watched their child participate in physical activity (Beets et al., 2010). This active participation by parents appears to be a key form of parental influence.



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Neglect and control reflect pressures by parents to prompt their child to be active without focusing on the child's interest (e.g., nagging the child; Mâsse et al., 2017). This strategy includes pressuring influences such as nagging, punishing, or using threats to prompt physical activity. Negative social control reflects parents nagging and pressuring them to be active and it does not appear to have a strong influence on physical activity behaviors unless it is perceived as supportive by adolescents (Wilson & Spink, 2010). Another aspect of this type of parenting strategy is that the parent does not encourage physical activity at all and leaves it up to the child to decide if they want to engage in physical activity or not (e.g., permissive strategies; Mâsse et al., 2017). While not directly looking at parenting strategies, parenting styles such as uninvolved and permissive styles may provide some idea of how this absence of strategies may relate to physical activity (e.g., Hennessey et al., 2010). One study reported that children (aged 6–11 years) were the least active when they had uninvolved parents (low demandingness and low responsiveness) when compared to other parenting styles (Hennessey et al., 2010). The strategies identified by Mâsse et al (2017) may have less of an impact on physical activity behavior than the other strategies.

While much of what we know about the different types of parental influences can be classified into these three broad types, it is still a relatively new taxonomy. There is a lot of support for the benefits of the autonomy supportive strategies and the structure that parents may provide (Beets et al., 2010; Gustafson & Rhodes, 2006). For more on discussion on autonomy in the context of self-determination theory, see Chapter 3 (Quested et al., 2021) and Chapter 32 (Kingston et al., 2021). In an effort to move our knowledge of parental influences forward, Mâsse et al. (2020) have also put forward a tool for measuring parental influences based on this classification system.

Learning Exercise Four

A parent is wondering how to help their child become more physically active. Based on the different types of parental influences, what recommendations would you give?

Group Dynamics in Physical Activity Settings

It is thought that as humans we have a fundamental need to belong (Baumeister & Leary, 1995). Physical activity settings allow us to satisfy this need to belong by interacting with other people. Much of our physical activity is done with others and many people prefer to be active with others.¹ For example, people may go for a walk or run with a friend, join a fitness class, or lift weights with a group of friends. While some of these settings may seem very much like a group (e.g., walking with a group of friends), others may look a little less like a group (e.g., a drop-in fitness class).

Groupness

Researchers have examined the degree to which these physical activity groups possess the characteristics of a true group (Spink et al., 2010). This is reflected in the concept of groupness, which is the degree that a collection of individuals perceives themselves to be a group (Spink et al., 2010). In structured exercise settings like group fitness classes or running clubs, perceiving that setting to be more like a group has been related to participating more often in that setting (Spink et al., 2010). Perceiving higher levels of groupness within a fitness class has been related to greater reports of enjoyment and effort (Graupensperger et al., 2019). In fitness classes, having higher perceptions of groupness was related to greater satisfaction for the basic needs of relatedness and autonomy² and subsequently was related to more physical activity (Evans et al., 2019). Because groups can address the need to belong and have positive effects on physical activity behavior, it is not surprising that research has focused on group environments, structure, and processes (for review, see Eys et al., 2019). The group environment (e.g., member characteristics), structure of the group (e.g., positions, norms, and roles), and group processes (e.g., group cohesion) all have the potential to influence the experience of people within physical activity groups (Eys et al., 2019).

Cohesion

One of the group processes that has received the most attention in the physical activity setting is cohesion (Eys & Brawley, 2018). Cohesion reflects the “dynamic process that is reflected in the tendency for a group to stick together and remain united” (Carron et al., 1998, p. 213). It is conceptualized with two dimensions. The first dimension relates either to the *task* such as getting a good workout in, or the *social* aspect such as enjoying the company of the others you are active with. The second dimension relates to either *attraction to the group* (i.e., the desire of an individual to be a part of the group) or *group integration* (i.e., the unity of the group as a whole). The combination of

¹ While many people prefer to engage in physical activity, others may prefer to participate in activity alone for a variety of reasons including personality types and even experiences self-presentational anxiety (Hausenblas et al., 2004).

² The basic needs of autonomy and relatedness pertain the psychological needs that are part of the self-determination theory. When the basic psychological needs are satisfied in an activity, a person is thought to develop more autonomous motivation (Deci & Ryan, 2000).

these two dimensions creates four types of cohesion, namely (a) attraction to the group - task, (b) attraction to the group - social, (c) group integration - task and (d) group integration - social.

Cohesion has been examined for decades in the physical activity area with a meta-analysis conducted in 1996 describing a positive effect of task cohesion on exercise adherence (Carron et al., 1996). Cohesion has been described as an important aspect of physical activity groups and it is thought to relate to many positive outcomes including adherence or attendance and satisfaction with the settings (Burke et al., 2014). Not all physical activity groups possess the same level of cohesion. Cohesion is influenced by a variety of factors, including similarity, leadership characteristics, and environmental factors such as group size (Chapter 25; Cotterill & Fransen, 2021; Eys & Brawley, 2018). Most of the research on cohesion in the physical activity domain has focused on structured groups like group exercise classes (Eys & Brawley, 2018). Although one study looked at cohesion in unstructured groups, such as people walking with groups of friends (Spink et al., 2014). In that study, adherence was related to higher levels of task cohesion and lower levels of social cohesion. With cohesion playing an important role in the dynamics of a group, researchers have looked at how to develop cohesion (Eys et al., 2019).

Developing cohesion through team-building is one strategy that has been used to promote physical activity participation (Eys et al., 2019). Some of the earlier work involved fitness class instructors' team-building with their group exercise classes (Spink & Carron, 1993). Researchers held a workshop for fitness instructors and guided them in creating a list of team-building strategies that they would implement in their own fitness classes. Examples of strategies focused on the group environment (e.g., neon shoelaces, slogans for the class), group structure (e.g., norms such as group goals), and group processes (e.g., partner work; Carron & Spink, 1993). In this early research, classes with team building had higher levels of task cohesion (attraction to the group-task) as well as fewer dropouts than the control classes (Spink & Carron, 1993). This teambuilding approach has been used in a variety of settings including fitness classes, physical activity clubs for youth, and exercise settings with older adults (Eys, et al., 2019). At the center of this team-building approach is having the leader identify strategies that will work in their setting for the following areas (a) group distinctiveness (e.g., developing a group name, t-shirt), (b) individual positions (e.g., spots for different intensities), (c) group norms (e.g., standards for work ethic or achievement), (d) individual sacrifices (e.g., asking individuals to help others), and (e) interaction and communication (e.g., incorporating partner work; Burke et al., 2014).

Social Identity and Self-Categorization Theory

Social identity is another factor related to the group that has the potential to influence an individual's behavior (Stevens et al., 2019). Social identity describes the identity that is formed when an individual derives part of who they are as a person (e.g., their self-concept) from being part of a specific social group (Tajfel, 1981). For example, someone may identify that being a member of a running club is part of who they are. They may describe themselves as part of a weight-lifting club and that identity is central part of who they are as a person. This concept of social identity has been receiving increasing attention in physical activity settings with it being investigated in organized running groups (Stevens et al., 2019) and exercise groups (Dunlop & Beauchamp, 2011; Steffens et al., 2019). In exercise group members, identifying with the group has been associated with higher attendance and effort exerted within the exercise classes (Steffens et al., 2019). Self-categorization theory is a related theory that focuses on factors that contribute to perceiving oneself as part of a group and identifying with other group members (Beauchamp, 2019). Beauchamp (2019) describes the potential importance of social identity and self-categorization in the physical activity context to promote physical activity adherence. Given the potential of social identity in influencing physical activity adherence, researchers are starting to examine how leadership behaviors and other group factors (e.g., groupness or cohesion) may relate to the formation of social identity (Eys et al., 2019).

Learning Exercise Five

Think about the settings where you have exercised with other people. How did being part of a group influence your experience? What is it you liked or did not like about being part of a group?

Enhancing Social Support and Social Influences

Interventions that promote social support have been shown to be effective, especially when used alongside other interventions that incorporate behavioral components such as goal setting and behavior contracts (Heath et al., 2012). There are numerous strategies to increase social support in efforts to promote physical activity. Some examples of social approaches include buddy systems, creation of walking groups, and establishing contracts between leaders and participants (Heath et al., 2012). These social strategies have been explored in families, worksites, communities, clinical settings, and schools (Heath et al., 2012).

Many physical activity interventions incorporate some aspect of social support as part of the intervention whether it is involving family, friends, or being conducted with a group. In a review of 26 randomized control trials, 22 of those studies included some aspect of social support as part of the intervention (Howlett et al., 2019). On average, these interventions had a positive effect on physical activity behavior and on maintaining that behavior change following that intervention. In another review of interventions that included both diet and physical activity for those at risk of type 2 diabetes, the inclusion of social support was one of the strategies that improved the effectiveness of interventions (Greaves et al., 2011).

One way to facilitate social support is to incorporate it as part of physical activity groups. For example, one study targeted social support through a variety of interpersonal strategies, including small group discussions about physical activity-related ideas and sharing experiences about how to get support and overcome barriers (McMahon et al., 2017). This study also included friendly comparisons of physical activity based on activity monitors as part of its interpersonal strategies (McMahon et al., 2017). Those who received this added interpersonal interaction reported more physical activity (McMahon et al., 2017). As described earlier, a common type of intervention used in physical activity settings is the team-building approach (Burke et al., 2014; Spink & Carron, 1993). This approach involves enhancing the group dynamics through targeting the group distinctiveness, structure, norms, individual positions and sacrifices, as well as interaction and communication (Burke et al., 2014). With team-building, the goal is to develop group processes like cohesion with the intent to increase adherence and attendance. Another group-based intervention that has received quite a bit of attention is the group-mediated cognitive behavioral intervention (Brawley et al., 2014). This approach uses group-dynamic principles to build the group and teach cognitive-behavioral strategies such as goal setting, barrier management, and self-monitoring. The key distinction with this approach is that the individual is weaned off the group so that they can become independent exercisers at the end. The group-mediated cognitive-behavioral intervention has been examined with a variety of populations and settings including cardiac rehabilitation, older adults, individuals with knee osteoarthritis, and postnatal moms (Brawley et al., 2014). Across these numerous studies, positive effects were seen for the group-mediated cognitive-behavioral approach for adherence (i.e., increasing physical activity levels), improving self-efficacy, and physical functioning (Brawley et al., 2014). There is a wide variety of strategies for using social support from others or peers to promote physical activity and many of these approaches appear successful (Heath et al., 2012). Some of these strategies may be promoting

interactions through physical activity groups, buddy systems, group discussions, and phone networks (Heath et al., 2012).



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Measuring Social Influences

Questionnaires are the most common tool for measuring social influences such as social support, parental influences, and group cohesion. A common social support questionnaire is called the Social Support for Exercise Survey that assesses support received from family and friends (Sallis et al., 1987). Example questions include whether family or friends: *exercised with me*, *discussed exercise with me*, and *helped plan activities around my exercise*. A newer measurement tool called the Physical Activity and Social Support Scale has been developed to assess different types of social support (Golaszewski & Bartholomew, 2019). This measure includes separate subscales that capture emotional, companionship, instrumental, informational, and validation support. For parental influence, a new measurement tool called the Parenting Practice Item Bank has been created to capture the three conceptualized parenting strategies described earlier, namely autonomy support, structure, and neglect and control (Mâsse et al., 2020). One of several tools to assess cohesion is called the Group Environment Questionnaire, which assesses the four dimensions of cohesion in a variety of sport and exercise groups (Carron et al., 2002). Although questionnaires are frequently used, there are some limitations of this form of assessment, including a potential self-report bias.

There have been several emerging strategies for assessing social influences including social network approaches (e.g., Prochnow et al., 2020), direct observation of social influences (e.g., Collins & Feeney, 2004; Herbison et al., 2020), and using “big data” from databases (e.g., Carpenter & Amaravadi, 2019). Social network approaches involve individuals describing the networks they are involved in (Prochnow et al., 2020). This may involve having individuals report about people in their networks (e.g., report on family, friends, classmates) or surveying an entire social network (e.g., a classroom at school, or a group exercise class). These networks can be evaluated for a variety of characteristics such as the number of connections a person has, the number of connections within a network, and similarity of the

people in a grouping (Prochnow et al., 2020). In a review of 28 social network studies, physical activity seemed to be an attribute that was shared *within* a grouping such that if a child's social network was physically active, they also were more likely to be active (Prochnow et al., 2020). A social network approach can be used to look at how interactions across a network of peers or family members influence physical activity behavior.

Another way to measure social influences is to use direct observations where interactions are recorded and then coded to reflect the type of influence. To gain a more objective measure of social support, researchers observed couples interacting and coded the behavior in terms of support provided (Collins & Feeney, 2004). In this study, researchers coded the type of support such as emotional support, and instrumental support in speech writing task. Audio recordings can also be used to capture the social interactions between individuals. Using an electronically activated recorder (EAR), researchers were able to explore youth hockey players' social interactions in a natural environment (Herbison et al., 2020). Recordings of interactions were then coded to classify the recordings into different types of interactions (Herbison et al., 2020). The use of observation and coding provides a more objective measure for classifying the interactions and, unlike a questionnaire, does not rely on what individuals recall or report.

Another way to gather information on social influences is to use "big data" from databases that exist through social network applications and wearable devices. One study used registered users from a web-based application that was used for creating fitness challenges, allowing people to connect with other users, participating in employer incentivized programs, or sharing results with others (Carpenter & Amaravadi, 2019). Both the number of friends reporting exercise as well as the average amount of friends' exercise were related to an individual's own reported level of exercise (Carpenter & Amaravadi, 2019). Others also used data from the Fitbit™ databases and application to look at the social networks on those applications (Stück et al., 2017). Being more active was associated with having a greater number of social ties on the network as well as those ties having a higher average activity level (Stück et al., 2017). These device-based measures and "big" datasets present an interesting avenue for future research examining social influences for physical activity, especially as the use of wearables such as smart watches increases.

The most common tools for measuring social influences are self-reported questionnaires such as scales for social support. Recently, there have been innovations in how to measure social influence including social network analyses, direct observation, as well as using big data. This increase in the diversity of ways to assess social influences will expand our understanding of how these relationships influence our physical activity behavior and our health.

Conclusion

We live in a social world and most of us interact with many people across the course of a day, whether they are family members, friends, coworkers, or even acquaintances at a gym. Given our fundamental need to belong (Baumeister & Leary, 1995), these social relationships are vital for our health and well-being (Umberson, et al., 2010). Social support refers to the resources that one perceives available or received from their social networks (Gottlieb & Bergen, 2010). There are many types of social support ranging from companionship support (e.g., being active together), emotional support (e.g., providing encouragement), and instrumental support (e.g., providing tangible assistance such as providing equipment). Receiving more social support is beneficial for participating in physical activity (Scarapicchia et al., 2017). There are several different frameworks for how social support may influence our physical activity and health. Social support is just one way that our social relationships may influence our behavior. Others, such as social norms, social control, and group processes also influence our physical activity behavior.

Chapter 11

Strategies to Facilitate More Pleasant Exercise Experiences

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Chapter Overview

There is broad acknowledgement that how we feel during exercise is a predictor of long-term adherence. In this chapter, we describe the evidence that leads us to focus on the promotion of pleasant exercise experiences as an important strategy for long-term exercise adherence. This is followed by a discussion of modifications that can be made to exercise itself (intrinsic) and to the exercise environment (extrinsic) with the aim of influencing how people feel during exercise. Exercise intensity is the predominant intrinsic strategy that we focus on given its central role in determining our affective responses to exercise. We discuss a range of extrinsic strategies including indoor and outdoor exercise, exercising with others, virtual reality, music, imagery, and exergaming, among others. We aim to provide a broad overview of the available strategies that researchers and practitioners can implement to help foster more consistently pleasant and enjoyable exercise experiences.

Introduction

When asking students to identify the benefits of regular physical activity, they will often recall numerous positive physical benefits of exercise such as improved cardiorespiratory and muscular fitness, and improved bone health. “Feeling good” is often a response when pressed to cite the psychological benefits of exercise, but such responses are often delivered reflexively with mantras such as “exercise makes you feel good” deeply ingrained. When probing for further critical insight, the notion of exercise inherently “feeling good” begins to be questioned. While exercise can elicit positive affective states among some, it is not a given that exercise will “feel good” for all. For some individuals, exercise is *not* a consistently pleasant experience, but it might not have to be this way. Researchers are examining different ways to positively shape how we feel about exercise.

Exercise is just one form of physical activity, but it is an effective way to achieve recommended levels of physical activity and can guard against the deleterious effects of sedentary behavior. Acquiring at least 150 minutes of moderate-intensity or 75 minutes of vigorous-intensity activity, and two bouts of muscle strengthening activities per week is achievable through structured exercise. Moreover, achieving these targets through one or two sessions per week (the so-called “Weekend Warrior”; Hamer et al., 2017, p. 2) can confer positive mental health outcomes.

As a society, we spend considerable time, effort, and financial resources to shift inactive people towards physical activity. However, if the experience of physical activity is unpleasant, then we have likely wasted much of that time, effort, and money, as unpleasant experiences are unlikely to promote long-term adherence. Efforts to engage people in physical activity will ultimately have limited impact on the broader positive outcomes of physical activity (e.g., longer lives, better quality of life, reduced financial burden on government) without long-term adherence.

A substantial portion of the literature relating to exercise interventions focuses on helping people to *initiate* exercise, whereas there is less focus on what to do once people *begin* exercising; that is, the strategies we can employ to influence how people feel *during* exercise. Perhaps many exercise professionals follow the “exercise makes you feel good” mantra without critically considering whether this is true and might believe that their job is done once they have “convinced” people to exercise. It is long established that exercise is effective at improving health but continued consideration of how we can help facilitate an enjoyable experience is needed.

One approach to exercise promotion is to understand which type of exercise an individual “likes” and advocate that they engage in that activity, but if an inactive individual does not like *any* mode of exercise (this seems possible, otherwise they would probably be doing it already), then this might be a fruitless task. Perhaps we would then fall back on the exercises that we know and suggest activities such as cycling, running, or swimming. It is very likely that an inactive person has already considered these activities and has decided against them. However, we know that these activities can be very effective at eliciting physiological changes that will lead to improved health, so perhaps we can look again at these activities and consider ways in which we can modify them to make them more enjoyable.

This chapter discusses intrinsic and extrinsic strategies to facilitate pleasant and enjoyable exercise experiences. We define intrinsic strategies as those that modify the components of exercise prescription: Frequency, Intensity, Time, and Type (FITT; see Kohl et al., 2020, p. 22). We focus on modifications to exercise intensity as it is a primary intrinsic factor that drives affective responses to exercise. Extrinsic strategies pertain to environmental manipulations of the exercise experience that fall outside of the FITT principles (e.g., listening to music during exercise). These extrinsic strategies are predominantly employed during exercise, but there are some strategies that could be implemented prior to an exercise session to influence how people feel during that session.



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The Importance of Pleasant Exercise Experiences

Imagine that you return from work in the evening and you have an array of choices for your activity. You could relax, browse the internet, watch television, stream movies, study, get prepared for your next day of work, or exercise. Why is it that studying and exercising are preferable for our long-term interests, but so often avoided in favor of an alternative?

During leisure-time, people often choose the activities that make them feel good and avoid activities that make them feel bad. Repeating experiences that feel good and avoiding experiences that feel bad is the basic premise of Hedonic theory (Ekkekakis & Dafermos, 2012). The pleasure and displeasure people experience has been recognized as an important predictor of behavior for millennia (see Ekkekakis & Dafermos, 2012). Although engaging in exercise may be of great benefit for our long-term health and well-being, the tendency of humans to maximize pleasure may lead them to suboptimal or nonrational decisions and behaviors (Cabanac & Bonniot-Cabanac, 2007). In the case of exercising, this tendency may result in a person choosing a sedentary, but suboptimal, alternative.

As with other behaviors, affective responses—the pleasure or displeasure experienced during exercise—are related to current and predictive of future exercise behavior (Williams et al., 2012). For example, Williams and colleagues (2012) were interested in examining the relationships between affective responses during and after short, 10-minute moderate intensity treadmill walks. During and after the walks, participants responded to the Feeling Scale (Hardy & Rejeski, 1989; see Chapter 12; Zenko & Ladwig, 2021) to indicate how good or bad they were feeling, as a measure of the pleasure or displeasure experienced. In addition, participants were interviewed about their level of physical activity behavior in the last seven days. Participants who felt more positive during exercise reported more physical activity cross-sectionally (i.e., affective responses during exercise were related to self-reported physical activity behavior measured on the same day of exercise) and longitudinally (i.e., affective responses during exercise predicted self-reported physical activity when measured six months later). Interestingly, the pleasure and displeasure experienced while participants were seated and resting after exercise was not related to physical activity behavior, either cross-sectionally or longitudinally.

These general findings were supported by a systematic review (Rhodes & Kates, 2015). Rhodes

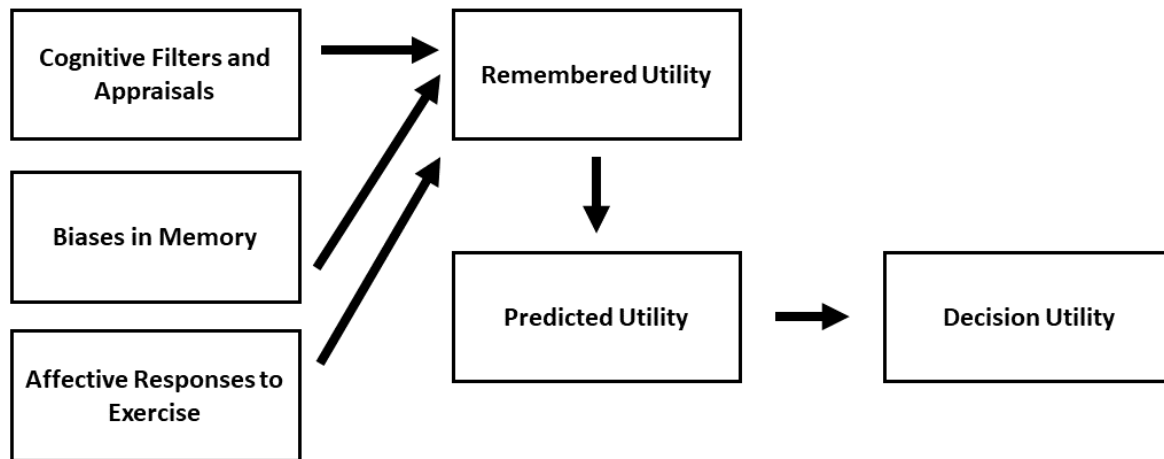
and Kates (2015) reviewed 24 studies, of which four assessed the relations between affective responses during exercise and physical activity behavior. In both adolescents and adults, affective responses during exercise were related to physical activity behavior, while the authors suggested that “the evidence does not support a relationship between postexercise affective response and subsequent physical activity performance” (p. 725). Overall, the outcomes of the systematic review supported the idea that maximizing pleasure experienced *during* exercise may be the most influential correlate of future physical activity behavior.

Memories of the Past, Decisions about the Present

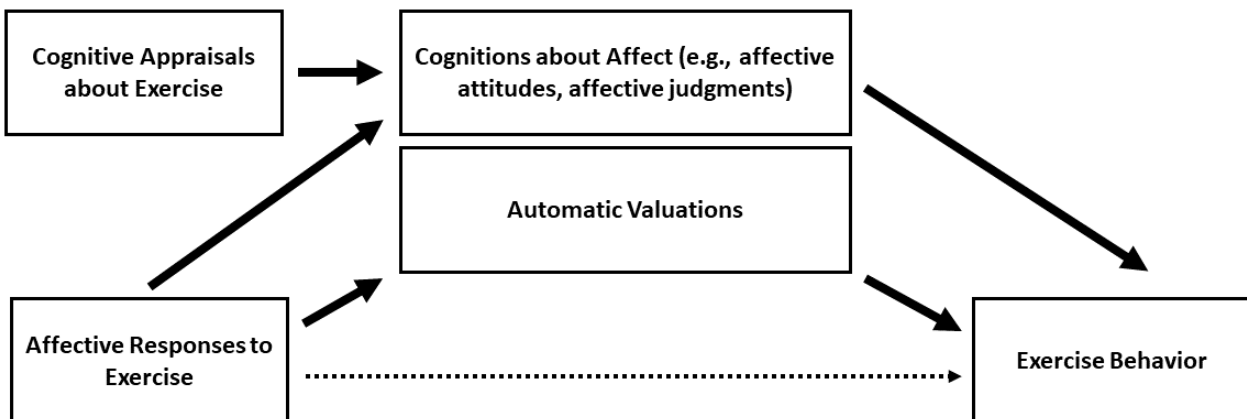
When making decisions and choices about behavior—such as the decision to exercise or not—it is important to acknowledge that decision makers may be biased, and the memories of pleasure and displeasure experienced during earlier exercise episodes may be distorted through cognitive filters and other biases in memory. The Nobel Laureate Daniel Kahneman and colleagues (Kahneman et al., 1997) distinguished between several types of utility, including experienced utility and decision utility. Experienced utility can be described as the hedonic experience of an episode, for example, the affective responses during exercise¹. An exercise experience that is more pleasant has higher experienced utility than a less pleasant (or more unpleasant) exercise session. Decision utility is inferred from the choices people are observed to make. For instance, if a person chooses a sedentary alternative such as television viewing instead of exercise, then television viewing would be said to have more decision utility. Interestingly, a large research literature suggests that decision utility is not a direct result of experienced utility. Instead, it is possible that an experience such as exercising can be remembered differently (e.g., as more or less positive than it actually was), and this remembered utility may inform predicted utility (i.e., predictions about how pleasant or unpleasant future exercise experiences will be), which in turn may inform decision utility (i.e., whether someone chooses to exercise, or not). As Kahneman and colleagues put it, the factors “that determine what people do are not pleasure and pain, but *memories* of pleasure and pain” (Kahneman et al., 1997, p. 385, emphasis added). This is illustrated in Figure 11.1, showing the theorized relations between affective responses to exercise, remembered utility, predicted utility, and decision utility. We suggest that some of the relationships depicted in Figure 11.1 may be conscious or unconscious.

Memories of an activity or event may be biased by many factors, such as the temporal characteristics of an exercise experience (described later), as well as cognitive filters and cognitive appraisals, including individual biases and cognitive appraisals about the importance of exercise (Karnaze et al., 2017), and individual differences such as personality. It is reasonable to suggest, for example, that a person with a strong optimism bias and tendency to predict future experiences will be positive will also be more likely to predict that engaging in an exercise bout will be pleasant and enjoyable. Future research is necessary to test this hypothesis. In short, cognitively mediated variables such as remembered utility (e.g., “How *did* the exercise make you feel?”), predicted utility, affective forecasts, or anticipated affective responses (e.g., “How *will* the exercise make you feel?”), and other affect-laden cognitive constructs such as affective attitudes and affective judgments may be influenced at least in part by previous affective responses to exercise experiences (also see Ekkekakis et al., 2018), but affective responses to exercise are not the only determinant of these cognitive constructs.

¹ When “utility” of exercise is conceptualized as the pleasure or displeasure associated with exercise, as is the case in this chapter, the terms *remembered pleasure* and *predicted pleasure* can serve as synonyms for remembered utility and predicted utility, respectively. An additional synonym for predicted utility is *forecasted pleasure*, and *affective forecasts* are predictions about future affective experiences.

Figure 11.1*Theorized Influences of Remembered Utility, Predicted Utility, and Decision Utility*

Note. Notice that affective responses to exercise, the focus of this chapter, are just one influence of remembered utility. Affective responses during exercise (an indicator of experienced utility) rarely, if ever, fully explain the remembered utility of exercise. Biases in memory and cognitive filters and appraisals also influence remembered utility.

Figure 11.2*Conceptualized Pathways in Which Affective Responses to Exercise Influence Future Exercise Behavior*

The dotted line in Figure 11.2 represents the overall relationship between affective responses to exercise and exercise behavior. Several mediators of the relationship between affective responses to exercise and exercise behavior are theorized. Here, we suggest that affective responses to exercise may impact future exercise behavior through affect-laden cognitions or cognitions about affect (e.g., thoughts about whether the exercise makes one feel prideful or shameful, embarrassed or accomplished, and expectations of the increased sense of energy or euphoria one may experience after exercising). These cognitions about affect are also expected to be influenced by messaging strategies and other techniques for changing cognitive appraisals about exercise (e.g., information and media campaigns about how exercise makes people feel good, and how exercise may reduce depression).

Indeed, affect-laden cognitive constructs have been shown to relate to exercise intentions (Helfer et al., 2015) and behavior (Geers et al., 2017; Kiviniemi et al., 2007). These suggested pathways are shown in Figure 11.2 and illustrated by the boxes labeled “Cognitive Appraisals about Exercise” and “Cognitions about Affect”.

Automatic Affective Valuations

We also adopt the theoretical predictions of the affective-reflective theory of physical inactivity and exercise (Brand & Ekkekakis, 2018; Chapter 4, Brand & Ekkekakis, 2021). According to affective-reflective theory, automatic affective valuations (Figure 11.2) are automatic positive or negative associations with physical activity or exercise. The positivity or negativity may be the “result of repeated experiences of emotions mediated by cognitive appraisals (e.g., pride, embarrassment) or a result of repeated exercise experiences of core affective reactions not necessarily mediated by cognitive appraisals (e.g., sense of physical reinvigoration, pain, bodily discomfort)” (Brand & Ekkekakis, 2018, p. 54). In turn, automatic affective valuations serve as a predictor of physical activity or exercise behavior. Therefore, based on the affective-reflective theory, automatic affective valuations serve as a link and mediate the relation between affective responses to exercise and exercise behavior (see Figure 11.2). Taken together, it seems imperative that practitioners should maximize the pleasure experienced during exercise because affective responses to exercise are predictive of future exercise behavior; this relationship is likely mediated by cognitions about affect and automatic affective valuations.

Learning Exercise One

Describe why having an exerciser experience exercise as pleasant or unpleasant is important, and how the *changing the experience of exercise* might influence behavior differently than providing information and education about the various benefits of exercise (e.g., improving cardiovascular health, reducing risk of disease).

Strategies to Facilitate More Pleasant Exercise Experience

How, then, can exercise professionals who understand hedonic theory and the importance of affective responses to exercise do their best to ensure that pleasure is maximized, and displeasure is minimized? Messaging strategies designed to give information about the benefits of exercise are thought to promote exercise behavior through cognitive appraisals (see Figure 11.2). To fully target affective responses to exercise, one must make the experience of exercise more pleasant (Ekkekakis et al., 2018). Here, we describe two broad strategies that are (a) focused on intensity (an Intrinsic factor) and (b) focused on situational and environmental (Extrinsic) factors.

Intensity and Affective Responses to Exercise

The dual-mode theory (DMT; Ekkekakis, 2005; 2009a) suggests that affective responses to exercise are influenced by cognitive factors, such as appraisals about exercise, goals, and perceptions of accomplishment, and interoceptive factors, such as physiological responses to exercise. According to the DMT (Ekkekakis, 2009a), when intensity is “moderate” (i.e., below the lactate threshold or ventilatory threshold), exercise is “consistently beneficial and it is essential for subsistence activities (hunting and gathering) and promotes the healthful effects of exercise without seriously challenging the maintenance of homeostasis or raising the risk of injury or exhaustion” (p. 82). “Heavy” exercise intensity is characterized by intensity that is slightly above the lactate threshold or ventilatory threshold to the level

of critical power, and exercise intensity beyond this point is considered “severe” (Ekkekakis, 2009a; also see Ekkekakis et al., 2005b). It is important to note that moderate, heavy, and severe exercise intensities in this context do not necessarily correspond to the definitions of moderate- or vigorous-intensity exercise according to the American College of Sports Medicine (e.g., 64–76% of maximum heart rate for moderate intensity; Garber et al., 2011). Instead, the domains of moderate-intensity, heavy-intensity, and severe-intensity correspond to levels of challenge to physiological homeostasis.

Since exercise at moderate intensities is “consistently beneficial” from an evolutionary perspective, moderate-intensity exercise is expected to result in pleasant affective responses for most people (Ekkekakis et al., 2005b). Exercise in the heavy domain is neither consistently beneficial nor dangerous, and therefore affective responses in the heavy domain are often highly variable, with cognitive factors being more closely related to affective responses during heavy exercise (Ekkekakis et al., 2005b; Ekkekakis, 2009a). Finally, exercise in the severe domain is associated with several physiological threats to homeostasis, including continuously rising blood lactate, decreased availability of energy resources, and, if prolonged, potential muscle damage (Ekkekakis et al., 2005b). Like moderate exercise, severe exercise is adaptationally significant. Whereas moderate exercise is considered to be adaptive, severe exercise is considered to be maladaptive and dangerous if prolonged. The DMT suggests that exercise at severe intensity is associated with displeasure for most people (Ekkekakis et al., 2005b; Ekkekakis, 2009a). The tenets of the DMT are well-supported in the literature (for a review, see Ekkekakis et al., 2011). Overall, the DMT (Ekkekakis, 2005; 2009a) suggests that practitioners should be cautious about prescribing exercise that is in the range of severe intensity; practitioners also take a risk when prescribing exercise that is in the heavy domain. These intensities are expected to elicit either (a) homogenous, consistent feelings of displeasure (severe intensity) or (b) heterogenous, mixed feelings of displeasure and pleasure (heavy intensity). To illustrate, Zenko and Ekkekakis (2019) had 95 participants exercise at a heavy intensity (i.e., at the intensity corresponding to the ventilatory threshold for each participant) while reporting their affective responses. Compared to affective valence (i.e., pleasure-displeasure) assessed shortly before exercise, 61.1% of participants felt less pleasant during exercise, and 25.2% of participants felt more pleasant. Put differently, prescribing heavy-intensity exercise may make some people feel worse, and, according to the DMT, prescribing severe-intensity exercise is expected to make most people feel worse, and ultimately may increase the risk of dropout.

Ekkekakis and colleagues (2011) suggested that the likelihood of exercise intensity to elicit pleasure or displeasure should be a primary consideration for practitioners, alongside exercise safety and exercise effectiveness. So how can exercise professionals prescribe intensity for pleasure, especially in the absence of equipment to assess the lactate threshold or ventilatory threshold for each client or patient? Promising strategies include allowing the exercisers to choose or self-select their own exercise intensity, using pleasure as a guide for prescribing intensity, and structuring the pattern of exercise intensity in a way that is expected to elicit positive remembered and predicted pleasure.

Self-Selected and Affect-Guided Exercise Intensity

In a review, Ekkekakis (2009b) noted that, when allowed to self-select their own exercise intensity, most (but not all) people select intensities that are both physiologically beneficial and not expected to elicit displeasure (also see Williams, 2008). Allowing people to choose their own exercise intensity may have other benefits that promote exercise adherence. For example, doing so may enhance autonomy (Vazou-Ekkekakis & Ekkekakis, 2009), one of the basic psychological needs proposed by self-determination theory. See Chapter 2 (Rebar et al., 2021) and Chapter 3 (Quested et al., 2021). It may also simplify the practitioner’s job of determining individual differences in preferences, recognizing that some people prefer and tolerate higher levels of exercise intensity more than others (see Ekkekakis et al., 2005a). Some students and professionals reading this, for example, may realize that they prefer high-intensity interval training and vigorous exercise experiences. However, students and professionals

should, at the same time, consider that "preferences for me do not mean preferences for thee." Prescribing exercise programming that we like but our patients or clients dread would not be advisable.

Williams and colleagues (2015; 2016) provided compelling experimental evidence in support of self-selected exercise intensity, or what they called self-paced exercise. In a randomized controlled trial, participants were enrolled in a six-month exercise program and randomly assigned to either (a) moderate-intensity exercise, based on heart rate (64–76% of maximum heart rate) or (b) self-paced exercise, in which the participants could choose their own intensity but were asked not to exceed 76% of their maximum heart rate as a safety precaution. All participants were insufficiently active and either overweight or had obesity. In addition, mode was not self-selected. All participants were asked to walk for their exercise. The self-paced exercisers walked about 26 more minutes per week than the exercisers who were prescribed moderate-intensity exercise (Williams et al., 2015). In another study, based on the same intervention, it was found that the self-paced exercisers had more positive affective responses to exercise, and these more positive affective responses mediated the relationship between the type of exercise program (i.e., self-paced vs. moderate intensity) and exercise adherence (Williams et al., 2016). In other words, allowing participants to self-select their own walking intensity evidently resulted in more positive affective responses to exercise, which in turn resulted in greater exercise adherence.

Another, more direct method of prescribing exercise that results in positive affective responses may be to simply guide exercisers to choose an intensity that feels good. This is affect-based, affect-regulated, or affect-guided exercise (Ladwig et al., 2017). Although experimental evidence linking this strategy to increased adherence is lacking, preliminary results requiring replication and extension appear promising (Parfitt et al., 2012; Zenko et al., 2020). With this approach, researchers or practitioners could use the Feeling Scale (Hardy & Rejeski, 1989) or other measure of affective valence as a guide to instruct exercisers to select only intensities that allow them to feel pleasant (i.e., more positive than neutral). This would, ideally, prevent participants from feeling unpleasant during exercise.

Restructuring Exercise Intensity

Researchers have also started to test the effects of restructuring exercise intensity to create more pleasant memories of exercise. Recall that remembered pleasure or remembered utility may be biased and not completely dependent on the overall average pleasure experienced during an episode of exercise. Instead, certain temporal characteristics may have disproportionate influence on how an exercise experience is remembered. Some of these characteristics include the end, or final moment of an experience, the peak, or most intense moment of an experience, and the trend or slope of an experience (i.e., whether an experience becomes more positive or more negative as time progresses; Ariely & Carmon, 2000; Kahneman et al., 1993). To illustrate how the “end” rule might influence the remembered pleasure or remembered utility of an experience, imagine that you’re enjoying dinner with friends and hosting a party at your house. For several hours, you enjoy conversation and company and are having a great experience overall. However, just as the dinner party is ending, one of your guests knocks over a fragile and sentimental family heirloom. The heirloom is destroyed just as guests are leaving your house. On average, you experienced pleasure for several hours. However, the end rule would suggest that the final moment of the dinner party—which was very unpleasant—disproportionately influences your overall memory of the experience. Now, you may recall the dinner party to be unpleasant, despite the fact that you experienced more pleasure than displeasure throughout the party.

Applied to exercise, the importance of the peak-end rule and the trend or slope suggests that experiences should have a more pleasant “peak” moment, end on a positive note, and improve throughout an experience (i.e., if participants start exercise feeling unpleasant, you should ensure that they feel more and more pleasant as exercise progresses). Zenko, Ekkekakis, and Ariely (2016) tested some of these concepts by having participants either (a) start at a high intensity and end at a low

intensity (thus starting from an unpleasant exercise intensity and ending at a more pleasant exercise intensity), or (b) start at a low intensity and end at a high intensity (thus starting from a pleasant exercise intensity and ending at an unpleasant, very challenging exercise intensity). Supporting the research on the trend (and to some extent, the end), the researchers found that exercisers who decreased intensity over time (and increased pleasure) remembered the experience to be more pleasurable, predicted future exercise experiences to be more pleasurable, and enjoyed the exercise more. Remembered pleasure and predicted pleasure were measured about 15 minutes, 24 hours, and seven days following exercise. The slope of pleasure (i.e., how much the exercise experience became more or less positive over time) explained 35–46% of the variance in remembered and predicted pleasure. These findings suggested that the trend of an experience—the slope of pleasure—was a major influence on how the experience of exercise was remembered.

While Zenko et al. (2016) tested this concept in the context of aerobic exercise (i.e., using a cycle ergometer), Hutchinson et al. (2020) replicated the findings in the context of resistance exercise. This time in a within-subjects design (where participants repeated both conditions), participants performed a resistance-training circuit that either increased or decreased in training load. In the decreasing-intensity condition, participants performed one circuit at 75% of their 1-repetition maximum, followed by one circuit at 65% of their 1-repetition maximum, and ending with 55% of their 1-repetition maximum. The pattern was reversed in the increasing-intensity condition. As expected, participants felt progressively more pleasant in then decreasing-intensity condition and reported more pleasure following exercise. The decreasing-intensity condition also resulted in more exercise enjoyment and remembered pleasure (both shortly after and about 24-hours after exercising).

It is likely that context matters here. For example, Stuntz and colleagues (2020) examined patterns of perceived exertion (also see Chapter 13; Hutchinson, 2021), feelings of accomplishment, and affective responses in the context of a practice session for college athletes. Importantly, the authors concluded that change in perceived exertion and change in pleasure did not predict remembered pleasure. Instead, feelings of accomplishment mediated the relation between change in exertion and change in remembered pleasure. When the athletes exerted themselves more and felt more accomplished, they also experienced greater remembered pleasure. Thus, the pattern of intensity is not the only predictor of remembered pleasure and remembered utility. As mentioned previously, individual cognitive appraisals or cognitive filters also play a role in shaping the memory of an experience. The research by Stuntz and colleagues (2020) suggests that cognitive appraisals of accomplishment may be particularly important for athletes in sporting contexts. Further research should continue to examine under which conditions the end, peak, and trend of an experience do and do not predict remembered pleasure, predicted pleasure, and ultimately exercise adherence.

Learning Exercise Two

You have just read about how intensity influences affective responses to exercise. Your task here is to design an exercise session to maximize pleasure and minimize displeasure by regulating exercise intensity. How do you do it? Justify your response.

Extrinsic Strategies

In the context of this chapter, an extrinsic strategy concerns manipulation of the exercise experience outside of frequency, intensity, duration, and type. In practice, strategies are often combined to create interventions, and in many instances more than one strategy is included in a research study which can make isolating the effects of a specific strategy difficult. For example, a person

attending a spin class could experience several extrinsic factors: the music played, the other participants in the class, the instructor, the room set up (e.g., mirrors, lighting), and each would influence a person's responses to the spin class. This is in addition to intrinsic factors such as exercise intensity, and other factors previously mentioned in this chapter. The interaction of these factors leads to difficulties in predicting responses to such situations. Nonetheless, researchers have sought to understand the affective responses to extrinsic strategies that could be employed during exercise.



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Indoor and Outdoor Exercise

For a moment, consider the physical environments in which you exercise. Are there blossoming trees, flowering plants, the scent of freshly cut grass, and birdsong? Are there mirrored walls, grey-metal machines, human body odors, and a local radio station playing in the background? The physical environment in which we exercise intuitively seems an important factor in determining our experience.

The environment in which people exercise is complex and includes numerous different sources of information (e.g., scenery, weather). However, the dichotomy of indoor or outdoor exercise can be useful when trying to understand affective responses to the physical environments where people exercise. Green Exercise is “activity in the presence of nature” (Barton & Pretty, 2010, p. 3947) and a review article by Gladwell et al. (2013) proposed that Green Exercise could be a powerful tool against physical inactivity. A study by Rogerson et al. (2016) sought to control for mode and intensity of exercise while examining the effects of a green outdoor setting, compared to a built indoor setting. Twelve participants cycled on an ergometer that was situated on a sports field or in a windowless laboratory with basic fixtures and fittings. The results of 15-min cycling at 50% heart rate reserve in these environments did not indicate any differences in enjoyment of exercise. Similarly, Turner & Stevinson, (2017) did not find that outdoor exercise promoted greater pleasure during high intensity exercise. They compared a 6km near-maximal intensity run on an indoor treadmill to a trail run through a large woodland area and found that Feeling Scale scores did not differ during or immediately after the ~15min run.

There has been support for the assertion that outdoor exercise leads to more positive affective responses. Lacharite-Lemieux et al. (2014) demonstrated that a 12-week outdoor exercise program comprising both aerobic and resistance components led to greater engagement with exercise and adherence in a sample of 23 postmenopausal women. Participants in this study also reported greater pleasure during outdoor exercise but this analysis did not include all exercise sessions as part of the program. Further work has explored the effects of outdoor exercise in women (e.g., Krinski et al., 2017;

Plante et al., 2007). Krinski et al. (2017) asked women with obesity to walk at a self-selected pace on a laboratory treadmill and on an outdoor athletic track and found that not only did participants walk further outdoors, they also reported greater pleasure during their walk. An important point to note was that pleasure declined from baseline in both indoor and outdoor conditions (a pattern previously reported in women with obesity by Ekkekakis et al., 2010), but the decline was not as sharp during outdoor walking. In a further study demonstrating the efficacy of outdoor exercise, a study with undergraduate women revealed that self-paced walking on a college campus was more enjoyable than walking on a treadmill in a health club (Plante et al., 2007). Focht (2009) compared an outdoor self-paced walk to a laboratory treadmill walk and found that both environments resulted in positive changes in core affective valence from baseline, but outdoor exercise yielded higher Feeling Scale scores, and that greater enjoyment was derived from the outdoor walk.

Consider again the outdoor environments that you might exercise in. There might be an outdoor run in the suburbs, a parkour circuit in a city center, a cycle ride in open countryside, or a swim in a reservoir. It seems plausible that each of these outdoor environments would elicit a different response as they each have different characteristics. Researchers have sought to examine different outdoor environments during exercise in attempts to further understand the role they might play. An early study in this domain explored different outdoor settings by comparing a park run to an urban run with 12 regular runners (Bodin & Hartig, 2003). The park run, through a large nature reserve, did not translate to significant differences in emotional response. In a different research design to many in this field, Dunton et al. (2015) measured affective responses during physical activity using ecological momentary assessment² and this captured participants exercising as they naturally would. The outdoor environments in the Dunton et al. (2015) study included at-home outdoors, at a park or trail, or other outdoor locations, but insufficient data for each of these outdoor categories prevented detailed analysis for each category. Nonetheless, outdoor exercise across these different environments resulted in less negative affect than indoor exercise. The benefits of a more “extreme” outdoor exercise environment have also been examined and a comparison between a coniferous forest and an urban park revealed that the more extreme outdoor environment of the forest led to greater pleasure reported immediately after exercise (Pasanen et al., 2018). There appears some preliminary evidence that not all outdoor exercise is the same, with some environments more effective at eliciting positive affective responses than others. However, the existing evidence is scant and there is a need to explore this topic with a greater range of outdoor environments considered. There are studies exploring different outdoor environments (e.g., Green Exercise, Blue Exercise), but the outcomes in these studies are often not captured with dimensional measures of affect or measures of enjoyment, and these outcomes are of central concern in this chapter.

Comparisons between indoor and outdoor exercise are popular but there have been equivocal findings regarding affective responses during exercise. A systematic review by Lahart et al. (2019) supported the view that acute bouts of outdoor exercise have favorable affective responses compared to indoor exercise but concluded that low quality evidence pervaded the field, and that further rigorous research is required.

Purposefully Selected Visual Stimuli

While the evidence for outdoor exercise compared to indoor exercise is tentatively positive, it is not always possible for people to exercise outdoors. Consequently, researchers have sought to

² Ecological momentary assessment allows participants to report thoughts, feelings, and behaviours close in time to the experience. As a method, it allows for more externally valid studies compared to lab-based studies as participants can continue with their usual routines. Prompts are provided to the participants to record what they have been doing and their psychological responses to it.

understand the effects of bringing the outdoors, indoors. Research into the effects of showing digital images and videos of the natural environment to people exercising indoors is an attempt to bring the benefits found in some outdoor exercise studies, to indoor exercise. Watching point-of-view video footage of traveling through a national park while cycling on an ergometer was not found to increase pleasure compared to a control condition (Jones et al., 2014). However, when the video footage was combined with a motivational music playlist, there were significant positive effects on pleasure compared to a control condition. The video footage played by itself (without motivational music) was not rated as any more pleasant than the control condition and it was postulated that the unnatural presentation of nature (e.g., without the other sensations of nature such as the wind in your face or hearing the rustle of leaves) was the reason why.

Beyond videos of natural environments, researchers have sought to understand the efficacy of other visual stimuli during exercise. Watching and listening to music videos while running on a treadmill led to greater pleasure when compared to listening to the music without the video, and a control condition (Hutchinson et al., 2015). Bigliassi and colleagues (2019) administered a range of visual stimuli (e.g., movie scenes and television series) based on participants preferences and found that, when delivered with auditory stimuli, pleasure was increased during a 10-min self-paced recumbent cycling task.

The video footage presented by Jones et al. (2014; see Figure 11.3), Hutchinson et al. (2015), and Bigliassi et al. (2019) was delivered to participants via a projection in front of the participant or on a television screen. It is apparent that visual stimuli by itself has been the subject of few studies, and the scant evidence available to date suggests that visual stimuli by itself has limited capacity to positively influence how we feel during exercise. Watching videos (e.g., music videos, or traversing natural landscapes) is common for many exercisers and has been implemented in gymnasias and on cardio exercise equipment for many years. However, recent advances in technology have permitted researchers to more easily examine the effects of visual stimuli delivered in a more immersive style.

Figure 11.3

Laboratory Set-Up for Showing Digital Footage of Nature While on a Cycle Ergometer



Note. Image credited to Dr. Leighton Jones

Virtual Reality and Immersive Environments. The recent emergence of low-cost consumer level headsets (e.g., Samsung Gear VR, Oculus Rift) has led to a resurgence of interest in how virtual reality and immersive environments can be used in an exercise context. While this new wave of research is in its infancy, there appears to be some use for the technology. Researchers have sought to use the headsets for two different types of visual stimuli: to display visual stimuli (e.g., music videos) in a more immersive way but lacking interaction, and as a means to display a “computer-simulated environment that aims to induce a sense of being mentally or physically present in another place” (Neumann et al., 2018, p. 183). This section focuses on recent understanding of virtual reality and pertains only to those studies that use a head-mounted display. Neumann et al., (2018) noted in their systematic review that although the term “virtual reality” was used in many studies, it was typical that researchers would use a computer screen or projector with participants. The recency of head-mounted displays being more available for exercise research means there are presently few studies in this area, though this number is growing.

Extending the lineage of work exploring digital images of nature while exercising indoors, Calogiuri et al. (2018) used a Samsung Gear VR headset to compare a 10-min walk along a river bank against video footage of the same walk observed in the head-mounted display. Participants walked on a non-motorized treadmill while viewing a 360° video but found the “real life” walk along the riverbank more pleasant and enjoyable. Alongside the quantitative measures of the exercise experience, participants also provided qualitative feedback about the experience. The issue of cybersickness was raised and this is likely a significant issue for exercise–VR researchers to overcome if the technology is to be used effectively; feeling nauseas and positive affective responses are not common bedfellows. The mode of exercise implemented in this study is also worth considering because the complete visual obstruction of the treadmill has been raised by Neumann et al. (2018) as hazardous and impractical. The comparison of “virtual nature” to “real nature” can have some use in enabling a greater understanding of the extent to which the benefits of real nature can be recreated. However, the benefit of virtual nature during exercise does not seem to be in replacing outdoor exercise, but rather when outdoor exercise in pleasant environments is not possible or feasible. Perhaps continued exploration and comparisons of different types of visual stimuli delivered through head-mounted displays, as opposed to their “real life” counterparts, will prove a fruitful endeavor.

In a change of mode from treadmill exercise, other studies using head-mounted displays have used cycle ergometers to explore the possible benefits during exercise. A study with inactive and overweight adults using a Samsung Gear VR during stationary recumbent cycling demonstrated efficacy for a highly immersive condition (Jones & Ekkekakis, 2019). A comparison between a control condition (no stimulus), watching music videos on a television screen while listening to the music through speakers (low-immersion), and watching the music videos in the head-mounted display while listening to the music via headphones (high-immersion) showed that being highly immersed increased pleasure during exercise. Bird et al. (2020) also used cycle ergometry and a Samsung Gear VR to examine whether a virtual cycling game (with or without a motivational music accompaniment) promoted a pleasant exercise experience. Their findings indicated that during a short bout of exercise (6-min), the virtual cycling game (Cycle: Le Tour by VirZOOM) with and without motivational music promoted greater pleasure and enjoyment than a control condition.

The application of VR and head-mounted displays during exercise has largely involved “traditional” modes of exercise to date (i.e., treadmill and cycle ergometry). However, the application of these emerging technologies with pre-existing modes of exercise might be doing a disservice to the possibilities that these technologies offer. The application of VR while exercising using newly developed devices such as omni-directional treadmills or the Icarus Pro flight simulator might permit exercisers and practitioners to re-imagine what is possible during exercise. Dębska et al. (2019) offered some preliminary data in support of using VR and newly developed equipment to facilitate physical activity,

but further rigorous research is required to understand the efficacy and suitability of such devices for exercise. This topic area appears ripe for extensive future research and there are opportunities for collaborative work between industry and academics.



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Learning Exercise Three

The exploration of virtual reality during exercise is in its infancy and offers numerous opportunities for researchers. For this creative learning exercise, design a research study wherein you test the efficacy of a newly created virtual environment using an exercise mode of your choosing. Ensuring a robust research design can be challenging, but the additional challenge here is deciding what that virtual reality environment will be and how it will link to exercise. Try to think beyond what has been examined so far and create a new virtual world or task that would be ideal for a pleasant exercise experience.

Music

So far in this chapter we have mentioned music alongside visual stimuli, but there is a more extensive literature exploring the effects of music singularly. Music is arguably the most popular accompaniment to exercise and is a topic that garners considerable interest from the public and researchers. The past decade has seen a marked increase in the quantity and quality of music in sport and exercise literature and a recent meta-analysis by Terry et al. (2020) included 139 studies. The meta-

analysis serves as a landmark paper in this topic area and would be a suitable starting point for any research on this topic. The role of music in a sporting context is discussed in Chapter 23 (Karageorghis et al., 2021), and this section introduces the use of music in an exercise context.

Music can be applied prior to exercise but the carry-over effects into exercise are not well understood. There are very few studies exploring whether listening to music beforehand can improve how people feel during exercise. Karageorghis et al. (2018) were interested in whether listening to different combinations of loud/quiet and fast/slow tempo music prior to an isometric contraction task (using a handgrip dynamometer) influenced how people felt immediately after the physical task. The loud and fast music (80 decibels and 126 beats per minute [bpm]) resulted in the highest affective valence scores following the task. Other work has examined the use of music before physical tasks, but without assessing affective responses (e.g., Crust, 2004). Pre-task music warrants closer research attention because influencing affective states prior to exercise might have beneficial effects for responses during exercise, but at present it appears that the most efficacious way for music to influence how people feel during exercise, is to listen during exercise.

Listening to music during exercise can be divided into two different applications: synchronous and asynchronous. Synchronous application is when an exerciser *consciously* moves in time to the beat of the music, whereas asynchronous is when there is *no conscious* synchronization between physical movement and the musical beat. In a synchronous music study, Karageorghis et al., (2009) asked 30 exercisers to walk to exhaustion on a treadmill while striding in time with a musical beat. The music tracks were all 125bpm and participants were asked to stride in time with the music “so that the foot of their dominant leg came into contact with the treadmill belt in time with the first beat of each bar” (Karageorghis et al. 2009, p. 25). A playlist of motivational music tracks, rather than motivationally neutral tracks, applied synchronously, resulted in the highest reported pleasure during the task compared to a no-music control condition. A study examining the effects of synchronous music use during circuit training revealed less negative affect with motivational and motivationally neutral music selections in female exercisers only (Karageorghis et al., 2010). The music tracks for the muscular endurance-type activities were slightly slower than the treadmill walking activity, with 120bpm deemed appropriate to match a suitable movement rate for circuit training exercises (e.g., standing squats). While these studies offer some support for applying music synchronously, the application of synchronous music has additional practical considerations. Establishing a suitable tempo for movement is highly dependent on the task and the individual; the two studies described here included healthy young adults but appropriate tempo for different populations might be different. Applying music asynchronously to exercise does not require the same practical considerations as matching music beats to human movement patterns.

A theoretical model proposed by Karageorghis (2016) described the antecedents, moderators, and consequences of music use in sport and exercise. Musical factors (e.g., tempo, rhythm, lyrics, idiom) form the antecedents and these are thought to be hierarchical in nature whereby tempo is the most influential musical factor over the consequences (e.g., positive affective states) of listening during an exercise task. On this premise, a lineage of work explored the role of tempo in understanding asynchronous music liking and affective responses across a range of exercise intensities. In their 2014 article, Karageorghis and Jones showed that music tracks ranging in tempo from 95 to 160 bpm resulted in greater pleasure compared to not listening to music across exercise intensities ranging from 40%–90% heart rate reserve (HRR). However, it was the “medium” tempo condition (115–120bpm) that elicited the most pleasure across the exercise intensities. This indicated that regardless of whether exercise intensity was low, moderate, or high, medium tempo music was the most suitable for their sample of young, healthy, and active participants. An important point to note is that pleasure was significantly lower at 90%HRR regardless of condition (in line with theoretical predictions), but the pattern of responses to the music conditions across intensities was consistent. In other words, all conditions led to

decreases in pleasure as intensity increased, but medium tempo music remained the most pleasant condition compared to the other conditions. However, the recent meta-analysis by Terry et al. (2020) found that music tempo did not significantly moderate affective responses, and additional work could be undertaken to further understand the role of tempo in determining affective responses.

Beyond explicit examination of music tempo, studies by Elliott and colleagues (Elliott et al., 2004; Elliott et al., 2005) examined the influence that an asynchronous application of motivational music had on affect during moderate-to-high intensity cycle ergometry tasks. The studies found a consistent result; music enhanced in-task affect compared to the no-music conditions. Although previous literature has found a diminished effect of asynchronous music at very high exercise intensities, several studies have explored the effects of listening to music asynchronously during high-intensity interval training (HIIT) and sprint-interval training (SIT). Stork et al. (2015) played music throughout a SIT session and found that self-selected motivational music did not significantly improve pleasure compared to a no-music control condition. Contrastingly, Stork et al. (2019) found that a motivational music playlist did elicit more pleasure over the course of a SIT session compared to a no-music control. As demonstrated by these results, the possible role for music during very high-intensity exercise is not fully understood.

The most recent addition to the music and exercise literature is *respite music*. Respite music is listening to music only during the rest periods of an interval exercise session and preliminary evidence suggests some benefit for this type of application. Listening to fast-tempo music (125–135bpm) during 3-min recovery intervals between 5-min treadmill exercise bouts was shown to increase pleasure compared to a no-music control condition (Jones et al., 2017). Similarly, Karageorghis et al. (2021) found that pleasure increased during the exercise bouts and recovery periods when listening to medium tempo music (120–125bpm), despite only listening to music during the recovery periods. However, listening to music only during the recovery periods appears less effective than playing music throughout an entire HIIT session (Jones et al., 2020). A consistent pattern in music and HIIT studies is that affect declines over the course of a HIIT session, but in some instances, music can attenuate that decline.

Listening to music post-exercise (*recuperative music*) has received comparatively less research attention than in-task applications yet offers some promise in regulating affective states immediately after exercise. There is typically an *affective rebound* following demanding exercise whereby people experience a positive affective response immediately upon cessation of exercise (Ekkekakis et al., 2011). Karageorghis and colleagues (2018) found that slow, sedative music (~71bpm) augmented this affective rebound over a 30-min period post exercise.

The extensive evidence base for applying music in an exercise context has provided a foundation for larger scale projects; however, such projects have not yet been conducted and the most significant next step for music and exercise research is to understand the role that music might play in determining continued engagement in exercise.

Exergaming

Combining computer games with physical activity—known as exergaming—has proved to be a popular strategy with many people, and a lucrative strategy for technology companies such as Nintendo and Microsoft. While Nintendo had significant commercial success with their Wii console, researchers have sought to understand whether exergames can be successful in positively changing how we feel during and after physical activities. The literature examining exergaming is vast and a common outcome measure is energy expended, or exercise intensity, with several review articles demonstrating that exergames, or active video games, can result in activity considered moderate-to-vigorous (e.g., Sween et al., 2014). Alongside energy expenditure, enjoyment is a frequently measured outcome, but the measurement of enjoyment in this field includes a wide range of instruments and the measurement of in-task affect is uncommon.

The Wii Fit has frequently been used as a way to examine exergames; Garn et al. (2012) asked

college-age students to engage in several exercises on the Wii Fit (e.g., Ski slalom) and participants reported that the experience was enjoyable. Examining a bespoke cycling game, McGloin and Embacher (2018) showed a relationship between enjoyment and “desire to ride again” (p. 498). However, there is a severe methodological limitation in those two example studies as there is no matched-control condition. For example, Garn et al. asked participants to report their enjoyment following bouts of exercise on a Wii Fit compared to responses based on the question stem “When I am physically active...”. This comparison between enjoyment of specific Wii fit exercises and a hypothetical bout of physical activity undermines the usefulness of the findings relating to enjoyment. Further, McGloin and Embacher (2018) did not examine enjoyment compared to cycling without the game. There have been few studies examining affective responses compared to a matched-control condition (e.g., Li et al., 2018). A comparison between Wii Fit exercises (e.g., Wii Bowling) and traditional exercises (e.g., indoor bowling) demonstrated more positive emotions following the exergaming condition using the Positive and Negative Affect Schedule (PANAS; Li et al., 2018). However, conceptual issues with the suitability of the PANAS have been raised previously (see Ekkekakis, 2013).

There appears to be a pattern of positive affective responses associated with exergames (measured with myriad approaches), but there is limited evidence demonstrating affective responses during exercise. As was described earlier in the chapter, it is the affective responses during exercise that appears most predictive of future engagement. To further understand the in-task responses during exergames, researchers could seek to collect core affective responses *during* game play. A recent meta-analysis indicated that active video games do not increase physical activity in young people at short-term follow-up (Oliveira et al., 2020), and it could be suggested that further examination of affective responses during exergaming might help to explain why.

Exercising with Others

As mentioned earlier in this chapter, there are several concurrent factors that influence affective responses when exercising. Studies that have explored the influence of exercising with others have often done so in combination with other strategies. An example of this is an examination of different types of partners when exergaming. Feltz et al. (2014) asked participants to complete exercises in the presence of virtual partners (human, a nearly human-like humanoid, and a hardly human-like software generated partner) or a no-partner control. There were no differences found in enjoyment between the conditions. Further, while participants worked harder in the experimental conditions, they did not report less enjoyment. The role of a virtual partner during a rowing task in virtual reality was examined by Murray et al. (2016) and was not found to influence in-task pleasure or enjoyment. There does not seem to be strong evidence to support the use of digital partners during exergaming or virtual activities if the aim is to enhance affective outcomes. Similarly, there does not appear strong evidence for exercising with “real” partners. For example, Rogerson et al. (2020) showed that running either individually or with a small group (4–5 others) in an outdoor environment did not lead to any changes in total mood disturbance, but they did not assess core affective responses during exercise or enjoyment afterwards. Plante et al. (2007) did measure enjoyment of women exercising alone or with others and found no differences in enjoyment across both indoor and outdoor exercise. Moreover, exercising with a friend during indoor cycling did not garner greater enjoyment than cycling without a friend (Plante et al., 2011). This is not to say that exercising with others does not lead to greater pleasure or enjoyment, just that it appears scant research has directly explored this. While numerous studies have explored other benefits of exercising with others, there are very few that have included responses pertaining to pleasure–displeasure and enjoyment.

The wider role of social support as a determinant of physical activity has received significant attention. It is generally regarded as an important factor towards initiating and maintaining physical activity and exercise programs. However, there are fewer studies showing the direct effects of exercising

with others on affective responses and further examination of this would bolster the evidence base for exercising with others as an effective strategy.

Imagery

The use of imagery in sport and exercise psychology is extensive and it remains a popular and effective tool in the sport and exercise psychologist's toolbox (see Chapter 20; Rymal & Hill, 2021). There is also evidence that it can be effective at positively shaping our exercise experiences. In two studies by Stanley and Cumming (2010a, 2010b), the efficacy of *enjoyment imagery* was demonstrated in participants exercising at moderate intensities on cycle ergometers. Enjoyment imagery comprised of asking participants to recall an exercise experience they found "really enjoyable", and to recreate this as vividly as possible (including the sounds, colors, other people present). The imagery script played during the exercise encouraged participants to "experience again in the present the sensations associated with that time" (p. 585, 2010b). The results indicate this strategy can positively influence in-task pleasure and enjoyment measured post-task. This was further demonstrated by Tempest and Parfitt (2013) who reported greater pleasure in participants listening to an enjoyment and energy-focused imagery script at an exercise intensity proximal to the ventilatory threshold, compared to a control condition. As with other strategies described in this section, imagery appears to have limited benefits at higher exercise intensities as participants in the Tempest and Parfitt (2013) study reported no differences at exercise intensities above the respiratory compensation point.

Compared to some of the other strategies in this section, imagery represents a "low-tech" approach and might be more suitable in a variety of situations. While the imagery script in these studies was delivered via a recording, it could be delivered "live" with likely similar outcomes. The studies described here included young students (~20 years old) as participants, but enjoyment imagery might also have a role to play with other sub-groups of the population and with those where technology-intensive approaches (e.g., VR) might not be entirely suitable.

Mindfulness

Mindfulness can represent a "no-tech" strategy and has been explored as an option for enhancing pleasure during exercise. Most of the strategies described in this chapter seek to promote a dissociative focus, but a mindful approach offers a different path to a pleasant exercise experience. Mindfulness and physical activity are discussed further in Chapter 14 (Cox & Ullrich-French, 2021). The role that mindfulness might play in enhancing affective responses to exercise has only recently begun to be examined. In a study where insufficiently active people were assigned to either a mindfulness condition, a podcast condition, or a "self-monitoring" condition, there were no differences in Feeling Scale scores between the mindfulness and podcast conditions (Gillman & Bryan, 2020). Both conditions were more pleasant than the associative self-monitoring condition, and this is in line with previous literature describing that a predominant focus on bodily sensations (associative focus) elicits more negative affective responses (e.g., Lind et al., 2009). In a similar study exploring the relative merits of mindfulness compared to a condition promoting dissociation (music) and a control condition, there were no statistically significant differences in pleasure reported during the self-paced exercise (Cox, Ullrich-French, Hargreaves, et al., 2020). The intensity of the self-paced exercise might help to explain the lack of difference between the experimental conditions and the control condition; the intensity ranged from ~35%HRR to ~55%HRR. This lower intensity exercise did not likely pose a challenge to regulating affect, and there is diminished "need" for additional strategies at such intensities (Lind et al., 2009) as affective responses are less likely to decline in this range with normal weight exercisers.

The capacity of mindfulness to alter feelings during exercise has also been examined without comparison to other strategies. A mindful treadmill walking condition elicited greater pleasure than a control condition (Cox et al., 2018), and a mindful yoga session resulted in greater remembered pleasure

than a neutral yoga session (Cox, Ullrich-French, Cook-Cottone, et al., 2020). Bigliassi et al. (2020) found a mindful approach to be more pleasurable and enjoyable than a “mindless” condition, and control condition, during a 200m self-paced walk in a park. While mindfulness does not require an electronic device, many of the studies to date have used mindfulness scripts delivered using electronic devices (e.g., iPod). Mindfulness has been described in terms of trait- and state-like qualities, and it might be that individuals with higher trait mindfulness are able to achieve greater pleasure from utilizing this strategy during exercise. Similar to imagery, it is a technique that might require prior training to be effective and further examination of how applicable these approaches are for “naïve” exercisers would shed further light on this strategy.

Other Strategies

The strategies described in this chapter are not exhaustive and there have been other attempts to purposefully influence how people feel during exercise. For example, Pottratz et al. (2020) used subliminal primes (e.g., happy, pleased, joyful) within music videos while participants walked briskly on a treadmill. The music videos with the primes (i.e., words displayed for 16ms every 10s) led to participants reporting more pleasure compared to the music videos without primes. The efficacy of watching music video during exercise has previously been demonstrated (e.g., Hutchinson et al., 2015; Jones & Ekkekakis, 2019) and the additional effects of subliminal primes warrants further research attention.

Other novel strategies include asking exercisers to alter their facial expressions (e.g., smiling or frowning; Brick et al., 2018; Philippen et al., 2012), to wear a transcranial direct current stimulation device (Baldari et al., 2018), and rinse their mouth with water rather than drink it (Shaver et al., 2018). However, the initial evidence on these strategies does not suggest they are particularly effective at improving how we feel during exercise.

Conclusion

This chapter has described how pleasant exercise experiences are of critical importance for adherence. Exercise intensity, a fundamental component of exercise, is a central determinant of affective responses to exercise, and we have described different ways in which exercise intensity can be manipulated to promote more pleasant feelings. We described several strategies (e.g., virtual reality, music, and imagery) that can be considered in addition to the fundamentals of exercise and highlighted some key experimental work supporting their use.

There are several strategies (e.g., imagery) that have not received a great deal of research attention regarding their capacity to positively influence pleasure–displeasure and enjoyment, but there is a foundation for these to be built on. There remains tremendous potential for innovative and creative strategies and it is hoped that those described herein represent only the start of the endeavors to promote more consistently positive exercise experiences.

Further Reading

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<https://doi.org/10.4324/9781315158501-20>
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Chapter 15

Exercise and Physical Activity for Depression

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Chapter Overview

Depression is a leading cause of global disease burden. Standard treatments include pharmacologics and psychotherapy; however, these treatments are not effective for everyone and are associated with several barriers and a substantial side effect profile. There is a need to identify alternative approaches that can be used to treat, or even prevent depression. Exercise and physical activity are two lifestyle behaviors that have been touted for decades as strategies used in the prevention and treatment of depression. The purpose of this chapter is to provide an overview of the scientific evidence supporting the role of these two lifestyle behaviors used in the prevention and treatment of depression. This chapter is organized by first providing insight into the burden of depression, how it is defined and diagnosed, and main treatment approaches. We then outline the scientific evidence for the role of exercise and physical activity in the prevention and treatment of depression. We also discuss plausible neurobiological mechanisms of the exercise-depression relationship and challenges associated with establishing such mechanisms. An overall summary is then provided and outstanding areas of inquiry for future research are highlighted throughout the chapter.

Depression

Prevalence and Burden

Major depressive disorder (MDD) is one of the most common mental illnesses, with point estimates from the Global Burden of Disease Study indicating a mean global prevalence of 3.6% each year between the years 1990 and 2019 (Global Burden of Disease Collaborative Network, 2020). MDD is a leading cause of disability around the world (Friedrich, 2017) that results in a loss of work productivity and increased morbidity (Cuijpers et al., 2014; Walker et al., 2015). MDD is also highly recurrent, with about 35% of individuals experiencing an additional MDD episode within the first year of recovery (Hardeveld et al., 2010). Even after 15 years following recovery, up to 85% of individuals experience another MDD episode. Therefore, it is unsurprising that MDD often exacts profound suffering and places a staggering burden on individuals, families, and society. Indeed, the total estimated economic burden (i.e., indirect and direct costs attributed to depression) of people with MDD was US\$210.5 billion in 2010, with ~US\$27.7 billion in direct costs due to medical and pharmaceutical services directly related to MDD treatment (Greenberg et al., 2015).

How is Depression Diagnosed?

Current diagnostic practices rely on the *Diagnostic and Statistical Manual, Fifth Edition* (DSM-5; American Psychiatric Association, 2013) and use specific symptoms as indicators of depression to reach a diagnostic threshold. To constitute a DSM-5 MDD diagnosis, an individual must endorse five of nine symptoms that are present for a minimum of 2 weeks and persist for most of the day, nearly every day (American Psychiatric Association, 2013). In terms of symptoms, individuals must endorse at least one of two core criterion symptoms: depressed or low mood (e.g., feelings of sadness) and/or loss of interest or pleasure in almost all typically or previously enjoyable activities (i.e., anhedonia). An individual must also endorse at least four other symptoms: (a) significant unintentional change in weight or appetite; (b) sleep disturbances (defined as insomnia or hypersomnia); (c) severe psychomotor agitation (e.g., restlessness) or retardation observable by others (e.g., slowed speech or movement patterns); (d) fatigue or low energy; (e) a sense of worthlessness or excessive guilt; (f) impaired ability to think or concentrate and make decisions; or (g) recurrent thoughts of death that include suicidal ideation and/or attempts. Any symptom combination including at least one core criterion symptom and four additional symptoms constitutes a MDD diagnosis. Importantly, these above-mentioned criteria are also used to inform and issue a MDD diagnosis in the United Kingdom (National Collaborating Centre for Mental Health and National Institute for Health and Clinical Excellence, 2010).

Treatments for Depression

Clinical practice guidelines recommend the use of pharmacotherapy (i.e., antidepressant drugs), psychotherapeutic approaches (e.g., cognitive-behavioral therapy [CBT]), or a combination of the two as first-line treatments for MDD (American Psychiatric Association, 2010; American Psychological Association-Depression Guideline Development Panel, 2019; National Collaborating Centre for Mental Health and National Institute for Health and Clinical Excellence, 2010). In terms of pharmacotherapy, antidepressant drugs, namely selective serotonin reuptake inhibitors (SSRIs), are among the most prescribed medications and work by blocking the reuptake of serotonin. Other classes of antidepressant drugs are also prescribed, including serotonin and norepinephrine reuptake inhibitors (SNRIs), and norepinephrine and dopamine reuptake inhibitors (NDRIs); all of which work by inhibiting the reuptake of their targeted neurotransmitter(s), thereby increasing its availability, and altering their levels in the brain.

Despite antidepressant drugs being among the most widely prescribed drugs globally, they are not consistently beneficial. For example, in the Sequenced Treatment Alternatives to Relieve Depression (STAR*D) study, which is the largest depression treatment trial ever conducted and funded by the U.S. National Institute of Mental Health, the effectiveness of antidepressant drug treatments was examined among 4,041 patients with a MDD diagnosis (age range = 18–75 years). In both specialty and primary medical care settings, patients were exposed to a tiered approach consisting of four separate treatment levels. The trial was structured such that patients who were unable to achieve remission—defined as becoming symptom-free following treatment—at one level would then move onto the next treatment level, which could consist of either a switch to a different antidepressant drug or a combination of antidepressant drug with add-on psychotherapy.

STAR*D study results indicated that between 28% and 33% of the patients achieved remission following an initial 14-week treatment with citalopram (Trivedi et al., 2006). After multiple treatment steps, Rush et al. (2006) reported a cumulative remission rate of 67%, which was computed by summing remission rates achieved across all trial stages. As highlighted by Ekkekakis (2021), it is important to note, however, that the actual remission rate is a bit unclear given that a re-analysis of the available data from the trial resulted in a remission rate of 2.7% when appropriately accounting for patients who dropped out or relapsed and remained in the trial at the end of the original trial duration (see Pigott, 2011, 2015). Regardless of the “true” remission rate, many patients were left symptomatic or clinically depressed following the STAR*D trial, which highlights the variable and limited effectiveness of antidepressant drugs in depression treatment. Furthermore, research has shown that even among patients who achieve remission using antidepressant drug treatment, long-term outcomes are uncertain, as estimates indicate that approximately 40% of remitted patients are likely to relapse within a two-year period (Boland & Keller, 2008).

The efficacy of antidepressant drugs to treat depression was also examined in a meta-analysis of six studies that evaluated the impact of antidepressant drugs in randomized placebo-controlled depression treatment trials approved by the U.S. Food and Drug Administration. After aggregating individual patient-level data from 434 patients receiving antidepressant drugs and 284 patients in placebo groups across six separate trials, Fournier et al. (2010) found that antidepressant drug efficacy varied as a function of depressive symptom severity. Only patients exhibiting severe symptoms of depression before enrolling in the trial experienced significant pre-to-post treatment reductions in depressive symptoms compared to those in placebo control groups. Antidepressant drugs were no more efficacious in reducing depressive symptoms compared to placebo control for patients experiencing mild-to-moderate symptom severity before treatment.¹ Despite their limited efficacy, there are also several barriers associated with antidepressant drugs, such as costs and a vast side effect profile, including but not limited to nausea, vomiting, diarrhea, dry mouth, constipation, and sexual dysfunction. These side effects are intolerable (Corponi et al., 2020) and a primary contributor for treatment discontinuation and poor patient compliance (Lader et al., 2004).

Various forms of psychotherapy, including CBT and behavioral activation, are broadly used to treat MDD and have shown effectiveness in reducing depression (e.g., Cuijpers et al., 2013). In the largest quantitative review conducted on the topic, Cuijpers et al. (2020) assessed whether psychotherapeutic approaches were effective in reducing depression and whether effects differed across the lifespan. The authors compiled data from 366 randomized controlled trials (RCTs) and found that psychotherapy was generally effective, resulting in moderate-to-large depressive symptom reductions among individuals with clinically diagnosed depression of all ages, with moderate-to-large

¹ Those interested in learning more about research regarding the use of antidepressant drugs in the treatment of depression should refer to review articles and commentaries by Kirsch et al. (2008), Kirsch (2014), Kirsch (2019), Möller (2008), and Turner et al. (2008).

effect sizes. As with antidepressant drugs, psychotherapy is associated with variable success rates, with many individuals often failing to adequately respond or derive antidepressant benefits to psychological treatments (DeRubeis et al., 2005). Moreover, there are several barriers to accessing psychotherapy for many individuals. Psychotherapy is costly and its effectiveness depends on some of the following: (a) availability and accessibility of mental health professionals (Olfson & Marcus, 2010); (b) characteristics of the mental health professional (Olfson et al., 2009); (c) the delivery of intervention (e.g., internet-delivered versus face-to-face versus group-based); and (d) individual preferences.

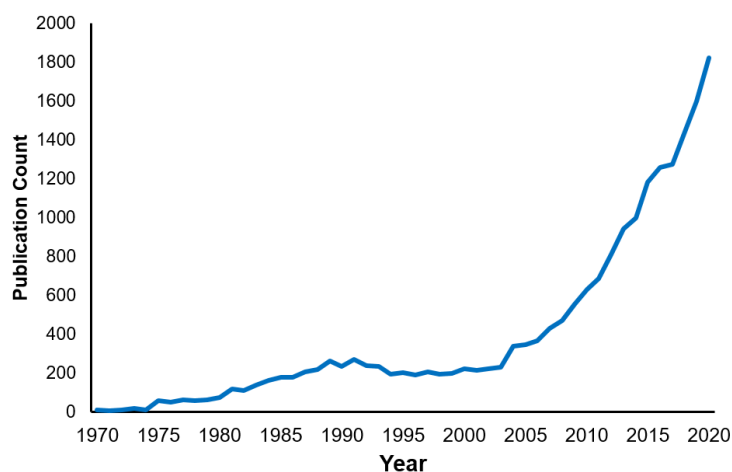
Purpose of the Chapter

Given the variable effectiveness and barriers associated with first-line treatments, there is an urgent need to identify alternative strategies or interventions that can be used to treat depression or prevent the occurrence of the disorder altogether. Exercise and physical activity are two lifestyle behaviors that have been touted for decades and even centuries in the management and treatment of depression. For example, ancient physicians and philosophers including Hippocrates (460 BCE–370 BCE), Suśruta (or Sushruta; 600 BCE) and Hua T'o from the Eastern Han Dynasty (25 CE–220 CE) recognized the importance of engaging in regular physical activity to maintain health. In particular, Hippocrates, considered by many to have been the first physician to prescribe exercise for a patient suffering from a mental health condition, believed that “eating alone will not keep a man well; he must also take exercise” (Jones, 1923; p. 229).

In the context of depression, exercise and physical activity did not gain significant traction in the scientific literature until seminal studies by William P. Morgan were published in the late 1960s to early 1970s. For example, Morgan conducted the first observational studies to indicate that patients with depression tended to have lower cardiovascular fitness levels compared to non-depressed healthy counterparts (Morgan, 1968, 1969; Morgan et al., 1970). These early efforts laid the foundation for decades of work to follow and since then, there has been burgeoning research interest in this area (see Figure 15.1), which has resulted in substantial progress toward establishing these two lifestyle behaviors as important strategies to ward off depression.

Figure 15.1

Annual Number of Research Publications on the Exercise and Physical Activity–Depression Relationship from 1970–2020



Note. The data used to create this figure were adapted from PubMed by using the search terms “exercise” OR “physical activity” AND “depression” between the years of 1970 and 2020.

In this chapter, we examine the research evidence related to exercise and physical activity² as prevention and treatment strategies used for depression. We will first summarize the evidence base of exercise and physical activity in the prevention of depression. Then, we will summarize the current evidence on the influence of exercise as a treatment for depression before offering insight into possible neurobiological mechanisms associated with the exercise-depression relationship. We highlight open questions that need to be explored in future research to help elucidate on the exercise, physical activity, and depression relationship throughout the chapter. Before presenting the evidence base, it is important to gain insight into how MDD and depressive symptoms are measured to understand conclusions that can be drawn from the evidence.



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Measurement of Major Depressive Disorder and Depressive Symptoms

Depression is characterized by substantial symptom heterogeneity, which makes its assessment difficult. For example, combinations of the nine DSM-5 criteria required for a MDD diagnosis results in 227 unique depressive symptom profiles (Fried & Nesse, 2015). Furthermore, Fried (2017) assessed content overlap of the seven most commonly used depression scales in scientific research and found 52 disparate symptoms, with little symptom overlap across instruments. Therefore, it is important to note that measurement of depression is a challenging endeavor because there is no single measure that can account for its complexity. MDD diagnoses are made using well-validated clinical interviewing instruments, while depression symptom severity is measured using self-report rating scales. Researchers often combine clinical interviews with rating scales to provide insight into the presence and nature of an individual's depression. For the purposes of this chapter, we provide a brief overview of some of the most prevalent diagnostic interviews and rating scales used in research to measure MDD and depressive symptoms.

² Physical activity is broadly defined as any bodily movement resulting in energy expenditure, while exercise is a subset of physical activity that is performed for the intention of improving or maintaining one or more aspects of physical fitness (Caspersen et al., 1985).

Diagnostic Interviews

Structured Clinical Interview for DSM-5 (SCID-5)

The SCID-5 is the most widely used structured diagnostic interview for assessing the DSM-5 disorders (First et al., 2015). Clinical psychologists and psychiatrists and other trained mental health professionals use the SCID-5 to assess for the presence of lifetime and current depressive disorders and other psychiatric disorders such as anxiety disorders (see Chapter 16; Schuch et al., 2021), psychotic disorders (see Chapter 17; Fibbins et al., 2021), substance use disorders, somatic symptom disorders, externalizing disorders and trauma-and stressor-related disorders. Research suggests that the SCID-5 has excellent reliability, sensitivity, and specificity for most of the psychological disorders with high prevalence, even when the clinical interview is conducted over the telephone (Osório et al., 2019). Due to its popularity, the SCID-5 has been translated into other languages and has been used in a variety of cultures.

Mini International Neuropsychiatric Interview (MINI)

The MINI is a short, structured diagnostic interview developed jointly by clinicians in the U.S. and Europe that follows diagnostic criteria outlined in the International Classifications of Disease and DSM-5 (Sheehan et al., 1997). The MINI is typically used in research to reduce the amount of time it takes to assess and diagnose depression. Research suggests that the reliability and validity of the MINI is on par with the SCID (Sheehan et al., 1997). The MINI can be conducted within 15 minutes and can serve as a short and accurate assessment of psychopathology.

Dimensional Measures of Depressive Symptoms

Depressive symptoms have been measured using a variety of self-report rating scales. Three commonly used rating scales of depressive symptoms are described below. These rating scales assess current depressive symptoms severity over a specified time interval and aid in determining whether treatments are effective in reducing depressive symptoms.

Beck Depression Inventory, Second Edition (BDI-II)

The BDI-II assesses the presence of depressive symptoms during a two-week period using 21 items scored on a 4-point Likert scale ranging from 0-3. Higher scores indicate greater severity of depressive symptoms (Beck et al., 1996). A score of 13 or higher on the BDI-II is generally an indication of the presence of a depressive disorder and the BDI-II has shown high sensitivity and specificity when utilized as a screening tool (Lasa et al., 2000). The BDI-II has shown good internal consistency and validity across different settings and populations (Steer et al., 1999, 2000).

Hamilton Rating Scale for Depression (HRSD)

The HRSD is a reliable assessment of the severity of depressive symptoms using 17 items scored from 0-4 (Trajković et al., 2011). A total score is obtained by summing all the items together. The HRSD provides categories that indicate the severity of depressive symptoms based on the total score. For example, a total score from 0-7 indicates normal levels of depressive symptoms, 8-16 suggests mild depression, 17-23 suggests moderate depression, and a score ≥ 24 indicates severe depression (Sharp, 2015).

Center for Epidemiologic Studies Depression Scale (CES-D)

The CES-D was originally developed to assess the number and frequency of depressive symptoms in the general population (Radloff, 1977). The CES-D exhibits strong psychometric properties, including high internal consistency and good divergent and convergent validity (Van Dam & Earleywine, 2011). The CES-D has also been validated for use in a variety of age groups, including adolescents and

young adults (Roberts et al., 1990). The CES-D has 20 items scored from 0-3. A total score is computed by summing all the items together, with possible scores ranging from 0-60. Higher scores indicate more symptoms and are weighted by frequency of occurrence during the past week. CES-D scores that are at least 16 or greater have been shown to identify individuals at risk for clinical depression (Lewinsohn et al., 1997).

In 2004, the original CES-D was revised by Eaton et al. to reflect depressive symptoms more accurately as indicated by the DSM. The CESD-R (i.e., Center for Epidemiologic Depression Scale Revised; Eaton et al., 2004) is also 20 items, with each item scored on a scale of 0-3. Higher scores reflect greater severity of depressive symptoms. Based on scoring, participants can be categorized into one of five possible depressive categories: meets criteria for major depressive episode, probable major depressive episode, possible major depressive episode, subthreshold depression symptoms, or no clinical significance. An advantage of using the CESD-R compared to other instruments is that it is free and accessible as a part of the public domain (see <https://cesd-r.com/>).

Exercise and Physical Activity in the Prevention of Depression

The potential for exercise and physical activity to prevent depression has been studied across a number of cross-sectional and prospective epidemiological studies. One of the earliest epidemiological studies to examine the relationship between physical activity levels and depression was conducted when Farmer and colleagues (1988) analyzed data from the Epidemiologic Follow-Up Study (1982–1984). Farmer et al. (1988) looked at the association between depressive symptomatology, as measured by the CES-D, and self-reported recreational and nonrecreational physical activity levels. Among 1,900 healthy adults between the ages of 25 and 77 included in analyses, increased depressive symptoms were related to lower levels of recreational physical activity. Interestingly, individuals engaging in little-to-no recreational physical activity were at almost a two-fold elevation in odds of experiencing increased depressive symptoms compared to those engaging in regular recreational physical activity. To determine whether recreational physical activity levels were a unique risk factor for increased depressive symptoms, the authors examined the influence of age, race, education, employment status, self-reported chronic health conditions, and household income on the relationship. After accounting for these variables, lower recreational physical activity levels still significantly related to increased depressive symptoms.

Goodwin (2003) extended this work by examining the relationship between regular physical activity levels and rates of MDD diagnoses among 5,877 Americans (~57% female) between the ages of 15 and 54 years who participated in the National Comorbidity Survey (1990–1992). Physical activity levels were assessed using a self-report item that asked participants, “How often do you get physical exercise—either in your job or in a recreational activity?”; MDD diagnoses were confirmed using a structured interview (i.e., the World Health Organization Composite International Diagnostic Interview). Goodwin found that regularly active individuals had a 25–38% reduced likelihood of having a MDD diagnosis compared to those who engaged in lower levels of physical activity. Importantly, this relationship remained significant even after accounting for important sociodemographic variables, such as age, gender, race, marital status, education, and income, suggesting that lower physical activity levels independently confer risk for MDD. There was also a significant dose-response³ relationship between frequency of physical activity and current prevalence of MDD in the sample, such that MDD was least prevalent among individuals who regularly engaged in physical activity (8.2%) compared to those who occasionally engaged in physical activity (11.6%), those who rarely engaged in physical activity (15.6%), and those who never engaged in physical activity (16.8%).

³ Dose-response refers to the amount of exercise or physical activity (duration, intensity, and frequency) that results in graded effects.

Gudmundsson and colleagues (2015) examined the relationship between physical activity and depression in a large-scale, prospective study of 676 Swedish women (M_{age} at baseline in 1974 = 53.4 years; $SD = 0.2$) followed over 32 years (1974–2005). Depression was measured using semi-structured interviews and the Montgomery-Åsberg Depression Rating Scale (Montgomery & Åsberg, 1979), which assesses the presence of symptoms and signs of depression over the past month. Physical activity levels were self-reported and measured using the Saltin-Grimby Physical Activity Level Scale (Saltin & Grimby, 1968), which indexes an individual's physical activity frequency and intensity. Four separate assessments occurred across the 32-year study in 1974, 1992, 2000, and 2005, which allowed the authors to test several hypotheses. First, the authors examined cross-sectional associations and showed that lower physical activity levels were associated with increased depressive symptoms at the baseline assessment in 1974. Next, the authors were interested in determining whether changes in physical activity levels coincided with changes in depressive symptoms over time. There was a significant relationship between physical activity levels and depressive symptoms, such that decreasing physical activity levels were associated with increased depressive symptoms over time. Lastly, the authors tested whether the relationship between physical activity and depressive symptoms was bidirectional and wanted to determine whether initial depressive symptoms could predict subsequent physical activity levels. Women reporting increased depressive symptoms at the 1974 baseline assessment reported less engagement in subsequent physical activity at later time points in 1992 and 2000, but not in 2005. This study provides evidence for a potential bidirectional relationship between physical activity and depression from a large and well-characterized sample of women followed over the span of three decades and suggests that physical activity may not only be a risk factor for depression but may also be a long-term consequence of depression.

There may be specific aspects of depression that could explain the prediction of lower subsequent physical activity from depressive symptoms. For example, one of the core criterion symptoms of depression is the loss of interest or pleasure in almost all typically or previously enjoyable activities (i.e., anhedonia). In a study by Leventhal (2012), the author examined whether anhedonia was related to physical activity participation levels in a sample of 157 undergraduate students ($M_{\text{age}} = 19.9$ years; $SD = 1.8$). Increased anhedonia was associated with lower physical activity levels, providing preliminary support for the notion that anhedonia, rather than other symptoms of depression, may be associated with lower physical activity participation. It is important to be cautious in interpreting these preliminary data. Leventhal used a normative sample of undergraduate students to examine the relationship between concurrent levels of anhedonia and physical activity. The sample was relatively homogenous in terms of physical and mental health status and was more physically active compared to the general population. Further, the causative nature of the relationship cannot be established since the study employed a cross-sectional design. Therefore, these findings are only suggestive, but provide a template for examining specific symptoms of depression (e.g., anhedonia) and whether they can explain physical activity participation in clinically depressed samples.

Other prospective studies have shown significant relationships between physical activity levels and depression. A ten-year cohort study of 424 patients with depression (M_{age} at baseline = 39.9 years; $SD = 14.1$; 54% female) found that higher physical activity levels were associated with lower concurrent depressive symptoms at four separate assessment points spanning 10 years (Harris et al., 2006). This evidence corroborates previous findings demonstrating that increased physical activity is associated with a reduced incidence of depression diagnoses and symptoms.

The preventive effects of physical activity on depression (i.e., symptoms and diagnoses) have also been summarized in reviews (Mammen & Faulkner, 2013; Teychenne et al., 2008). Teychenne and colleagues (2008) performed a literature review of observational and interventional research on the relationship between physical activity and depression in adults. The authors were interested in identifying the influence of physical activity characteristics (i.e., dose, domain, and physical activity

setting) on depression outcomes. There were inverse associations between physical activity and the likelihood of developing depression. Even lower doses of physical activity (i.e., 1.5 hours of moderate-intensity leisure time physical activity per week) demonstrated a protective effect against developing depression. Leisure-time physical activity was associated with a decreased likelihood of depression, while there was insufficient evidence regarding the role of other physical activity characteristics on depression.

In 2013, the Mammen and Faulkner review aimed to update the findings from Teychenne et al. by reviewing high quality research, which was determined using criteria from the Critical Appraisal Skills Programme (CASP; see <https://casp-uk.net/>) for prospective studies. Mammen and Faulkner aimed to address three important questions: (a) do baseline physical activity levels prevent follow-up depression?; (b) is there a specific dose of physical activity that protects against depression?; and (c) do physical activity levels over time impact risk for subsequent depression?. To address their first question, Mammen and Faulkner (2013) examined studies including assessments spanning at least two separate time intervals (range of follow-up period = 1–40 years) among nonclinical community samples of women/girls and/or men/boys between the ages of 11 and 100 years. Twenty-five of 30 reviewed studies supported the notion that baseline physical activity levels protect against the incidence or onset of subsequent depression. To address their second question, the authors found that engaging in as little as 10–29 min of daily physical activity could reduce subsequent risk for depression by 10% (e.g., Lucas et al., 2011), while other research showed that being physically active just one-to-two or more times per week was associated with a 40% reduced risk for depression (Bernaards et al., 2006; Hamer et al., 2009). To address their third question, the authors found that engaging in less physical activity over time was associated with an increased risk of developing depression compared to those who either maintained or increased their activity levels over the same time frame. These findings led Mammen and Faulkner to conclude that there was sufficient support for the preventive effects of physical activity on depression, even when physical activity is performed at low levels.

A few years later, Schuch et al. (2018) conducted a meta-analysis of 49 prospective studies that included 266,939 individuals (median proportion of males = 47%) with physical activity and depression measurements at baseline and a follow-up assessments (mean follow-up period = 7.4 years; range of follow-up period = 2–15 years) across studies. The authors found that individuals of all ages (youths, working-age adults, elderly persons) who engaged in more physical activity had 16.3% decreased odds of developing future depression. These findings were observed across geographical regions around the world (Asia, Europe, North America, and Oceania) and across genders.

Exercise programs may also have preventive effects. In 2020, Hu and colleagues performed a systematic review of eight meta-analyses that comprised a total of 134 individual studies examining the role of exercise interventions in the prevention of depression. In their review, Hu et al. (2020) examined the impact of exercise on depression across the lifespan among the general population and found moderate depressive symptom reductions among children, adolescents, adults, and the elderly. Although exercise interventions decreased depressive symptoms among the general population, the authors were unable to examine whether the incidence of MDD could be prevented, as no research has specifically investigated this relationship. A notable finding was that low-intensity exercise was comparable to higher intensities of exercise in reducing depressive symptoms; however, the authors noted that this evidence is of low-quality and needs to be investigated further.

The above-mentioned evidence highlights the preventive effects of physical activity and exercise on depression and symptoms of depression; however, the following questions remain: (a) does physical activity *causally* protect against—or decrease risk for—depression; or (b) does depression *causally* reduce physical activity? These two questions formed the basis for an investigation by Choi and colleagues (2019). To answer these questions, the authors used bidirectional Mendelian Randomization

(MR),⁴ a genetically informed method for establishing causality in large-scale observational studies by testing whether genetic variants in a potential underlying trait (or risk factor) are associated with an outcome of interest. In bidirectional MR, researchers can evaluate whether the underlying trait causes the outcome and vice versa. In the Choi et al. (2019) study, the authors performed MR analysis in one direction (i.e., physical activity to depression), and then performed the analysis in the opposing direction (i.e., depression to physical activity) using genetic variants associated with each trait.

Choi et al. (2019) extracted physical activity and depression data from the U.K. Biobank Study (Klimentidis et al., 2018; Wray et al., 2018) to examine relationships between physical activity and depression. Physical activity was assessed using device-based (accelerometry) and subjective (self-report) assessments and MDD was assessed using structured clinical interviews to make a lifetime MDD diagnosis, clinician-administered checklists, or medical record reviews (see Wray et al., 2018 for details). When testing the causal pathway of physical activity to depression, the authors found that each 1SD unit increase in accelerometer-assessed physical activity was associated with a 26% reduction in risk for developing MDD. This 1SD increase in accelerometer-assessed physical activity was reported as being equivalent to replacing time spent engaging in sedentary behaviors (e.g., sitting) with 15 min of vigorous physical activity, 60 min of moderate physical activity, or a combination of light and more vigorous activities. Notably, this relationship was only evident for the accelerometer-assessed measurements of physical activity, not self-reported physical activity, suggesting that measurement artifacts associated with different self-report instruments (e.g., cognitive biases, memory recall bias) could influence inferences drawn from studies examining the physical activity-depression relationship. When testing the causal pathway of depression to physical activity, the authors found no associations, regardless of how physical activity was measured. Thus, there seems to be more evidence supporting the causal pathway of physical activity to depression rather than the reverse. That is, physical activity may offer protection against the development of depression and symptoms of depression.

Overall, the available evidence indicates that physical activity and exercise are effective strategies for preventing depression. Although some research suggests that the relationship may be bidirectional (e.g., Gudmundsson et al., 2015), the study by Choi and colleagues (2019) indicates that there may be more robust evidence supporting the preventive role of physical activity on depression.

Exercise and Physical Activity in the Treatment of Depression

Initial research on exercise as a treatment for depression was conducted in the 1980s and 1990s. These findings were first summarized in a quantitative review by North and colleagues (1990) who found that exercise alleviated depressive symptoms and demonstrated comparable effectiveness—and at times increased effectiveness—to traditional treatments (e.g., antidepressant drugs and psychotherapy). A major limitation of their review was the heterogeneous composition of the patient samples. That is, the review included both nonclinical and clinical samples. Nonetheless, this meta-analysis was the first to quantitatively demonstrate the potential benefits of exercise for reducing depressive symptoms and laid the foundation for growing interest in exercise as a treatment for depression.

In the following sections, we outline the evidence examining exercise as a treatment for depression and document the potential utility for exercise to be used either as a stand-alone or complementary intervention to existing depression treatments. To date, research has examined the effects of exercise on depression using short- (acute) and long-term (chronic) interventions. Acute

⁴ For additional reading on MR and how it can be used in physical activity studies to strengthen causal inferences, refer to the research by Byrne et al. (2017) and Choi et al. (2019), respectively. Zheng et al. (2017) also provide an overview of the different types of MR studies that can be used to evaluate causality in observational epidemiological studies.

interventions have focused primarily on providing short-term relief of specific symptoms, such as elevating low mood. Chronic interventions have assessed whether exercise can be used to effectively treat or resolve depression (i.e., the disorder) and reduce total depressive symptom severity. Below, we discuss the evidence related to both types of exercise interventions and their role in the treatment of depression.



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The Role of Acute Exercise in the Treatment of Depression

Acute (single bouts) exercise will not necessarily resolve an individual's depressive symptoms, but it has well-characterized effects on regulating mood and has the potential to manage symptoms associated with depression. For example, Yeung (1996) reviewed the effects of acute exercise on mood from studies conducted between the years of 1976 and 1995 and found support for acute exercise-induced mood benefits among normative, healthy samples. In 2005, Bartholomew et al. were interested in determining whether these acute exercise-induced mood benefits could generalize to clinical samples. Using a mixed within-between subjects experimental design, the authors examined the impact of a single bout of aerobic exercise on mood and psychological well-being among 40 adults diagnosed with MDD ($M_{\text{age}} = 38.1$ years; range = 18–55; 25 female). Twenty participants completed a 30-min bout of moderate-intensity brisk walking, while the other 20 participants completed a seated rest control condition. Mood and well-being measures were assessed at four separate time points: before the condition (i.e., baseline), 10-min post, 30-min post, and 60-min post. Compared to the control condition, moderate-intensity exercise improved psychological well-being and the mood state of vigor at all postcondition assessments.

Meyer, Koltyn, and colleagues (2016a) examined the impact of acute exercise on mood in a sample of 24 females diagnosed with MDD ($M_{\text{age}} = 38.6$ years; $SD = 14.0$). The authors were interested in determining whether postexercise mood improvements were influenced by exercise intensity prescription. In a within-subjects crossover design, participants performed three different exercise intensities and a seated rest control condition on separate days. Exercise intensities consisted of light,

moderate, or hard and were performed at a rating of perceived exertion [RPE] of 11, 13, and 15, based on the original 6–20 RPE scale (Borg, 1998), respectively, while the control condition consisted of quiet rest on a stationary bicycle. Compared to the control condition, a single, 20-min bout of cycling, regardless of intensity, significantly improved depressed mood at 10- and 30-min following exercise cessation.

Meyer, Ellingson, et al. (2016) examined whether other exercise characteristics, such as a self-selected, preferred versus prescribed exercise intensity, would influence the postexercise mood effects. Using the same study sample as above, the authors had 24 females perform an additional 20-min bout of aerobic exercise at a self-selected, preferred intensity. Mood outcomes following the preferred exercise intensity were compared to the light, moderate, or hard intensity session that was closest in terms of RPE to the exercise they did during their preferred session (i.e., if the participants exercise at a low intensity during the preferred session, then their response to that session was compared with the response to the low-intensity during the prescribed session). Although the prescribed exercise session resulted in slightly favorable effects relative to the self-selected, preferred exercise bout, these effects were small and nonsignificant, highlighting that single bouts of aerobic exercise at any intensity may benefit depressed mood for up to 30- and 60-min postexercise.

Taken together, these findings support the idea that acute exercise can influence specific mood states (e.g., depressed mood and vigor),⁵ at least in the short-term, among individuals with MDD. Given that one of the core criterion symptoms of MDD is low or depressed mood, acute exercise may be particularly effective for managing these core symptoms. Additional research has shown that acute exercise impacts other disrupted psychological processes in depression. Acute aerobic exercise can attenuate self-reported negative affect among individuals who recovered from MDD (Mata et al., 2013). Separate studies have shown that acute aerobic exercise can diminish reactivity to sad stimuli (Brush, Olson, et al., 2020) and enhance positive emotional reactivity processes (Brush, Foti, et al., 2020) among adults at risk for developing MDD (i.e., those high in depressive symptoms), respectively. Therefore, the utility of acute exercise in the treatment of depression may be most evident in the management of symptoms and psychological processes commonly disrupted in depression. Acute exercise may be most effective in managing symptoms of depression while other treatments and their effects can manifest, which can take weeks to months (e.g., Knubben et al., 2007; Legrand & Neff, 2016).

Several questions remain in the acute exercise-depression relationship. The time-course of the postexercise mood benefits are not clear. The Bartholomew et al. and the Meyer et al. studies documented benefits anywhere from 30-60 min postexercise, however, whether the effects are sustained beyond 1 hr is unknown. Studies have primarily assessed aerobic forms of exercise performed for approximately 20-30 min. Research is needed to determine how long the postexercise benefits last, whether they relate to clinical outcomes (e.g., depressive symptom reduction), and whether other exercise characteristics (e.g., modality and dose) impact the effects of acute exercise on mood among individuals with clinical depression. Further, it is unknown how acute exercise compares to other brief interventions (e.g., brief behavioral activation treatment for depression [BATD]; Gawrysiak et al., 2009; Lejuez et al., 2001, 2011) in managing symptoms of depression. Research that addresses these items can

⁵ In acute exercise studies, the Profile of Mood States (McNair et al., 1971) has been the primary instrument used to assess mood in the context of depression, which has its limitations. As highlighted by Ekkekakis and Zenko (2016), the POMS was not designed to capture the global construct of “mood”. Instead, it decomposes mood into six distinct states: (a) tension; (b) depression; (c) anger; (d) vigor; (e) fatigue; and (f) confusion. There are conceivably other mood states that are not being measured by the POMS; therefore, when interpreting studies that use the POMS, it is important to note which specific mood states are altered through exercise to draw valid and accurate inferences. Refer to Ekkekakis (2013) and Ekkekakis and Zenko (2016) for a discussion on problems associated with using the POMS in exercise psychology research.

help “push the envelope” for the role of acute exercise in the treatment and management of depression.

The Role of Chronic Exercise in the Treatment of Depression

There is a long history of research examining the effects of chronic (longer-term) exercise on depression. Numerous RCTs have examined the effects of exercise for individuals with elevated depressive symptoms. There has also been an accompanying rise in the number of meta-analyses and systematic reviews on this topic (e.g., Cooney et al., 2013; Ekkekakis, 2015; Schuch, Vancampfort, et al., 2016).

In one of the most widely cited and influential RCTs, James Blumenthal and colleagues (1999) randomized 156 men and women with MDD (aged 50 years and older) to one of three conditions: (a) an aerobic exercise training program, (b) standard antidepressant treatment with sertraline (i.e., SSRI drug), or (c) combined exercise plus sertraline treatment. For the aerobic exercise training program, three, supervised sessions per week were performed over a span of 16 consecutive weeks in a group setting. Participants were prescribed walking or jogging at a vigorous exercise intensity that ranged from 70–85% of an individual’s heart rate reserve (HRR).⁶ Although participants assigned to the sertraline treatment alone experienced a more rapid initial antidepressant response, there were no significant differences in depressive symptom reduction between groups by the end of the 16-week trial, indicating that each treatment resulted in similar-sized reductions in depressive symptoms. Of the 156 patients who entered the trial, 60.4% of patients in the aerobic exercise training program, 68.8% of patients in the sertraline treatment condition, and 65.5% of patients in the combined exercise and sertraline treatment no longer met criteria for a DSM-IV⁷-defined diagnosis of MDD post-treatment.

The authors examined long-term outcomes following each of the three treatments in a subsequent study that followed these same patients six months post-intervention. At the follow-up visit, participants assigned to the aerobic exercise group displayed significantly lower rates of depression (~30%) compared to the other two treatment groups (sertraline: 52%; combined exercise plus sertraline: 55%; Babyak et al., 2000). At the time that this study was published and to the present day, this finding has been critically important in providing some of the strongest evidence for aerobic exercise as both a stand-alone and complementary treatment to standard antidepressant drugs. Notably, the aerobic exercise treatment group experienced lower relapse compared to participants in the other two treatment groups that included antidepressant drugs, providing the first evidence for the use of aerobic exercise to resolve depression over the long-term (6 months following the end of treatment). A limitation of this study was the lack of a placebo control group, which does not permit conclusions about the specific effects of each treatment arm. Therefore, Blumenthal and colleagues (2007) performed a follow-up investigation that added a placebo group to their previous study design. In this study, 202 participants (~76% female) diagnosed with MDD were randomized to the following groups for 16 consecutive weeks: (a) home-based aerobic exercise, (b) supervised exercise in a group setting, (c) sertraline, or (d) a placebo pill. After treatment, 41% of the participants achieved remission and all active treatment groups tended to have greater remission rates compared to the placebo pill group (home-based aerobic exercise: 40%; supervised exercise in a group setting: 45%; sertraline: 47%; and placebo pill: 31%). These studies have helped establish preliminary support for the efficacy of exercise treatment for MDD.

⁶ HRR is computed as the difference between an individual’s maximal and resting HR.

⁷ The American Psychiatric Association periodically updates the DSM to reflect the most current and cutting-edge research literature on psychopathology. The most recent addition (i.e., DSM-5) was released in 2013. The first version of the DSM-IV was released in 1994.

There has been a longstanding interest in determining the presence of a dose-response relationship between exercise and reductions in depressive symptoms (Rethorst & Trivedi, 2013). In 2000, a scientific symposium sponsored by Health Canada and the U.S. Centers for Disease Control and Prevention was held to determine whether a dose-response relationship exists between physical activity and multiple health-related outcomes, including depression. At the time, experts indicated that aerobic exercise programs lasting at least 6–12 weeks were consistently associated with depressive symptom reductions comparable in magnitude to those observed following antidepressant drug treatment; however, they noted that there was insufficient evidence to make definitive conclusions regarding a dose-response relationship between exercise and depression.

Dunn et al. (2005) conducted the first RCT designed specifically to assess the dose-response relationship of exercise for depression. Dunn et al. randomized 80 patients with mild-to-moderate depression severity to two different doses of aerobic exercise (7 kcal/kg/week, low dose [LD] or 17.5 kcal/kg/week, public health dose [PHD]) performed at two different frequencies (three or five days per week) or to an attention-controlled placebo group of flexibility training. Exercise intensity was self-selected by participants assigned to the exercise conditions. There was a significant dose-response effect. A greater depressive symptom reduction (47% decrease in HRSD score) was observed in patients in the PHD group compared to the LD (30% decrease in HRSD score) and stretching control groups (29% decrease in HRSD score). Further, the treatment response following the PHD prescription was comparable to depressive symptom reductions that are typically observed following other depression treatments, including antidepressant drugs and CBT.

Additional evidence supporting a dose-response relationship was examined in a study examining the effects of a long-term exercise training program on depression among women (Chu et al., 2009). The authors randomized women with high levels of depressive symptoms to a low- or high-intensity aerobic exercise condition or to a stretching program for 10 weeks. Participants in the aerobic exercise groups met for one 30–40 min supervised session and then were asked to complete three-to-four additional unsupervised exercise sessions during the week. During the unsupervised sessions, participants in the exercise groups were permitted to choose their preferred mode of aerobic exercise, such as aerobic dancing, walking, or biking. This approach may have served to enhance motivation and adherence to exercise, since participants were afforded greater autonomy in their own exercise. All three groups demonstrated a significant reduction in depressive symptoms at the end of the 10-week intervention. After controlling for pre-treatment depressive symptoms (as measured by the BDI-II), depressive symptoms were significantly lower following the intervention for the high-intensity aerobic exercise group compared to the low-intensity and stretching groups, suggesting a potential dose-response effect that favors higher aerobic exercise intensity prescriptions for depressive symptom reduction.

Trivedi and colleagues (2011) extended these findings by showing that aerobic exercise training can also be effective as an augmentation or complementary treatment for patients with treatment-resistant depression. The authors enrolled patients who failed to achieve remission following at least six weeks of treatment with SSRIs and prescribed one of two 12-week exercise doses: to patients: a low dose (4 kcal/kg/week) or high dose (16 kcal/kg/week).⁸ Among 126 men and women, those assigned to the higher dose of exercise showed a trend for increased remission rates compared to participants who were assigned to the lower dose of exercise (28.3% for the high-dose group compared to 15.5% for the low-dose group). Higher doses of aerobic exercise may more effectively treat depression and be a viable augmentation strategy for patients with depression who have failed to respond to antidepressant drugs.

⁸ The low dose prescription of 4 kcal/kg/week is equivalent to approximately 45 min of moderate-to-vigorous intensity exercise per week, while the high dose prescription is equivalent to approximately 180 min of moderate-to-vigorous intensity exercise per week.

It is important to note that the authors found that those assigned to the lower exercise exhibited better adherence rates compared to those assigned to the higher dose. This suggests that lower doses of moderate-to-vigorous intensity aerobic exercise may be more tolerable and acceptable for individuals with depression aiming to begin an aerobic exercise program.

Most of the work on the effects of exercise as a treatment for depression has focused on the effectiveness of aerobic exercise. Indeed, a meta-analysis that aggregated findings across 25 RCTs ($M_{\text{length}} = 10.29$ weeks; range of RCT length = 3–32 weeks)⁹ indicated large reductions in depressive symptoms following exercise compared to nonactive interventions (Schuch, Vancampfort, et al., 2016). The extent to which other exercise modalities have utility in the treatment of depression has received less attention.

Resistance exercise training is known for its many health benefits, especially for increasing muscular strength, muscle mass, endurance, and/or power (U.S. Department of Health and Human Services, 2008). Research has also examined the potential antidepressant effects of resistance exercise training. Singh et al. (1997) assessed whether 10 weeks of resistance exercise training significantly reduced depression among 32 older adults over the age of 60 years ($M_{\text{age}} = 71.3$ years; $SD = 1.2$) with diagnosed major/minor depression or dysthymia.¹⁰ Intervention groups consisted of a supervised, high-intensity resistance exercise training program performed three times per week or an attention-control group. Participants in the resistance exercise group performed upper and lower body exercises for three sets of eight repetitions at 80% of an individual's one-repetition maximum (1-RM).¹¹ Following the intervention, participants in the resistance exercise group reported a 59% reduction in depressive symptoms compared to a 26% reduction in the attention-control group. Participants were encouraged to continue their resistance exercise regimen at least twice weekly for another ten weeks upon completing the intervention. After this follow-up period, ~73% of the participants in the resistance exercise group achieved remission, while only ~36% of the attention-control participants achieved remission (Singh et al., 2001). This evidence served as preliminary support for the antidepressant effects of resistance exercise in both shorter and longer periods of time.

Similar to the aerobic exercise literature, Singh and colleagues were interested in examining potential dose-response effects of resistance exercise training on depressive symptoms. In a subsequent study, the authors randomly assigned 60 older adults between the ages of 60 and 85 to a high (80% of 1-RM), low (20% of 1-RM), or treatment-as-usual comparator condition for eight weeks (Singh et al., 2005). High-intensity resistance exercise training resulted in larger antidepressant effects compared to the low-intensity and treatment-as-usual comparator groups. Interestingly, the authors examined the proportion of individuals who adequately responded to treatment, which was defined as a 50% reduction in the HRSD score from pre-to-posttreatment. The results revealed a 61% response rate in the high-intensity group compared to 29% and 21% in the low-intensity and treatment-as-usual groups, respectively. This evidence suggests that resistance exercise training may not only reduce depression, but higher intensities may be more effective.

Noting that the majority of meta-analyses conducted on the exercise and depression relationship has focused on aerobic forms of exercise, Gordon and colleagues (2018) aimed to address

⁹ One RCT included in the Schuch, Vancampfort, et al. (2016) meta-analysis did not report the trial length. Therefore, the mean and range of trial lengths reported include 24/25 RCTs that were included in the authors' meta-analysis.

¹⁰ Dysthymia is no longer considered a diagnosis in the DSM-5. It was removed from the DSM-Fourth Edition (IV) and replaced by persistent depressive disorder. Minor depression was considered a Depressive Disorder Not Otherwise Specified in the DSM-IV.

¹¹ Resistance exercise training intensity is often based off an individual's 1-RM, which is defined as the maximal amount of weight an individual can lift in a single repetition (Garber et al., 2011).

this gap in the literature by performing the first quantitative review of the effects of resistance exercise training on depression. Gordon et al.'s meta-analysis combined effects from 33 RCTs, which included a total of 1,877 participants, and found that resistance exercise training resulted in a moderately-sized reduction in depressive symptoms, regardless of sociodemographic characteristics. It is important to note that this meta-analysis combined findings from studies that included both normative and clinical samples. When examining only those with clinical levels of depression (i.e., at least mild-to-moderate depressive symptoms), the resistance exercise training had even larger antidepressant effects compared to those with subclinical depression. There was no specific impact of type of resistance exercise training protocol on depression outcomes.

Collectively, the evidence generally supports the notion that both aerobic and resistance forms of exercise are effective treatments for depression. This has been shown in studies examining exercise training as a monotherapy or complementary treatment to other treatments, namely antidepressant drugs. The antidepressant effects of exercise are at least as comparable to antidepressant drug treatments and over the long run, there is evidence to suggest that exercise may result in more favorable reductions in depressive symptoms and disorders compared to antidepressant drugs (Babyak et al., 2000). It is important, however, for future studies to continue to examine the long-term impact of exercise on depression to fully understand how long effects are sustained. Most of the research to date has been conducted in adults with depression, while the impact of exercise training in other populations that are substantially impacted by depression, including children and adolescents, is less clear. Future studies are needed in other vulnerable populations to determine whether effects are generalizable. Overall, exercise should play an important role in the treatment of depression as a stand-alone and/or complementary treatment.

Plausible Neurobiological Mechanisms of the Antidepressant Effect

Establishing neurobiological mechanisms of the antidepressant effects of exercise has historically been challenging. Depression is characterized by substantial heterogeneity; estimates indicate that anywhere from 119 (Park et al., 2017) to 681 (Akil et al., 2018) to even 1,000 (Fried & Nesse, 2015) unique combinations of depressive symptoms could meet MDD diagnostic criteria. Research attempting to explain the mechanistic effects of exercise on depression often struggle to accommodate for the diverse symptom profiles and presentations across individuals, which makes it difficult to design studies aimed at establishing the mechanisms implicated in the antidepressant effects of exercise. In addition to the heterogeneity, exercise is associated with a wide range of neurobiological effects (see Dishman et al., 2006 for an extensive overview). When considering these factors, it is unsurprising that our general understanding of the mechanisms underlying the antidepressant effects of exercise is unclear. It is likely that exercise improves mood and other specific symptoms in the short-term and reduces depression in the long-term through numerous mechanisms because of its wide range of effects. Indeed, experts have proposed several neurobiological mechanisms (Kandola et al., 2019; Schuch, Deslandes, et al., 2016). In this section, we provide a very brief overview of some of the neurobiological mechanisms that have been assessed in acute and chronic exercise studies. We only focus on neurobiological mechanisms that are hypothesized to be implicated in the pathophysiology of depression and have been studied in relation to changes in depressed mood or depressive symptoms.

Neurobiological Mechanisms Studied in the Context of Acute Exercise

In a series of secondary data analyses, Meyer and colleagues were interested in exploring potential neurobiological mechanisms implicated in the mood-enhancing benefits of acute exercise. In three separate reports, Meyer and colleagues examined whether serum concentrations of brain-derived neurotrophic factor (BDNF; Meyer, Koltyn, et al., 2016b), endocannabinoids and related lipids (Meyer et

al., 2019), and cytokines (Perez et al., 2020) could be altered by acute bouts of aerobic exercise and whether alterations were associated with acute changes in mood states. In each of these studies, blood was drawn before and within 10 min after each condition to determine changes in BDNF, endocannabinoids and related lipids, and cytokines. Specific descriptions related to each analysis are outlined below.

In Meyer, Koltyn, et al. (2016b), 24 women with MDD performed 20 min of aerobic exercise at light, moderate, and hard exercise intensities; participants also completed a quiet rest condition. While acute exercise significantly improved depressed mood and increased BDNF levels regardless of exercise intensity, these findings do not explain the mechanism, as authors failed to observe a relation between changes in depressed mood and BDNF levels. In Meyer et al. (2019), 17 women with MDD exercised for 20-min at a moderate- (prescribed exercise intensity equivalent to a RPE of 13) or self-selected, preferred-intensity of aerobic exercise (chosen exercise intensity: $M_{RPE} = 12.5$; range = 8.5–16). Moderate-intensity aerobic exercise significantly increased endocannabinoid levels (i.e., anandamide) and a related lipid (i.e., oleoylethanolamine). Most notably, changes in anandamide and oleoylethanolamine were associated with changes in mood states, suggesting that endocannabinoids and related lipids may contribute to the mood-enhancing effects of acute exercise in MDD. No such relationships were observed following the self-selected, preferred intensity of aerobic exercise. Differences in findings may be attributed to the substantial variability in how participants selected and performed their preferred exercise intensity, with some individuals exercising at lower intensities than others (see wide range of self-selected intensities above). This suggests that lower intensities of exercise may not be sufficient to induce changes in the endocannabinoid system, or alternatively, other mechanisms might be at play. In 2020, Perez et al. examined whether serum concentrations of interleukins (ILs) 6 and 8 and tumor necrosis factor- α (TNF- α) could be altered by a 20-min bout of aerobic exercise. Participants completed the same experimental conditions as the Meyer, Koltyn, et al. (2016a) study. Perez and colleagues found that hard exercise increased IL-6, IL-8, and TNF- α among 19 women with MDD, while the light and moderate exercise intensities failed to modulate these inflammatory markers. The authors also examined whether changes in each of these inflammatory markers related to changes in depressed mood but failed to find any significant relationships. The authors hypothesized that there is a possibility that acute increases in inflammatory markers following hard exercise could have potential for prompting chronic inflammatory adaptations; however, limited conclusions can be drawn from these studies with longer-term studies needed to test this possibility. Other limitations of these data include the secondary nature of the analyses, as well as the relatively small samples including only women.

In addition to this empirical evidence, Schuch, Deslandes, and colleagues (2016) propose other plausible neurobiological mechanisms that may be modulated by acute exercise, including increased levels of atrial natriuretic peptide, brain natriuretic peptide, copeptin, and growth hormone; however, currently, there is a dearth of evidence to make strong conclusions about neurobiological mechanisms sensitive to change through acute exercise in depression.

Neurobiological Mechanisms Studied in the Context of Chronic Exercise

Chronic exercise studies have examined potential neurobiological mechanisms associated with the antidepressant effects of exercise. These studies have focused on assessing whether changes in specific neurobiological mechanisms are associated with depressive symptom change in response to exercise treatment. To date, chronic exercise studies have focused mostly on oxidative stress markers, BDNF, and inflammatory markers.

In Schuch et al. (2014), the authors conducted a RCT evaluating the effects of exercise as an add-on treatment to treatment-as-usual on thiobarbituric acid-reactive substances (TBARS) and BDNF among 26 severely depressed inpatients. Fifteen individuals were randomized to an exercise plus treatment-as-

usual condition, while the other 11 participants were allocated to a treatment-as-usual comparator condition. Participants performing the add-on exercise completed a 16.5 kcal/kg/week dose three times per week throughout their hospitalization stay. Across the hospitalization stay (exercise participants: $M_{\text{duration}} = 21.63$ days; $SD = 4.5$; control participants: $M_{\text{duration}} = 23.82$ days; $SD = 5.7$), participants in the exercise group performed ~9 total sessions. Serum TBARS levels increased for the exercise group, while no effects were found for serum BDNF levels, indicating that TBARS may be a potential neurobiological mechanism implicated in the antidepressant effects of exercise for participants with depression. It was unclear from this study whether change in TBARS was associated with changes in depressive symptom improvements. Importantly, these patients were also receiving treatment-as-usual which consisted of antidepressant drugs and/or electroconvulsive therapy. It is possible that exercise may impact serum TBARS levels when combined with other forms of treatment. Future studies specifically designed to test whether TBARS is a neurobiological mechanism of the effects of exercise for depression are needed.

Lavebratt and colleagues (2017) examined whether serum IL-6 levels could be altered by 12 weeks of exercise and whether changes corresponded to changes in depressive symptoms. In their study, 116 patients (age range = 18–64 years) completed light ($n = 48$), moderate ($n = 36$), and vigorous ($n = 32$) exercise conditions for three 60-min exercise sessions over 12 weeks. Reductions in IL-6 levels were associated with depressive symptom reductions, indicating that IL-6 may be a potential mechanism of the antidepressant effects of a chronic exercise training program. The authors did not examine other inflammatory markers or mechanisms. It remains unknown whether other mechanisms could also account for the effects.

These studies suggest that there may be a role of specific oxidative stress and inflammatory markers in the antidepressant effects of exercise training. Overall, the available data for neurobiological mechanisms of the antidepressant effects are too premature to make any definitive conclusions.

Conclusion

MDD is a prevalent and burdensome mental illness that significantly impacts individuals of all age groups across the globe. Antidepressant drugs and psychotherapeutic approaches are first-line treatments for MDD; however, these treatments are associated with several barriers, have variable effectiveness, and have vast side effect profiles. Exercise and physical activity are two lifestyle behaviors that can be used to effectively prevent and treat depression. Cross-sectional and longitudinal research indicates that engaging in regular physical activity is associated with a diminished risk of current and future depression. Even engaging in small amounts of physical activity can reduce an individual's risk for depression. A large body of evidence has examined exercise programs in the treatment of depression. Acute exercise has shown to exert powerful effects on mood and other disrupted psychological processes in depression, which may have implications for using exercise as a strategy to help alleviate or manage symptoms (e.g., low mood or deficits in positive or negative affect) in the short run. In the long-term, treatment studies have shown that aerobic and resistance forms of exercise training result in moderate-to-large effect sizes in depressive symptom reduction (Gordon et al., 2018; Schuch, Vancampfort, et al., 2016). Establishing neurobiological mechanisms of the antidepressant effects of exercise has been challenging due to the heterogeneous nature of depression and the wide range of neurobiological effects of exercise, but several mechanisms have been proposed and tested. In conclusion, research suggests that consistent physical activity can reduce the risk of developing depression and acute and chronic exercise treatment programs are efficacious in improving mood and symptoms of depression.

Learning Exercises

1. Clinicians issue a diagnosis of MDD based on the presence of at least five symptoms that are present over the same two-week time period. According to the Diagnostic and Statistical Manual, Fifth Edition (DSM-5) of the American Psychiatric Association (2013), which two symptoms are required for a diagnosis? Do both symptoms need to be present in order to be issued a diagnosis? Identify the remaining symptoms that may also be present to constitute a MDD diagnosis.
2. Can physical activity be useful in protecting against the development of depression or does depression reduce physical activity levels? What evidence is there to support your answer?
3. Can exercise be used to treat depression? What evidence is there to support your answer?
4. How does exercise compare with antidepressant drugs in the treatment of depression? What are the long-term depression outcomes of either treatment?
5. Does a specific dose of exercise elicit larger antidepressant effects compared to other doses of exercise? Use the available evidence to support your answer.
6. Most treatment studies examining the effects of exercise on depression have focused on aerobic forms of exercise. Does resistance exercise have utility in the treatment of depression? Use evidence to support your answer.
7. Several neurobiological mechanisms have been proposed to underlie the antidepressant effects of exercise. Identify at least two mechanisms that change in response to exercise. Indicate the dosage of exercise that was used in terms of type, frequency, intensity, and duration.
8. A friend of yours knows that you are taking a class in sport and exercise psychology. They have recently been issued a MDD diagnosis and are currently considering different treatment options. Your friend wants you to share your insights on how to manage and alleviate their depression. Based on the evidence outlined in this chapter, provide insight into different treatment options in terms of treatment outcomes and efficacy in reducing depression over the long-term. Which treatment would you recommend?
9. What type of exercise can individuals do in the short-term versus long-term to manage and reduce their depression? Use scientific evidence to support your answer.

Chapter 16

Physical Activity and Exercise for the Prevention and Management of Anxiety

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Chapter Overview

Anxiety disorders are a leading cause of the global disability burden. The mainstays of treatment for anxiety disorders are pharmacological and psychological interventions. Although a growing number of pharmacological treatments are available, about the half of patients do not respond adequately and require additional approaches for the prevention and treatment of anxiety disorders. Observational studies have demonstrated that physical activity is associated with a lower risk of incident anxiety at a population level. Randomized controlled trials have found that exercise, a subset of physical activity, can reduce anxiety and stress symptoms in people with anxiety disorders. The anxiolytic effect of exercise is observable following a single bout of exercise or a training program of aerobic and/or resistance training over several weeks. The mechanisms underlying the anxiolytic effects of exercise are not fully understood, but some neurobiological factors, such as changes on neurogenesis, inflammation, endocannabinoids, and regulation of the autonomic system, are potential candidates. In addition, psychological factors, such as reduction of anxiety sensitivity, increases in self-esteem, and mastery may help to explain this effect. In the present chapter, we provide an overview of the use of physical activity and exercise for the prevention and treatment of anxiety.

Physical Activity and Anxiety

Anxiety disorders are highly prevalent disorders in most cultures across the world (World Health Organization [WHO], 2017). In 2015, an estimated 3.6% of the global population had anxiety disorders. Some variation can be seen in different countries across the world, ranging from 2.2% (Vietnam) to 9.3% (Brazil; WHO, 2017). Anxiety disorders are the sixth leading cause of global disability with considerable economic costs (WHO, 2017).

The core features of anxiety disorders include persistent, intense, and excessive fear or worry (American Psychiatric Association [APA] 2013; WHO, 1992). Each disorder has its specific presentation, but the most common symptoms of anxiety across all disorders are feeling restless, wound-up or on-edge, fatigued, irritable, muscle tension, with difficulties concentrating, difficulties in controlling feelings of worry, and sleep problems. Although anxiety symptoms are present in daily life, for a clinical diagnosis these symptoms must (a) persist for a certain period of time, which varies from disorder to disorder (e.g., 6 months for generalized anxiety disorder [GAD]); (b) cause significant problems in areas of life (social interactions, school, and work); and (c) not be a consequence of substance use. The diagnosis must be made by a psychiatrist or psychologist. In addition to the profound burden on mental health and well-being, people with anxiety disorders experience poorer physical health (Firth et al., 2019). For example, people with anxiety disorders are at increased risk of cardiovascular disease (Batelaan et al., 2016), diabetes (Smith et al., 2018), and metabolic syndrome (Tang et al., 2017). Also, anxiety disorders are associated with a 77% increase in all cause-mortality risk and a 177% increase in cardiovascular mortality risk (Denollet et al., 2009).

The main treatments for people with anxiety disorders consist of antidepressants and psychotherapies (Katzman et al., 2014). While antidepressants are typically more efficacious than placebo (de Vries et al., 2018), the benefits are mostly modest for those with mild-to-moderate symptom severity, and eventually, the benefit-risk ratio may be unfavorable for these patients due to the side effects (de Vries et al., 2018). Also, adherence to antidepressant treatments is not optimal, and dropout rates can be four times higher compared to dropout with placebo pills (Cipriani et al., 2018) mostly due to common side effects, including weight gain, increased diabetes risk, and sexual dysfunction. Some of these side effects can potentially contribute to poor physical health for people with anxiety disorders. Psychological therapies, such as cognitive behavioral therapy, have moderate positive effects for people with anxiety disorders (Carpenter et al., 2018), but without improving physical health. In addition, dropout from psychotherapies can be up to 29%, which is substantially higher than control groups (17%; Carpenter et al., 2018).

Given the considerable individual and societal burden of anxiety disorders, there is an urgent need to identify modifiable risk factors and additional intervention strategies to mitigate this burden. Emerging evidence indicates physical activity (PA) and exercise can reduce the risk of developing anxiety disorders and be useful strategies for the treatment of anxiety disorders by reducing anxiety symptoms. In the present chapter, we provide a brief overview of the current evidence for (a) the role of PA and exercise as protective factors against incident anxiety, and (b) the use of PA and exercise as therapeutic strategies for people with anxiety disorders. We highlight the use of exercise as a strategy for acute management of symptoms, the effects of different types of exercise training, neurobiological mediators between exercise and anxiety, and issues related to exercise prescription, adherence, and dropout.

Can Physical Activity Protect Against Incident Anxiety?

Incident anxiety refers to the occurrence or development of new cases of anxiety. Cross-sectional studies have shown that people with higher levels of PA present a decreased risk of having anxiety. For example, cross-sectional evidence using data from the World Health Survey, accounting for 237,964 individuals from 47 countries, demonstrated that those with low levels of physical activity were

at 32% increased risk for having anxiety (odds ratio [OR], 1.32; 95% confidence interval [CI], 1.17–1.47; Stubbs, Koyanagi, et al., 2017). Also, meta-analytical findings reveal that those spending more time in sedentary behavior are 48% more likely to have anxiety, even when the analyses are adjusted for sociodemographic and health-related factors (OR, 1.48; 95% CI, 1.25–1.75; Allen et al., 2019).

A limitation of cross-sectional studies is the inability to infer directionality. However, these associations seem to be bidirectional. For example, there is a higher prevalence of low physical activity (da Silva et al., 2014; Stubbs, Koyanagi, et al., 2017) and high sedentary behavior in people with anxiety disorders (de Wit et al., 2011). Likewise, data from prospective cohort studies, which followed people free from anxiety for at least one year, demonstrate that people with higher self-reported PA levels are at 26% lower odds of developing meaningful anxiety symptoms and disorders (OR, 0.74; 95% CI, 0.62–0.88; Schuch et al., 2019). However, the evidence that PA protects against anxiety is mostly based on self-reported questionnaires that are more likely to suffer from social desirability and recall bias (Schuch et al., 2019). More recently, some evidence has shown that people with low cardiorespiratory fitness and muscular strength, two objectively assessed measures of physical capacity largely influenced by PA levels, have 60% higher risk (95% CI 1.14–2.11) of presenting anxiety compared to those with higher cardiorespiratory fitness and muscular strength (Kandola et al., 2020).

Exercise as a Treatment for Anxiety

Exercise can be used to alleviate anxiety symptoms in people with anxiety disorders. The effects of exercise on anxiety symptoms can be both acute, following a single bout of exercise, as well as chronic, following a period of weeks of intervention.

Acute Bouts of Exercise for Managing Symptoms

Current evidence has demonstrated that a single exercise bout results in a small reduction in state anxiety (Ensari et al., 2015). However, this notion has changed over the last five decades. In the 1970s, some believed that exercise would elicit increased anxiety symptoms in those prone to anxiety attacks/panic episodes (O'Connor et al., 2000). This fear seems to have been borne from an experiment conducted by Pitts and McClure (1967), who demonstrated that the infusion of sodium lactate induced symptoms of anxiety in people with anxiety neurosis. It is well-known that exercise results in an increased secretion of lactate. Therefore, the increase in lactate levels caused by exercise was thought to trigger panic attacks. This fear was reinforced because exercise results in bodily responses similar to somatic symptoms of anxiety, such as sweating, increased heart rate, and respiratory frequency. However, these findings were not supported by later evidence. A study in people with panic disorders observed only 5 panic attacks in 444 exercise bouts (1.13%) performed in the laboratory (O'Connor et al., 2000). In addition, further evidence has demonstrated that exercise is not only safe for people with panic disorders but also helps to alleviate acute state anxiety symptoms (Ströhle et al., 2009).

Exercise Training for Managing Symptoms

Diverse studies have attempted to discuss and meta-analytically synthesize the evidence on the anxiolytic effects of exercise training in healthy people (Conn, 2010), people with multiple chronic conditions (Herring et al., 2010), and people with anxiety disorders or elevated anxiety symptoms (Aylett et al., 2018; Bartley et al., 2013; Jayakody et al., 2014; Stubbs et al., 2017). Overall, small effects were observed for people with and without chronic health conditions (Conn, 2010; Herring et al., 2010). For people with anxiety disorders or elevated anxiety, there was some divergence in the magnitude and direction of the effects.

Bartley et al. (2013) synthesized data from seven randomized controlled trials and found no effect of exercise in people with anxiety disorders. However, this meta-analysis included studies comparing exercise versus relaxation plus paroxetine (Wedekind et al., 2010), strength exercises

(Martinsen et al., 1989), or cognitive behavioral therapy (Hovland et al., 2013). The inclusion of such studies in a meta-analysis is problematic because the control group “interventions” are largely effective. As such, to become effective, exercise effects would have to overcome the treatment effects of the control groups. For example, in the Wedekind et al. (2010) study, the authors compared the effectiveness of four interventions, namely (a) paroxetine + exercise; (b) paroxetine + relaxation; (c) placebo + exercise; and (d) placebo + relaxation, showing no differences between groups. However, all groups improved significantly and with a large effect on anxiety symptoms (paroxetine + exercise, Cohen’s $d = 2.33$; paroxetine + relaxation, Cohen’s $d = 2.00$; placebo + exercise, Cohen’s $d = 1.53$; placebo + relaxation, Cohen’s $d = 1.87$). In the study of Martinsen et al. (1989), aerobic exercise (walking and jogging) was compared to a combination of strength exercises, flexibility, and relaxation. At the end of the intervention, both groups improved without having any statistical differences between them. Lastly, Hovland et al. (2013) compared cognitive behavioral therapy versus exercise and found that cognitive behavioral therapy, a well-established and known intervention, is more effective than exercise for anxiety symptom reduction. The inclusion of such comparison groups likely results in a reduction of the estimated effect of exercise.

To address this, the authors explored the role of the type of control group in a subgroup meta-analysis, showing that exercise has a large effect compared to placebo or waitlist control groups. Stubbs et al. (2017) revisited the same topic in a more recent meta-analysis that included six randomized controlled trials. In this meta-analysis, studies did not include comparisons against other forms of exercise or other established treatments (e.g., antidepressants or psychotherapies). Stubbs et al. (2017) found a moderate effect of exercise on anxiety (SMD, -0.58; 95% CI -1.0, -0.76). The literature investigated the effects of exercise training in multiple anxiety disorders, and some heterogeneity in the effects can be observed across disorders. This can potentially mean that exercise may be more effective for some anxiety disorders compared to others. Due the small number of studies, it is not clear if the differences in the effects between studies are related to the diagnosis or to other methodological aspects, such as weekly frequency, exercise intensity and type, trial duration, comparison group, or other factors related to the sample, such as age, gender, symptom severity, or use of other treatments. For example, the seminal study lead by Broocks et al., (1998) found a very large effect of aerobic exercise on anxiety symptoms in patients with panic disorder. The exercise sessions were held three to four times per week, for eight weeks, while Herring et al. (2012) in another seminal study found a moderate effect of strength training, held two times per week, for 6 weeks in people with GAD. A brief summary of some existent studies by anxiety diagnosis is given in Table 16.1.

Exercise Prescription

The American College of Sports Medicine (ACSM) recommends the use of the frequency, intensity, time, and type (FITT) principle as a strategy to guide the exercise prescription (American College of Sports Medicine, 2013). However, the literature exploring how the FITT characteristics relate to the anxiolytic effects of exercise in people with anxiety disorders remains scarce.

Table 16.1*Randomized Controlled Trials on Exercise as a Treatment for Anxiety*

Study	Disorder	Sample	Exercise type	Frequency/ Duration	Add-on	Control/ comparison	Mental health outcomes
Abrantes et al. (2017)	OCD	<i>n</i> = 28 exercise/ <i>n</i> = 28 control	Aerobic Exercise	1x week/12 weeks	Usual treatment	Health Education	No differences between groups (both improved)
Bischoff et al. (2018)	PD	<i>n</i> = 39 Exercise group/ <i>n</i> = 38 control	Aerobic Exercise	2x week/6 weeks	CBT	Low-intensity exercise	↓Anxiety symptoms
Broocks et al. (1998)	PD	<i>n</i> = 16 Exercise group/ <i>n</i> = 15 placebo	Aerobic Exercise	3 to 4x week/8 weeks	None (monotherapy)	Placebo pill	↓Anxiety and depressive symptoms
Gaudlitz et al. (2015)	PD	<i>n</i> = 24 Exercise group/ <i>n</i> = 23 control	Aerobic Exercise	3x week/8 weeks	CBT	Low-intensity exercise	↓Anxiety symptoms
Goldstein et al. (2018)	PTSD	<i>n</i> = 21 Exercise group/ <i>n</i> = 26 control	Resistance Exercise + Aerobic exercise	3x week/12 weeks	About 40% of participants were taking psychiatric drugs	Waitlist control	↓PTSD symptoms

Table 16.1 (continued)

Herring et al. (2012)	GAD	<i>n</i> = 10 aerobic/ <i>n</i> = 10 resistance/ <i>n</i> = 10 control	Aerobic Exercise / Resistance Exercise	2x week/6 weeks	Some patients were also using antidepressants	Waitlist control	↓Worry symptoms
MarCom et al. (2008)	PD, GAD, SP	<i>n</i> = 38 exercise/ <i>n</i> = 36 control	Aerobic Exercise	Unknown frequency /9– 10 weeks	GCBT	Educational sessions	↓Anxiety, depressive, and stress symptoms
Plan et al. (2020)	GAD	<i>n</i> = 17 exercise/ <i>n</i> = 16 control	Aerobic Exercise (High intensity interval training)	Every second day/12 days	Stable doses of antidepressants and pregabalin were allowed	Low-intensity exercise	↓Anxiety, depressive, stress symptoms, somatic and worry symptoms
Powers et al. (2015)	PTSD	<i>n</i> = 5 exercise/ <i>n</i> = 4 control	Aerobic Exercise	1x week/12 weeks	ET	ET Alone	↓PTSD symptoms
Rosenbaum et al. (2015)	PTSD	<i>n</i> = 42 exercise/ <i>n</i> = 39 control	Resistance Exercise + Aerobic exercise (walking)	3x week/12 weeks	Usual treatment	Usual treatment alone	↓PTSD, depressive, stress symptoms

Note. Abbreviations: CBT = Cognitive Behavioral Therapy, ET = Exposure Therapy, GAD = Generalized Anxiety Disorders, GCBT = Group Behavioral Therapy, OCD = Obsessive Compulsive Disorder, PD = Panic Disorder, PTSD = Post-traumatic Stress Disorder, SP = Social Phobia

Most of the literature is based on aerobic exercises. However, a recent meta-analysis has shown that nonaerobic forms of exercise, such as resistance training, are also capable of reducing anxiety symptoms in healthy people and people with a chronic condition (Gordon et al., 2017). Early head-to-head trials comparing aerobic versus nonaerobic exercises found no difference between the two interventions (Martinsen et al., 1989). When taken together, there is evidence that both aerobic and nonaerobic exercises are effective.

The optimal dose of exercise intensity, frequency, or volume remains unclear. Most randomized controlled trials investigating this topic have shown the effectiveness of moderate-intensity exercise (Stubbs, Vancampfort, et al., 2017). Still, there is a paucity of trials investigating whether the same benefits could be achieved with lower intensities. The optimal weekly frequency and optimal weekly volume for achieving the greatest reductions in anxiety symptoms is also unclear.

In order to maximize exercise engagement and reduce dropout, clinicians and exercise professionals should consider that exercise should be rewarding and enjoyable, and autonomous motivation seems to be a key element for exercise engagement and maintenance (Vancampfort et al., 2015). Autonomous motivation leads someone to engage in exercise behavior for its own sake and intrinsic rewards, such as challenge and enjoyment (also see Chapter 3; Quested et al., 2021). Accordingly, every aspect related to the exercise prescription should consider personal preferences and make the exercise the most enjoyable. Hence, it might be viable to prescribe moderate exercise intensities or self-selected exercise intensities. For example, exercising at self-selected intensities and intensities below the ventilatory threshold are more likely to be linked to positive affective responses. In contrast, higher intensities are more likely linked to a negative affective responses in people with lower physical activity levels (Brand & Ekkekakis, 2018; also see Chapter 11 [Jones & Zenko, 2021]; Chapter 12 [Zenko & Ladwig, 2021]). In addition to intensity, it is also important to consider other personal preferences such as music listening, time of day, the presence of friends or family, and exercise setting. All these aspects should be considered and discussed with the exerciser. Including the exerciser in the decision-making process should increase the sense of autonomy (Vancampfort et al., 2015).

Potential Mediators (Mechanisms) of the Anxiolytic Effects of Exercise

Potential mechanisms underpinning the anxiolytic effects of exercise may be related to changes in biological markers and psychosocial/behavioral factors. Some biological markers are candidates to explain the effects of exercise. We do not intend to provide an exhaustive list; instead we briefly mention those with supporting evidence. It is believed that brain-derived neurotrophic factor (BDNF) potentially counteracts the impact of stress hormones on the hippocampus (Suliman et al., 2013). However, this protective mechanism seems to be impaired in people with anxiety, given that BDNF levels are reduced in this group (Carbone & Handa, 2013). Exercise, in turn, releases the secretion of BDNF in humans (Szuhany et al., 2015). A previous study has found an increase in BDNF serum levels following a single exercise bout (Strohle et al., 2010). However, no study has evaluated the BDNF response to long-term interventions in people with anxiety disorders (to the best of our knowledge). A second potential explanation is the exercise-related regulation of the autonomic system (AS) functioning. Some people with anxiety have impaired AS functioning, marked by decreased heart rate variability (Alvares et al., 2016). In turn, exercise improves AS functioning in patients with anxiety and AS dysfunction (Asmundson et al., 2013; Gaul-Aláčová et al., 2006). Lastly, the endocannabinoid hypothesis proposes that the endocannabinoid system is responsible for linking and integrating the perception of external and internal stimuli with psychological outcomes, such as anxiety, allowing the adaptation to its constantly changing environment (Lutz et al., 2015). Acutely, exercise promotes an increase on anandamide peripheral levels, an endocannabinoid in women with major depressive disorder, and this increase was associated with lower state anxiety (Meyer et al., 2019). However, there is a paucity of

studies showing that long-term anxiolytic effects of exercise are related to adaptations, instead of transient increases, in endocannabinoid levels.

One potential psychosocial mechanism relates to anxiety sensitivity. The systematic and deliberate exposure to somatic symptoms, such as increased heart rate, respiratory frequency, and sweating through exercise, has been shown to reduce the fear of bodily sensations related to anxiety (Smits et al., 2008). Second, physical activity and exercise have been associated with increased self-esteem, improved self-concept, and increased self-efficacy. For example, a study with young college women demonstrated that PA is inversely associated with symptoms of social phobia, GAD, and obsessive-compulsive disorders. Among the factors that explained these associations were self-concept and self-esteem (Herring et al., 2014). Lastly, exercise might lead to an increased sense of mastery and self-efficacy, especially when associated with marked increases in physical capacity and functioning, body composition changes, or when mastering a new set of techniques (Elavsky, 2010).

Conclusions

PA can reduce the risk of anxiety symptoms and disorders in the general population. In addition, among people with anxiety, acute and repeated exercise is safe and can alleviate symptoms. Exercise also has cardiovascular and metabolic benefits that may reduce the physical health risks that are associated with anxiety disorders. The mechanisms related to the anxiolytic effects of exercise are not fully understood but likely include both neurobiological and psychosocial factors. Exercise prescription should consider the participant's motivations, expectations, barriers, and preferences to promote autonomous motivation and long-term adherence.

Learning Exercises

1. Is physical activity associated with a reduced risk of incident anxiety? What is the magnitude of the effect?
2. Does exercise induce panic attacks?
3. Can exercise be used as a complementary treatment for anxiety disorders?
4. Explain how exercise could reduce anxiety.

Further reading

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Chapter 17

Physical Activity and Severe Mental Illness

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Chapter Overview

Severe mental illness refers to a group of disorders that significantly impair a person's psychosocial functioning. Psychosocial functioning is a person's ability to engage in activities of daily life (work, socializing, and play) in a way that is meaningful. Although many other mental disorders can be severe, the term *severe mental illness* includes diagnoses of schizophrenia, bipolar disorder, and major depressive disorder. People living with severe mental illness can experience marginalization due to societal stigma in addition to poor physical health and reduced life expectancy. The impact of these disorders on a person's physical health will be explored throughout this chapter, along with the beneficial role that physical activity can play in their prevention and management.

The concept of *lifestyle psychiatry* will be explored, outlining the importance of addressing modifiable risk factors for people living with severe mental illness. In addition to physical activity, this chapter will also examine the vital role of sleep on a person's physical and mental health, the role of exercise professionals within mental health systems, and the process of integrating physical activity interventions as part of routine mental health treatment in real-world settings.

What is Severe Mental Illness?

Many mental health conditions can be considered as *severe* for the people who experience them, as well as their support networks. However, the phrase *severe mental illness* refers to a specific group of mental health disorders including schizophrenia, bipolar disorder, and major depressive disorder, that significantly impact a person's functioning in daily life and in some cases, may be associated with a distorted sense of reality. Severe mental illness can also be defined by the length of duration and the subsequent disability it has on a person's life.

Here we outline some severe mental illness diagnoses as well as common symptoms, to provide a brief context for how physical health can be subsequently impacted and the role that physical activity can provide as part of a treatment plan. Examples of severe mental illnesses are schizophrenia, bipolar disorder, and major depressive disorder.

Schizophrenia

This complex disorder can impact a person's mental health, physical health, and cognitive function. Lifetime prevalence for developing schizophrenia affects approximately 1% of the population (Kessler et al., 2005). It can produce intense periods of psychosis and negatively impact long-term functioning. Symptoms of schizophrenia are generally divided into three broad categories: *positive symptoms*, such as hallucinations or delusions, which are "added" to someone's experience; *negative symptoms*, describing loss of experience, which can include social withdrawal, decreased enjoyment in activities, and reduced motivation; and finally, *cognitive impairment*, which is associated with changes in memory, attention, cognition, and executive function.

Bipolar Disorder

Bipolar disorder significantly impacts a person's mood or *affect* and is characterized by periods of elevated moods or *mania* where people can present with increased energy, impulsivity, and reduced sleep. This can be followed with interchanging significant depressive episodes. The lifetime prevalence for developing this disorder is approximately 0.6% (Merikangas et al., 2010). Bipolar disorder can also include experiences of psychosis.

Major Depressive Disorder

Major depressive disorder is a prevailing and debilitating mental health condition and is one of the leading contributors to the global burden of disease. Major depressive disorder has approximately a 16% lifetime prevalence (Kessler et al., 2005) and accounts for 8.2% of the global years lived with disability in 2010 (Ferrari et al., 2013). Symptoms that are commonly experienced with major depressive disorder include persistent low mood and reduced interest in activities or hobbies, in addition to poor sleep, changes in appetite, and in some cases suicidal ideation. People living with major depressive disorder may also experience declined social functioning, such as difficulties with maintaining relationships and employment.

Psychosis

Psychosis describes a group of symptoms which are experienced by people with a psychotic disorder (e.g., schizophrenia). Symptoms of psychosis impact an individual's thoughts and perceptions and are defined by a distortion in a person's perception of reality. This can include senses including smell, thoughts, visions, tactile sensations, and beliefs. Most people that develop a psychotic disorder will experience symptoms prior to 25 years of age (Morgan et al., 2012). It is important to recognize that psychosis is *a symptom of a mental disorder*, rather than an illness itself. While we do not fully

understand how psychosis develops, various factors can contribute to a psychotic episode including family history, mental or physical illness, extreme stress, trauma, and substance use.

Psychosis, which is not to be confused with psychoticism, typically develops in youth or adolescence and is typically accompanied by marked decline in functioning, changes in behaviour including difficulties concentrating, sleep disturbances, withdrawing from social supports, disorganized speech, depression, and anxiety. The first episode of psychosis or *FEP* is when someone experiences psychosis for the first time. Symptoms of psychosis can include *hallucinations*, *delusions*, and *disordered thinking and cognition*.

Hallucinations

Hallucinations occur when someone experiences sensations inconsistent with reality. Hallucinations are most commonly auditory (e.g., hearing voices), but may also be visual (e.g., seeing shapes or figures), in tactile form (e.g., experiencing sensations or feelings), olfactory (e.g., smelling odors that are not present), or gustatory (impacting taste).

Delusions

Delusions refer to beliefs that are not based on reality and may impact individual actions or functioning. Delusions can vary depending on the individual and the illness. Delusions may, for example, be paranoid, grandiose, somatic, persecutory, or romantic in nature.

Disordered Thinking and Cognition

Speech and thought patterns may become confused and mixed-up. People may experience problems with memory and other cognitive process (e.g., concentration). This can result in people behaving or communicating differently.

Early Psychosis and Ultra-High Risk

Those with *at-risk mental states* or *ultra-high risk* refers to a group of individuals who are at high risk of developing psychosis; this is often referred to as the *prodromal period* (Galletly et al., 2016). To meet criteria for ultra-high risk, individuals may exhibit a number of factors including genetics or family history, the presence of some brief limited intermittent psychotic symptoms (BLIPS), or attenuated (sub-threshold) symptoms of psychosis (Yung et al., 2005). Youth in these categories will be diagnosed by an experienced mental health clinician using a validated assessment tool (Nelson, 2014). Early intervention is a critical period, for both physical and mental health outcomes, to improve the future well-being for youth experiencing psychosis for the first time (Shiers et al., 2014).

Health Risks for People with Severe Mental Illness

People living with severe mental illnesses face high rates of premature mortality compared to members of the general population (Baxter et al., 2016). The major contributor to this early mortality is poor physical health, often linked to non-communicable diseases. In particular, people with severe mental illness have a 1.4–2.0x increased risk of developing cardiometabolic diseases compared to people without mental illness (Firth et al., 2019). This accounts for approximately 1/3rd of the premature mortality (John et al., 2018). This is associated with higher rates of diabetes, obesity, metabolic syndrome, and respiratory complications. Additionally, people living with severe mental illness face high levels of disability, with schizophrenia being the 12th largest cause of years lived with a disability worldwide (Vos et al., 2017).

Impact of Modifiable Risk-Factors

People living with severe mental illness are at high risk of cardiometabolic disease impacted, in part, by multifactorial lifestyle risk factors including cigarette smoking, physical inactivity, consuming poor diets, and experiencing poor sleep (Firth et al., 2020).

Cigarette Smoking

People with severe mental illness are more likely to smoke cigarettes, with approximately 1 in 2 people a smoker compared to approximately 1 in 10 people in the general population (Gilbody et al., 2015). People with severe mental illness are more dependent on nicotine and have increased difficulty engaging in smoking cessation programs compared to those in the general population (Ashton et al., 2013; Malpass et al., 2009).

Physical Inactivity

People with severe mental illness are less likely to be physically active than the general population and struggle to meet recommended physical activity guidelines of a minimum of 150 minutes of moderate-intensity physical activity per week (Vancampfort, Firth, et al., 2017). There are differences in physical activity engagement between differing severe mental illness diagnoses; people with bipolar disorder engage in more physical activity than people with schizophrenia or major depressive disorder. Symptomology and medication side-effects can affect engagement (this will be explored in greater detail later in this chapter). People with severe mental illness are more sedentary than the general population and on average are sedentary between 8–12 hours of the day (Vancampfort, Firth, et al., 2017). High levels of sedentariness may be further complicated by factors associated with the mental health disorders such as social isolation and obesity (Vancampfort, Firth, et al., 2017).

Poor Diet

Compared with the general population, people with severe mental illness tend to have diets that are low in nutritional quality and have a higher caloric intake. Medication side effects can increase a person's desire to eat; desire for energy-dense, low-nutrient foods is particularly increased (Teasdale et al., 2017).

Sleep and Mental Illness

As in the general population, sleep is a significant component of maintaining good physical and mental health and represents a growing facet of mental health research (Firth et al., 2020). When sleep is impaired, it can lead to weight gain, increasing a person's risk of the development of obesity and metabolic abnormalities. For people with severe mental illness, sleep disturbances are a highlighted risk factor that can lead to the development of psychiatric symptoms in addition to the exacerbation of known symptoms. For people with a mental disorder, up to 90% report abnormal sleep behaviour, which can include difficulty initiating sleep or insomnia (Abad et al., 2005). Symptoms of severe mental illness such as paranoia can impact sleep quality (Afonso et al., 2014), and medications can impact sleep duration and as such decrease sleep quality (Correll et al., 2011). Given the prevalence of poor sleep within this population group, providing interventions that promote good sleep practices is paramount. Importantly, regular physical activity is associated with improvements in self-reported sleep quality in people with mental illness (Lederman et al., 2018).

Social Determinants of Health

High rates of early death and cardiometabolic risk are prevalent in people with severe mental illness even once lifestyle risk factors are accounted for (Hamer et al., 2008). This suggests that physical health is influenced by external factors including the provision and availability of health care,

socioeconomic status, and geographical location. People with severe mental illness are less likely to have access to and engage in quality health care services compared to the general population. Social determinants of health, which includes employment and financial insecurity, homelessness, poor education, and poverty, increase a person's risk of developing mental and physical illnesses (Marmot, 2005).

Learning Exercise One

Describe some common symptoms of the main diagnoses of severe mental illness and include how these symptoms may affect a person's daily functioning.

Current Treatments for Severe Mental Illness

Clinical approaches to managing symptoms of severe mental illnesses vary according to symptom presentation, comorbidities and medical history, demographic factors, and severity of symptoms. People with severe mental illness may experience a combination of treatments including medications (typically antidepressants and antipsychotic medications), talking-based therapies (such as cognitive behavioural therapy), and occupational and social recovery therapies. Physical activity is becoming increasingly recognized as an adjunct treatment for severe mental illness, which will be explored throughout this chapter.

Impact of Anti-Psychotic Medications

The high rates of cardio-metabolic conditions in people with severe mental illness are in part due to the side-effects of antipsychotic medications that are prescribed to treat people with severe mental illness. These medications, whilst effective in reducing symptoms of severe mental illness and reducing risks of readmissions to hospital, have significant negative effects on physical health including the cardiometabolic, endocrine, and neuromotor systems.

Cardiometabolic

Weight gain is a commonly reported side effect from many antipsychotic medications, and increases a person's risk of developing metabolic syndrome, diabetes, and cardiovascular complications (Alvarez-Jimenez et al., 2008). Almost 80% of people who commence antipsychotic medication will experience clinically significant weight gain (more than 7% of their initial weight) within their first year of beginning the medication (Correll et al., 2009).

Endocrine

An abnormally high level of prolactin, or *hyperprolactinemia*, is a side effect of antipsychotic medications (Peuskens et al., 2014). This may cause unwanted side-effects such as sexual dysfunction and menstruation issues in people taking this medication (Knegtering et al., 2003).

Neuromotor

Side effects are more common of the older, first generation *typical* antipsychotics and may include muscle spasms, restlessness, and involuntary movements which can be socially stigmatizing for people it affects and can be associated with a reduced quality of life (Angermeyer et al., 1999; Briggs et al., 2008).

Learning Exercise Two

Outline how a person's physical health may be impacted if they are receiving treatment for schizophrenia.

Benefits of Physical Activity for Severe Mental Illnesses

A robust body of evidence exists documenting the bidirectional relationship between physical activity and mental health (Firth et al., 2019). In addition, physical activity is increasingly recommended as a routine component of treatment for people with severe mental illness, to protect and promote physical and mental health.

Physical activity is effective in improving cardio-metabolic health through various mechanisms including reduced blood pressure, improved glucose metabolism, and increased cardiorespiratory fitness (Henson et al., 2013). Cardiorespiratory fitness is a key risk factor contributing to the premature mortality in people with severe mental illness (Vancampfort, Rosenbaum, Probst, et al., 2015). Cardiorespiratory fitness is the ability of the heart, circulatory system, and lungs to supply oxygen to the skeletal muscles during physical activity. In the general population, improvements in cardiorespiratory fitness can reduce mortality by up to 13% and impact cardiometabolic risk reduction by 15% (Kodama et al., 2009). For people with severe mental illness, reduction in cardiometabolic risk following participation in an exercise program can occur in as little as 8 weeks of structured exercise (Firth et al., 2015; Vancampfort, Rosenbaum, et al., 2017; Vancampfort, Rosenbaum, Ward, et al., 2015). Additionally, it has been demonstrated that lifestyle interventions incorporating both physical activity and dietary components are effective in preventing the medication-induced weight gain associated with anti-psychotic medication (Alvarez-Jimenez et al., 2008; Curtis et al., 2016; Shiers et al., 2014). Severe mental illness-specific mental health benefits of physical activity are outlined below.

Schizophrenia

Physical activity, particularly performed at a moderate-to-vigorous intensity, can lead to improvements in both positive and negative symptoms for people with schizophrenia (Stubbs et al., 2018). Additionally, global cognition (which refers to a person's functioning and working memory) improves following physical activity engagement. These cognitive improvements can occur with as little as 90 minutes of physical activity weekly that is of moderate-to-vigorous intensity (Firth et al., 2017). Structured exercise that elicits increased fitness is correlated with increased brain volume, which may impact neurogenesis (the process by which new neurons are formed), through plasticity in the volume of the hippocampal brain region (Pajonk et al., 2010; Scheewe et al., 2013).

Major Depressive Disorder

Physical activity is effective in improving mood and well-being in people with depression following a single session of exercise, particularly those that are of moderate or higher intensity (Schuch, Deslandes, et al., 2016). Additionally, regular participation in physical activity in healthy people is beneficial for reducing future risk of developing depression (Firth et al., 2018). Studies indicate that physical activity is effective in treating mild-to-moderate depression with effects similar to traditional treatments including antidepressant medication and with fewer negative side-effects (and many adjunct positive effects; Schuch, Vancampfort et al., 2016). For people with major depressive disorder, physical activity and exercise is recommended as adjunct to standard care by the World Health Organisation (World Health Organization, 2018). While the mechanisms for how physical activity improves depressive

symptoms are not fully clear, research indicates that physical activity can improve abnormal biomarkers associated with major depressive disorder; this can include cortisol reduction and decreases in chronic inflammation (Kandola et al., 2019). For more on the relationship between exercise and depression, see Chapter 15 (Brush & Burani, 2021).

Bipolar Disorder

Early evidence related to the role of physical activity for people with bipolar disorder suggests that physical activity is protective for cardiovascular disease; however, more evidence is needed regarding the impact on mood symptomology (Vancampfort et al., 2018). Research suggests that physical activity may be beneficial for mood fluctuations for people with bipolar disorder due to potential links to improvements to the brain's functional connectivity (Douw et al., 2014). Cardiorespiratory fitness is associated with health-related quality of life in people with bipolar disorder for both physical and mental health domains (Vancampfort, Hagemann, et al., 2017).

Fitness vs. Fatness

People with severe mental illness are more likely to be overweight and obese, which can also impact participation in physical activity, due to not only reduced fitness, but also perceived pain and poor self-esteem (Mishu et al., 2019; Vancampfort et al., 2011). Achieving significant weight loss is difficult for the general population with only one in 210 males and one in 124 females achieving a healthy BMI (under 25 kg/m²) without surgery (Fildes et al., 2015). Exercise professionals and mental health clinicians should encourage people living with mental illness to engage in physical activity regardless of their body weight and perceived body image. It is important to note that without specialized dietary input, exercise is likely to be ineffective in achieving weight loss for people with severe mental illness (Firth et al., 2015). While this may seem disheartening for people, particularly when trying to engage in physical activity, research tells us that improving fitness can reduce cardiometabolic risk and improve mental health, sleep quality, functional capacity, and cognition, regardless of change in weight. This can be a useful educational and motivational strategy for clinicians to assist clients in increasing physical activity levels.

Learning Exercise Three

Discuss (a) the benefits of physical activity in people with schizophrenia and (b) the mechanisms of how physical activity affects depressive symptoms.

Barriers to Engagement in Physical Activity

Low engagement in physical activity is prevalent in the general population and this may be influenced by many factors, including socioeconomic status, low confidence and knowledge about exercise, and lack of enjoyment (Stutts, 2002). Further, people living with severe mental illness experience additional barriers to being physically active that can be categorized into physical health, mental health, and social factors (Firth et al., 2016).

Physical Health

The high rates of co-morbid physical health conditions can negatively impact engagement in physical activity (Firth et al., 2019a). Obesity, associated chronic pain and musculoskeletal problems, can make physical activity difficult and less enjoyable, thus impacting exercise participation (Firth et al., 2016). Low cardiorespiratory fitness, seen in people living with severe mental illness, can also impact a

person's ability to engage in long periods of physical activity, particularly if aerobic in nature. Other issues including low energy and lethargy as a result of antipsychotic medication can be impactful (Firth et al., 2016).

Mental Health

Common barriers from a mental health perspective include depressive symptoms, stress, low motivation, and acute psychotic symptoms (hallucinations, delusions and paranoia, avolition, anhedonia). Additional reported barriers include feeling unsafe when exercising or concerns about being injured, and low interest in exercise (Firth et al., 2016). Long-term engagement in physical activity is impacted by a person's autonomous motivation, that is, an internal desire to participate (Vancampfort et al., 2015e). Given the unique barriers that people with severe mental illness face regarding physical activity participation, developing individual strategies specific to members this population group that target improvements in intrinsic motivation (e.g., health coaching and motivational interviewing) is paramount when aiming to increase engagement (Farholm & Sørensen, 2016).

Social Factors

Strong social support networks assist in the uptake and regularity of physical activity, and individuals with severe mental illness are more likely to experience social isolation which can affect engagement (Daumit et al., 2005). Additionally, the financial costs, lack of resources, and time pressures associated with exercising are common reasons why people with severe mental illness may not engage in physical activity (Chapman et al., 2016; Firth et al., 2016; Stanton et al., 2015).

Learning Exercise Four

What are the key barriers for people with severe mental illness engaging in physical activity?



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Contraindications to Exercise

Unless a person with mental illness has been diagnosed with severe physical issues, physical activity and exercise is generally safe and effective to prescribe, particularly if of low-to-moderate intensity (Stubbs et al., 2018). The unique barriers to engaging in physical activity should be taken into consideration when promoting activities and an individualized approach (including referral to an exercise professional) should be encouraged.

Prior to beginning structured exercise programs, particularly if they are of high intensity, a person with mental illness should be screened for absolute and relative contraindications to engaging in physical activity, as per guidelines recommended by the American College of Sports Medicine (American College of Sports Medicine, 2017). The Physical Activity Readiness Questionnaire (PARQ) is a series of short questions that identifies any absolute contraindications to exercise (Thomas et al., 1992). Additionally, people experiencing higher levels of mental distress or poor mental health symptoms may need additional support to engage in appropriate and safe exercise. Like the general population, people with mental illness should be screened for typical contraindications to exercise (such as hypertension or a recent cardiac event). See Table 17.1 for a list of precautions that should be considered specific to this population (Firth et al., 2016).

Table 17.1

Precautions to Consider for People with Severe Mental Illness

Potential Issues	Precautions and Considerations
Fatigue	Prolonged tiredness and poor concentration are common for people and are influenced by factors including side effects of medication, sleep troubles, and symptomology. An individualized approach to an exercise plan tailored to times when a client may be more energetic is ideal.
Social anxieties	Public places and larger groups of people can act as a deterrent to engaging in physical activity. Setting small goals around exercise and establishing a routine is important initially, and gradually progressing may help with familiarity for clients
Side-effects of medications	In addition to extreme tiredness, other side effects of taking anti-psychotic medication can include visual problems, increased sweating and salivation, and issues with balance. Close monitoring of the client is important throughout the exercise session and adapting to the client's changing needs will assist with increasing their confidence.
Psychotic symptoms	While symptoms such as auditory and visual hallucinations may present complications to engaging in physical activity, they do not preclude a person from participating. Instead, choosing activities that are lower intensity in nature should be considered. Exercises and activities such as body-weight resistance training, walking, and light stretching may be helpful. If people are presenting as manic or with extreme elevated moods, limiting distractions including music or television is encouraged. Exercising in a quiet space without people around can also be helpful.

Learning Exercise Five

What are the key safety concerns for people with mental illness engaging in structured physical activity?

Motivation and Physical Activity

Physical inactivity is not just an issue for people with mental illness as many people in the Australian general population also do not meet the recommended guidelines for physical activity (150 minutes of moderate or 75 minutes of vigorous physical activity weekly; Australian Bureau of Statistics, 2015). In Australia, a little more than half of adults meet the guidelines for physical activity (Australian Institute of Health and Welfare, 2018). Less than half the population in the United States meet the recommended levels of physical activity (Carlson et al., 2015) and rates of inactivity are higher in Europe, where almost 60% of adults do not engage in appropriate levels of activity (Oja et al., 2010). Given the robust evidence demonstrating the benefits of physical activity in improving the physical and mental health of people living with severe mental illness, it is important to examine the ways in which people can be supported to engage in exercise throughout the life course. Examining people's internal perceptions and motivation to exercise is a vital component of physical activity prescription (Chapman et al., 2016; Firth et al., 2016; Vancampfort, De Hert et al., 2015; Vancampfort et al., 2016).

Negative symptoms that are associated with schizophrenia can be associated with lower levels of motivation towards physical activity (Vancampfort, Stubbs, et al., 2015). These low levels of motivation may be related to a person's own priorities towards exercise and their physical health, past experiences with engagement, or beliefs about structured physical activity. Competing interests and priorities including work, study, and family can also be factors that affect motivation, in addition to socioeconomic factors such as lack of access to resources (Ussher et al., 2007). Education about how to engage effectively and safely in structured exercise and fostering self-determined motivation through building self-confidence and a sense of competency can improve participation. People may have had previous negative experiences with exercise, which may affect future performance (Ladwig et al., 2018). The influences that may dissuade people with mental illness from engaging in physical activity should be discussed by their health care providers to formulate an exercise plan that is based on the person's interests and preferences.

Self-Determination Theory

Examining the research into what motivates people to exercise is important for understanding how to support those with severe mental illness with long-term healthy behaviours (for more discussion on physical activity and exercise behavior, see Chapter 2 [Rebar et al., 2021], Chapter 4 [Brand & Ekkekakis, 2021], and Chapter 5 [Delli Paoli, 2021]). Utilising the self-determination theory (Deci et al., 2008) can be an effective way of facilitating long term enjoyment in physical activity (see Chapter 3; Quested et al., 2021). This theory is regularly used by many health professionals, including exercise specialists, to ascertain factors that guide someone's motivation to engage in a particular behaviour (e.g., exercise). It is applicable in the well-being and recovery of people with severe mental illness and can be used in the identification of external and internal forms of motivation. This theory examines specific basic psychological needs including autonomy, relatedness, and competence, and when applied to exercise can help people understand the reasons why they may or may not exercise. The theory states that *intrinsic* motivation is more conducive with exercise participation. Self-determination has been identified as important in people with schizophrenia for engaging in long-term exercise participation (Vancampfort, Stubbs, et al., 2015). So how does this work in practice? Motivational

interviewing can be used with clients that have a mental illness to assist in promoting positive attitudes towards exercise and to help in discovering what motivates them to engage (Farholm & Sørensen, 2016).

Transtheoretical Model of Health Behaviour Change

The transtheoretical model (Table 17.2) suggests that making changes to health behaviours are typically achieved through five stages of change (Prochaska et al., 1997). These stages of change can assist in guiding conversation, education topics, and eventually exercise prescription (Prochaska et al., 1994).

Table 17.2
Stages of Change Based on the Transtheoretical Model

Stage	Stage Description
Pre-contemplation	During this period, people are not generally considering the practical steps to beginning exercise, so it is a good time to provide basic education. Identifying what is most important to them is of importance and gradually relating these goals back to the benefits of physical activity may be appropriate. As a clinician, you may then be able to get a level of understanding as to the client's thoughts towards physical activity.
Contemplation	As a clinician you can use this time to provide more detailed education and challenge beliefs that the person may have about what participating in physical activity might look like. Motivational interviewing should continue.
Preparation	During this stage, the client should engage in goal setting around future planned physical activity, planning next steps, and exploring barriers and strategies to overcome challenges.
Action	The client should begin regular physical activity and ideally in an activity that is enjoyable to them. Work with your client to identify ways in which they might adhere to their schedule. These may include the use of activity trackers or finding a friend or partner to exercise with. Reflecting on past experiences can be helpful to manage mood and psychological states. Providing positive reinforcement should be a focus of the clinician. Working towards maintaining current physical activity levels should be a goal of this period. This includes increasing goal setting to long-term periods in addition to reviewing goals to overcoming obstacles to participation.
Maintenance	At this stage, the client should be able to maintain their current physical activity levels independently and identify what are the internal motivating factors to their participation. New activities can be introduced and a plan to return if lapses occur can be developed.

Throughout these stages, it is important to openly discuss and listen to the client's needs and goals so that a recovery-focused approach to exercise prescription can occur (Slade, 2010). Understanding that symptoms of a person's mental illness may mean that the process may not be linear, and flexibility may be necessary. It can often take multiple attempts before someone succeeds in adopting lifestyle changes for the long term and as such, these stages can often be cyclical in nature. These components should be key in the education provided to clients to manage expectations and assist the client in adherence to the physical activity program.

A typical approach of health professionals when educating people in lifestyle behavior change is to initially educate and then persuade them to adopt new behavior through prescriptive directions. However, this approach is not typically effective as the simple act of providing information does not result in long-term behavior change. *Health coaching* is a method of utilizing the motivational theories previously discussed (such as the self-determination theory and the transtheoretical model) and practically applying them in a real-world context (Table 17.3).

Table 17.3

Consider the Ways in Which People with Mental Illness Might Benefit from a Motivational Health Coaching Approach to Physical Activity Education Compared to a Traditional Model

Traditional Interviewing Style	Health Coaching
Clinician is the expert in health topic	Client is the expert in their own life experience and is respected to make autonomous decisions
Clinician give advice and prescribes solutions to solve problems	Client identifies areas of health they would like to work on, and clinician works through options available providing information as requested
Clinician decides that client is at the stage to make health changes	Client leads change and clinician uses strategies to increase client confidence and self-determination
Clinician focuses on why things are not being achieved and solutions to this	Clinician provides positive encouragement at whatever stage client is in

Learning Exercise Six

Imagine a person with schizophrenia has been referred to you for lifestyle intervention. Utilising the concepts of the self-determination theory, outline the key concepts of your interview and physical activity prescription, given the person is in the preparation phase of the transtheoretical model of health behaviour change.

Exercise Professionals in Mental Health Services

There is overwhelming evidence supporting the benefits of physical activity for people living with severe mental illness, yet there are simultaneously significant barriers to participation. As such, embedding exercise professionals within mental health services to promote the integration of exercise interventions has been of growing interest to expand upon the treatment options available to people with severe mental illness.

Exercise professionals are experts in physical activity prescription, particularly for people living with chronic disease (including mental illness) and provide assessment, prescription, and delivery of exercise and physical activity interventions (Lederman et al., 2016; Stanton et al., 2018; Stubbs et al., 2018). Throughout Europe, physiotherapists have performed these roles and in the United States there are a growing number of clinician exercise physiologists working in mental health settings (Fibbins et al., 2019; Stubbs, Probst, et al., 2014; Stubbs & Rosenbaum, 2018).

Australia has an increasing number of exercise professionals working in private and public health systems, particularly in mental health services. Accredited Exercise Physiologists (AEPs) are university-qualified health professionals that lead physical activity interventions for people living with mental

illness. There are robust referral schemes that people can access for evidence-based exercise counselling and prescription through universal-health care. AEPs have growing representation in many mental health care settings including community health centers and inpatient wards. They work to provide specialized services to mental health clients delivering individualized interventions in addition to upskilling the mental health workforce to be more knowledgeable regarding the benefits of lifestyle interventions.

In response to this rapidly growing industry, exercise professionals specializing in mental health are represented internationally by peak bodies and organizations. A joint consensus statement by peak professional organizations throughout Australia, the U.S, the U.K, and New Zealand determined that integrating physical activity interventions lead by exercise specialists was key to improving the physical health of people living with mental illness (Rosenbaum et al., 2018). The International Organization of Physical Therapists in Mental Health (IOPTMH) supports exercise professionals and upholds them as experts in lifestyle management and integral members of the multidisciplinary health team (Stubbs, Soundy, et al., 2014b). Exercise and Sports Science Australia (ESSA) is the peak organization that represents and provides accreditation to AEPs within Australia and promotes their services to mental health bodies and the wider community. Promotion through these organizations is critical to promoting the services of exercise professionals and as such there have been emerging recommendations for exercise to be included as part of standard mental health care treatment. This is represented in documents published by the American Medical Society for Sports Medicine, Canadian Network for and Anxiety Treatments (CANMAT), the National Institute for Health and Care Excellence (NICE), and the Mental Health Commission of NSW (Chang et al., 2020; Mental Health Commission of New South Wales, 2016; Stanton et al., 2014).

The Importance of Culture and Organizational Leadership

Despite the overwhelming evidence demonstrating the need for integrated physical activity interventions in mental health settings, there are significant challenges to their practical implementation. Traditionally, physical health in mental health settings has not been prioritized despite key principles, processes and standards to do so enshrined in international policies and guidelines (Shiers et al., 2014).

Clinician and workplace culture has been identified as a novel and targeted way of bridging the traditional siloed structure between physical and mental health. There is evidence suggesting that health workers that partake in healthy lifestyle behavior are more likely to promote such behaviors to their patients and refer them to exercise professionals (Rosenbaum et al., 2018; Stanton et al., 2015). As such, designing workplace lifestyle interventions, incorporating exercise and diet, and providing them to mental health staff may be a key strategy to improve outcomes for patients. Such interventions in mental health settings are feasible and acceptable (Fibbins et al., 2018) and do not negatively impact productivity. Additionally, improvements to workplace satisfaction, morale, and workforce capacity can be improved through such interventions (Forsberg et al., 2008; Hjorth et al., 2016; Tucker et al., 2016).

A Real-World Lifestyle Intervention for Youth with Mental Illness

The Keeping the Body in Mind (KBIM) program (Curtis et al., 2016) is a lifestyle intervention implemented as part of routine clinical care in a public mental health service in Sydney, Australia. The service includes weekly individualized consultations with a physical health team including a nurse, exercise physiologist, dietitian, and peer support worker. There is a free onsite gym that is serviced by student exercise physiologists, in addition to cooking and sports groups.

Initially evaluated as a pilot research project, youth with psychosis were offered this innovative program over a 12-week period and results were compared to another mental health service offering standard care (medication and psychotherapy).

The KBIM program successfully prevented antipsychotic related weight gain following the intervention while participants in the control group gained almost 8kg in weight and 7.0cm in waist circumference over the same time. Additional improvements were also evident, including improved cardiorespiratory fitness and significant improvements in discretionary food intake and increases in diet quality. As a result, the pilot program was implemented as part of standard care and now delivers interventions by multiple KBIM teams employed on a full-time basis across a large mental health district.

Conclusion

Physical activity interventions should be a key component of the standard treatment for people living with severe mental illness. These interventions can be safe, are highly acceptable, and result in improvements in both physical and mental health outcomes. Exercise professionals should be included within multidisciplinary mental health treatment teams and be referred to by primary care physicians and mental health team members. The uptake of such services is vital to improving health outcomes for people living with severe mental illness.

Further Reading

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Within performing arts settings, typically services can include individual clients accessing support via self-referral or referral from a professional company/training institution/health-care provider; the facilitation of group workshops on key performance-psychology topics or sessions that focus on collective group issues; and the provision of consulting advice to management and administration about addressing psychological issues, recommended solutions, curriculum, or the development of specific programs (see Moyle 2019a). Similar to sporting settings, being an embedded SEPP practitioner in an organisation or program has been observed to be more effective than being an external consulting psychologist. However, usually reduced funding within performing arts settings that can be allocated to SEPP services can limit the scope of practice and service provision that can be achieved. In such circumstances, prioritisation of identified servicing needs is critical.



Photo credit: James Dillon, QUT Dance

Exercise

When focusing upon the exercise side of sport and exercise psychology, there are a range of career opportunities across medical, health, and occupational related settings. SEPP practitioners often work in hospitals and clinics as part of a multidisciplinary health team dealing with such issues as pain management, weight-loss, smoking cessation, injury rehabilitation, including occupational rehabilitation regarding supporting individuals recovering from physical or psychological injury to return to employment. Additionally, many SEPP practitioners will work as the sport and exercise psychologist within a physiotherapy practice, assisting in the case management of individuals undertaking sports injury rehabilitation programs.

Other Areas of Professional Practice

Training in SEPP provides a strong foundation and set of skills that can position graduates to apply their knowledge to areas beyond sport yet related through a focus on supporting performance and enhancing human potential. For example, sport and exercise psychologists may work in career counselling, given a significant part of supporting athletes and performing artists is with transition—including entry into, during, and exiting out of their careers. Working in employment services is another sector where the specific skills that SEPP practitioners develop are very beneficial, whereby facilitating

group workshops in topics like self-confidence, motivation, and goal-setting compliment individual sessions with job seekers in exploring strengths, skills, and career aspirations. Furthermore, depending upon the training pathway and the tailored experiences students undertake, SEPP practitioners are valued within business and corporate settings with their ability to support individuals, teams, and leaders obtain the best performance and outcomes possible.

Further Training and Endorsement

Depending upon what type of career opportunities and experiences SEPP practitioners end up having, alongside what regulatory framework is in place, there are further training and endorsement options that can be explored. For example, SEPP practitioners interested in clinical sport psychology can look to upskill in clinical areas through short courses and professional development activities to further enhance their understanding, knowledge, and skills related to those areas. Additionally, in Australia as an example, if psychologists wish to obtain a second endorsed area of practice (e.g., clinical psychology), they can choose to undertake a bridging program that meets the registration and endorsement requirements for that additional field of practice (see Figure 33.2). Further areas of endorsement also require sport and exercise psychologists to meet continuing professional development for each specific area of practice. Another avenue for further training and development includes ongoing supervision, which can involve formal supervision from a more experienced psychologist, as well as peer supervision both individually and in groups. Ensuring that you continually reflect upon your practice or psychological work (i.e., teaching and research), is an integral part of being an effective psychologist.

Key Tips: Developing Your Career

When thinking about developing your career as a future sport and exercise psychologist, the following key tips may be helpful:

- Explore and be open to all possibilities, particularly the unexpected ones. Whilst you might have a dream of working with an Olympic sport or professional sporting team, opportunities to work as a psychologist or SEPP practitioner in a range of settings will only assist to further develop your skills, knowledge, and experience. Rounding out your experience in psychology simply makes you more employable and able to deal with, and work, in a variety of settings.
- Reach out to psychologists who are doing what you would like to do. For the most part, more experienced psychologists are only too willing to invest a bit of time answering questions from psychologists-in-training. Remember, they were once students too who were also nervous and excited about what their career might look like. Offer to take them for a coffee/virtual coffee as a thank you for their time. Be professional and make the effort to follow-up and stay in touch. One day they might end up being your colleague or friend!
- Find a mentor. A mentor might be a supervisor, or they might be someone in the profession that you admire or has the type of career you aspire too. Mentors can provide career advice, not just content advice, and can provide you with introductions to key people and networks.
- Invest in your ongoing development. Your learning does not stop with graduation or registration. Ensuring that you are an effective psychologist means that you will continually be developing your knowledge, and skills. Additionally, do not limit your life-long learning solely to psychology topics or for continuing professional development points. There are a range of areas that will be of benefit to you as a SEPP practitioner, that are not always covered in your psychology courses (e.g., leadership development, marketing, presentation skills, proposal development, business acumen, professional report writing of the nonpsychological type).

Conclusion

Being a sport, exercise, and performance psychologist is an exceptionally rewarding career. It provides variety, different challenges, and presents opportunities to work across a range of different settings to support people be the best that they can be. Investing time at the beginning of your training to be strategic in your choices and tailoring your experiences along the way, will enable you to more likely achieve the type of SEPP roles and work that you would most like to do.

Learning Exercises

1. In the country you wish to practice as a SEPP practitioner, what is the name of the regulatory body or professional organisation that is responsible for governing this process?
2. Identify what training and accreditation pathways these organisations state are available to become a qualified/licensed SEPP psychologist.
3. Check whether your current/future undergraduate and/or graduate/postgraduate program is listed as an accredited course in order to meet the training requirements to become a SEPP psychologist.
4. Review your current/future undergraduate and/or graduate/postgraduate programs to identify internship/placement opportunities. When do these occur within your training pathway (e.g., Semester 2, Year 2)? Are these internships/placements organised for you or do you have to identify/approach organisations yourself?
5. Reflect upon the type of client populations that you are most interested in/not interested in working with. Why? What would be the benefits of working with both of these populations in the context of becoming a SEPP practitioner?
6. What SEPP topics are you most interested in undertaking researching about? Who are the leading researchers (globally) within these areas?
7. What type of Vacation/Summer Semester schemes are available at your University/College regarding opportunities to be involved in research projects during the break?
8. Identify which SEPP related professional associations in your country have activities that are available in your local area or via virtual platforms. Attend at least two events and connect with at least one other SEPP practitioner and/or student per event to start forming your professional network outside of your own University/College.